Enraf Auxitrol Marine
158, Bureaux de la Colline
92213 Saint Cloud cedex
France

Tel. + 33 (0)1 49 11 65 75
Fax. + 33 (0)1 49 11 65 76
Email : contact@enrafmarine.fr

MANUFACTURING FILE

FINAL

OWNER :	Maks Denizcilik
HULL :	NB 1434
SHIPYARD:	ANADOLU

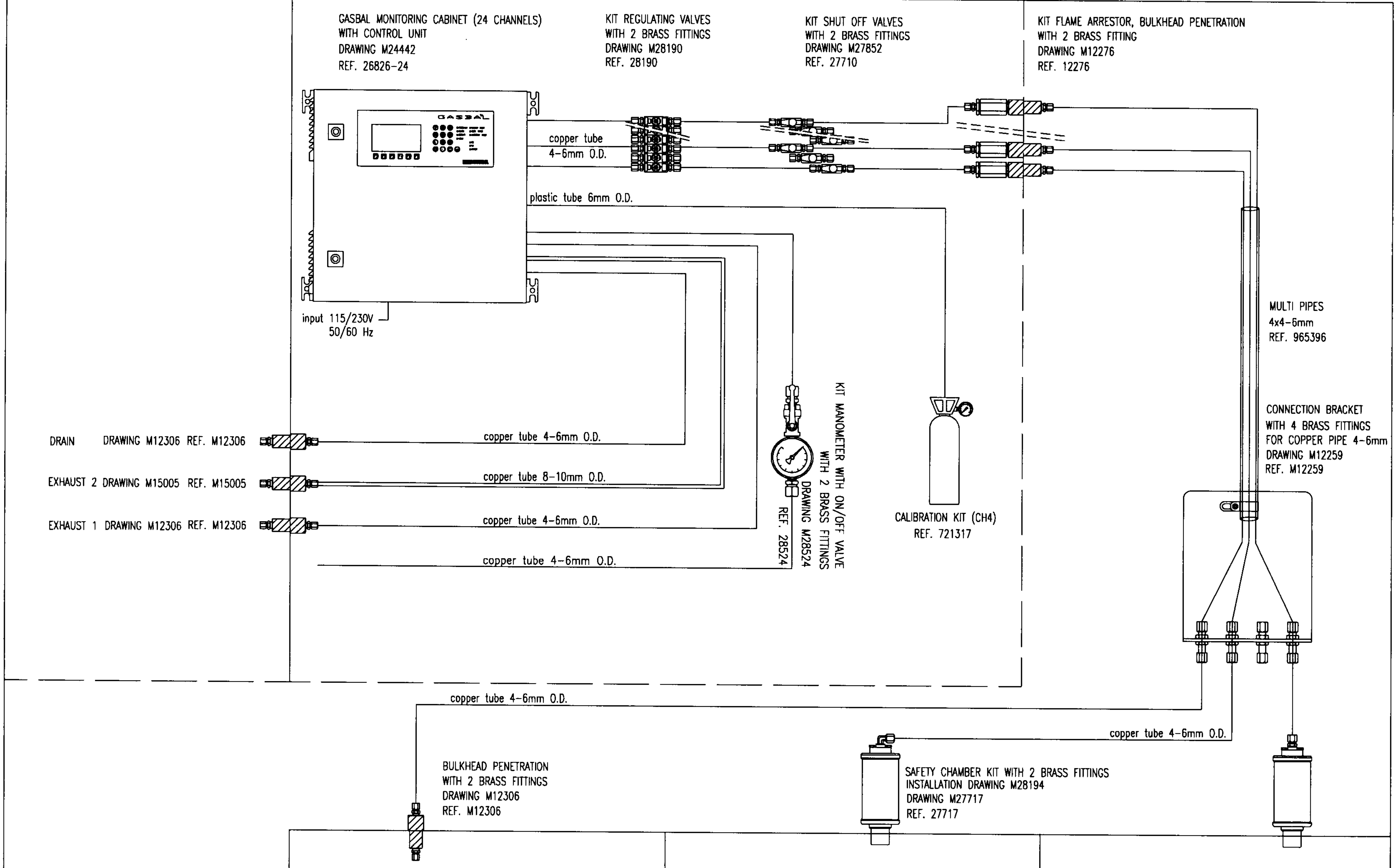
ENRAF AUXITROL MARINE Ref. :900981
Please remind this ENRAF AUXITROL MARINE Ref. N° for spare parts and any inquiries.

92213 Saint Cloud cedex
France

Tel. + 33 (0)1 49 11 65 75
Fax. + 33 (0)1 49 11 65 76
Email : contact@enrafmarine.fr

"FINAL" - CONTENTS

1



GAS DETECTION SYSTEM
MAKS DENIZCILIK - ANADOLU NB1434

900981GB-00						
SHEET	MOD	DATE	DESCRIPTION	DRAWN	CHECKED	APPROVED

ENRAF AUXITROL MARINE

158, BUREAUX DE LA COLLINE
92213 SAINT CLOUD CEDEX FRANCE
FAX: +33 (0)1 49 11 65 76
TEL: +33 (0)1 49 11 65 75
contact@enrafmarine.fr

ENRAF AUXITROL MARINE

ENRAF AUXITROL MARINE SAS
Bât. 59, Rue ISAAC NEWTON
ZA Port Sec Nord
18000 BOURGES (FRANCE)

ORDER ACKNOWLEDGEMENT

Our Reference : **900981**

Printed the : **21/01/2004** Page **1 / 3**

Agent : PE-GU MARITIME LTD
Currency : EURO
Fax :

Id CEE : FR 86 444 871 933

Shipping Address	Customer Address 99894
MAKS DENIZCILIK TICARET VE SANAYI A.S. MEHMET AFKAN SOKAK N° 7 81020 KOSUYOLO ISTANB TURKEY	**MAKS DENIZCILIK TICARET** VE SANAYI A.S. MEHMET AFKAN SOKAK N° 7 81020 KOSUYOLO ISTANB TURKEY

Customer Ref. / Date	Payment mode	Shipping by	Payment term
confirmation 04/12/2003	VIREMENT		PER ADVANCE

Line	Item / Designation	U/M	Qty Ordered	Unit Price	Due Date	Net Amount
1	**26826-24** ARMOIRE GASBAL 24V. av. unité contrôle Ref.26826-24 - P/N.M24442 Notes: Including: -Penetration for Drain and Exhaust (2) -Manometer with valve	UN	1.000		30/01/04	
2	**27717** TOP TANK SYSTEM (gasbal) 4-6 copper tube Ref.27717 - P/N.M27717	UN	17.000		30/01/04	
3	**M12306** TRAVERSEE ET SYST.GAZ 2x4-6 cuivre REF. M12306 - P/N. M12306	UN	2.000		30/01/04	
4	**34051** TRAVERSEE ET + FLAME ARRESTOR 2x4-6 cu	UN	19.000		30/01/04	
5	**28190** KIT ROBINET DE REGUL. 1/4" GASBAL LAITON Ref.28190 -	UN	19.000		30/01/04	
6	**27710** KIT VANNE D'ARRET 1/4"G GASBAL Kit shut-off valve with 1/4"G brass fittings Ref. 27710 - P/N M27710	UN	19.000		30/01/04	

ENRAF AUXITROL MARINE

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Customer Ref. / Date	Payment mode	Shipping by	Payment term
confirmation 04/12/2003	VIREMENT		PER ADVANCE

Line	Item / Designation	U/M	Qty Ordered	Unit Price	Due Date	Net Amount
7	**30811** TR Gas Test Bottle	UN	1.000		30/01/04	
8	**M12259** EQUERRE CONNEXION GASBAL (tube cu. 4-6) REF. M12259 - P/N. M12259	UN	4.000		30/01/04	
9	**965396** POLYTUBE CUIVRE GAINE PVC NOIR COMPOSE DE 4 TUBES DIAMETRE EXT=6mm, EPAISSEUR=1, EN LONGUEUR DE 210m MAXI PAR TOURET POLYTUBE CUIVRE GAINE PVC NOIR COMPOSE DE 4 TUBES DIAMETRE EXTERIEUR = 6 MM, EPAISSEUR = 1 MM EN LONGUEUR DE 210M MAXI PAR TOURET REF. 965396	MT	300.000		30/01/04	
10	**965410** Tube cuivre gaine PVC diamètre EXT=6mm, épaisseur=1mm en couronne de 125m (0, +1) Copper pipe 1 x 4/6 mm with sheath REF. 965410	MT	300.000		30/01/04	
11	**28219** KIT RECHANGE SYSTEME GASBAL -1 Pump piston -2 Pump gasket -1 Pump membrane -2 Electro valve -1 Filter cartrige -2 Flame arrestor -1 Stop valve -1 Regulating valve -2 Fuse 1.6A	UN	1.000		30/01/04	

ENRAF AUXITROL MARINE

ENRAF AUXITROL MARINE SAS
Bât. 59, Rue ISAAC NEWTON
ZA Port Sec Nord
18000 BOURGES (FRANCE)

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Customer Ref. / Date	Payment mode	Shipping by	Payment term
confirmation 04/12/2003	VIREMENT		PER ADVANCE

Line	Item / Designation	U/M	Qty Ordered	Unit Price	Due Date	Net Amount
12	**20959** COMMISSIONING CM	UN	1.000		30/01/04	

Notes de commande:
29500 DWT Tanker Gas Detection System

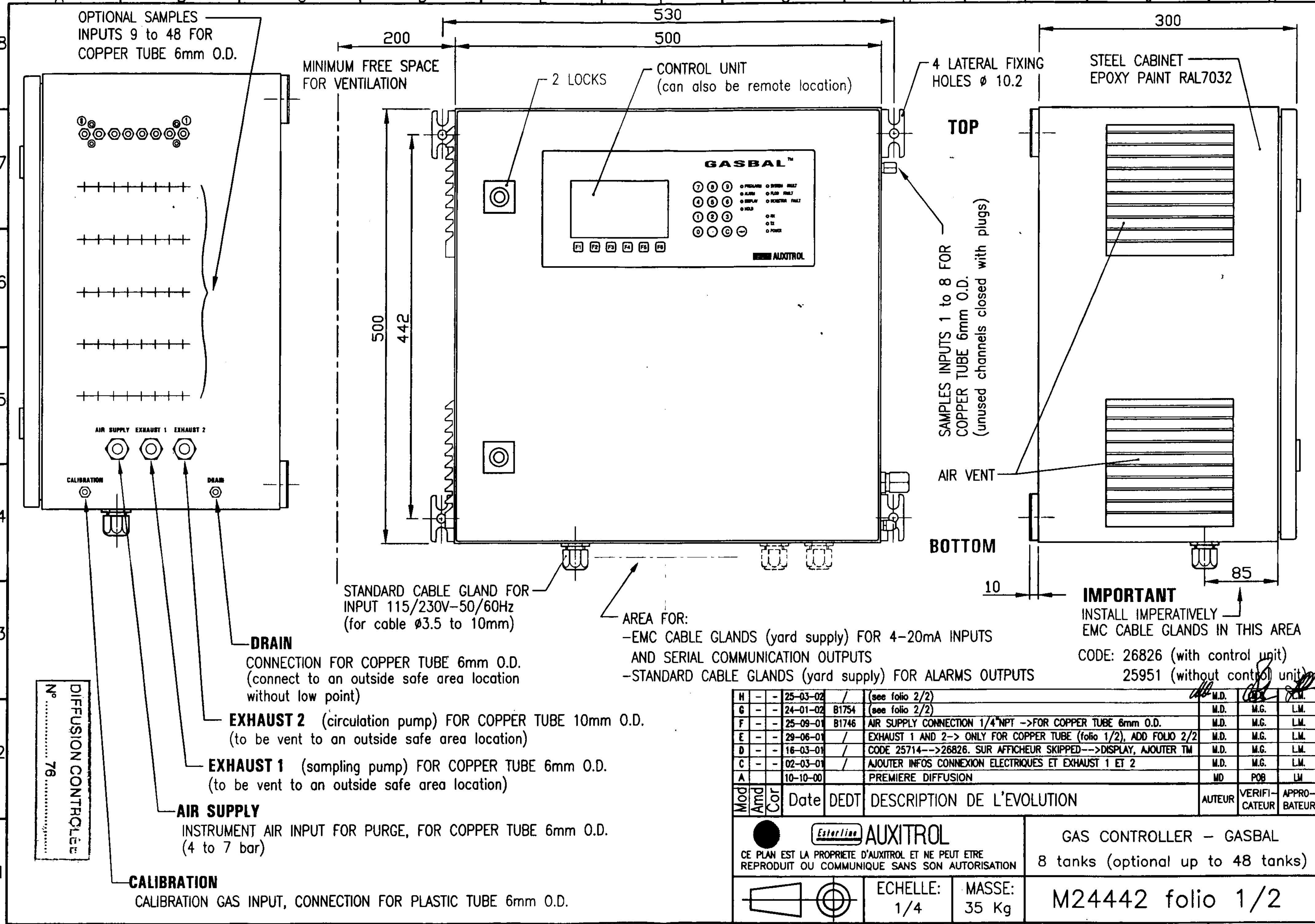
PAYMENT : CASH AGAINST DOCUMENTS

Adresse de facturation / Invoicing address
ENRAF AUXITROL MARINE Bât 59, rue Isaac NEWTON
18000 BOURGES (FRANCE)

HT Amount :	
Customer Discount :	0.00
Net Amount :	
Packing charges :	0.00
Shipping :	0.00
VAT :	0.00
Advance payment :	0.00
TTC Amount :	

2

2-1



Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFICATEUR	APPROBATEUR
H	-	-	25-03-02	/	(see folio 2/2)	M.D.		L.M.
G	-	-	24-01-02	B1754	(see folio 2/2)	M.D.	M.G.	L.M.
F	-	-	25-09-01	B1746	AIR SUPPLY CONNECTION 1/4"NPT ->FOR COPPER TUBE 6mm O.D.	M.D.	M.G.	L.M.
E	-	-	29-06-01	/	EXHAUST 1 AND 2-> ONLY FOR COPPER TUBE (folio 1/2), ADD FOLIO 2/2	M.D.	M.G.	L.M.
D	-	-	16-03-01	/	CODE 25714-->26826. SUR AFFICHEUR SKIPPED-->DISPLAY, AJOUTER TM	M.D.	M.G.	L.M.
C	-	-	02-03-01	/	AJOUTER INFOS CONNEXION ELECTRIQUES ET EXHAUST 1 ET 2	M.D.	M.G.	L.M.
A			10-10-00		PREMIERE DIFFUSION	MD	POB	LM

Esterline AUXITROL

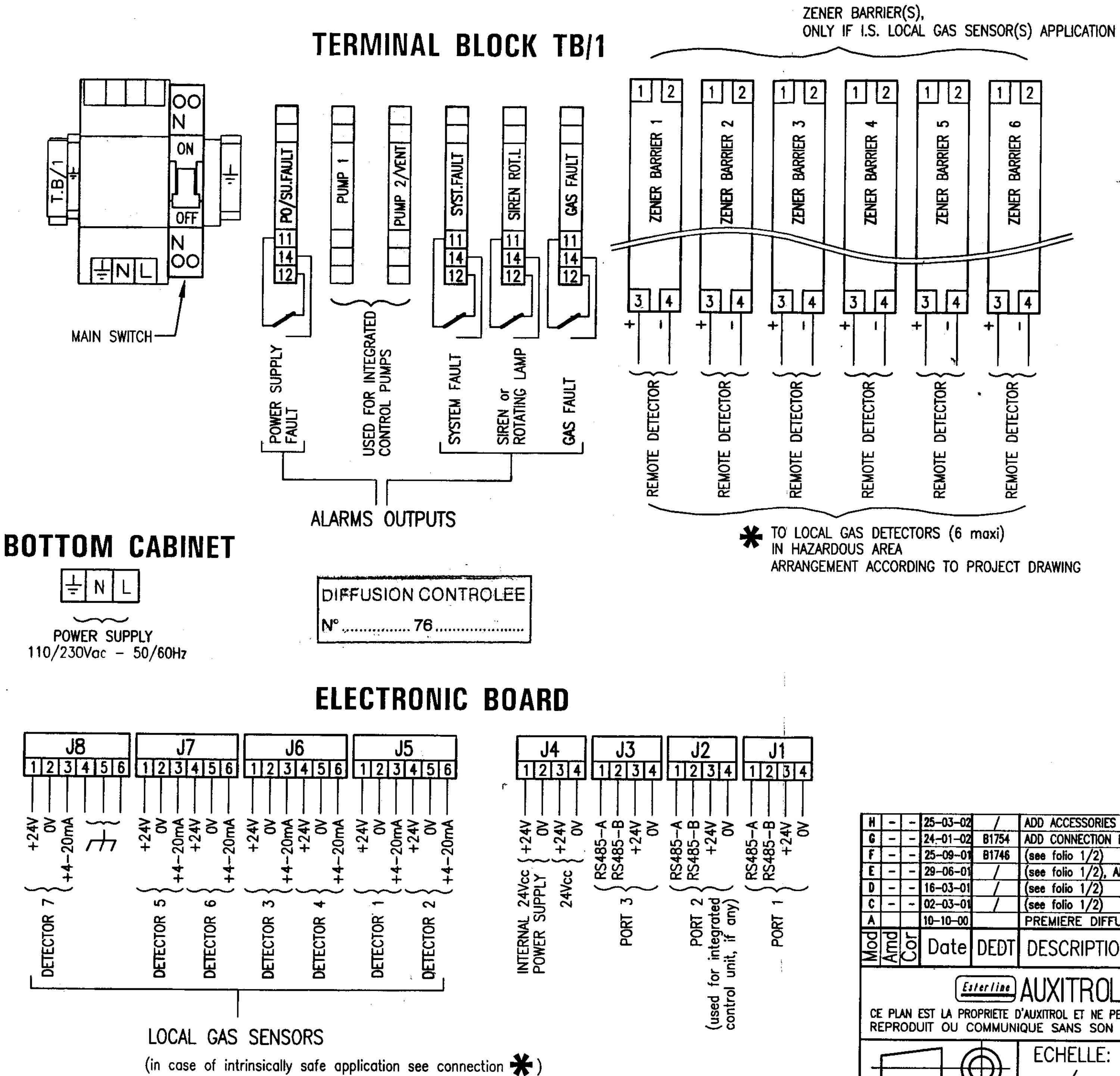
GAS CONTROLLER – GASBAL

8 tanks (optional up to 48 tanks)

ECHELLE: 1/4

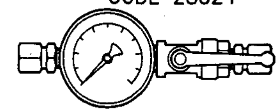
MASSE: 35 Kg

M24442 folio 1/2



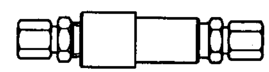
ACCESSORIES SETS

1x MANOMETER WITH VALVE
CODE 28524

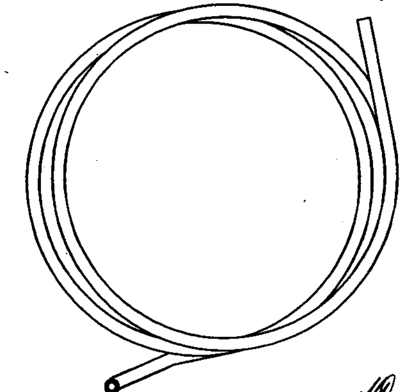


2x ø6mm BULKHEAD PENETRATION
FOR SAMPLING EXHAUST 1 AND DRAIN
CODE M12306

1x ø10mm BULKHEAD PENETRATION
FOR CIRCULATION EXHAUST 2
CODE M15005



CALIBRATION TUBE (1 meter)
CODE 720840



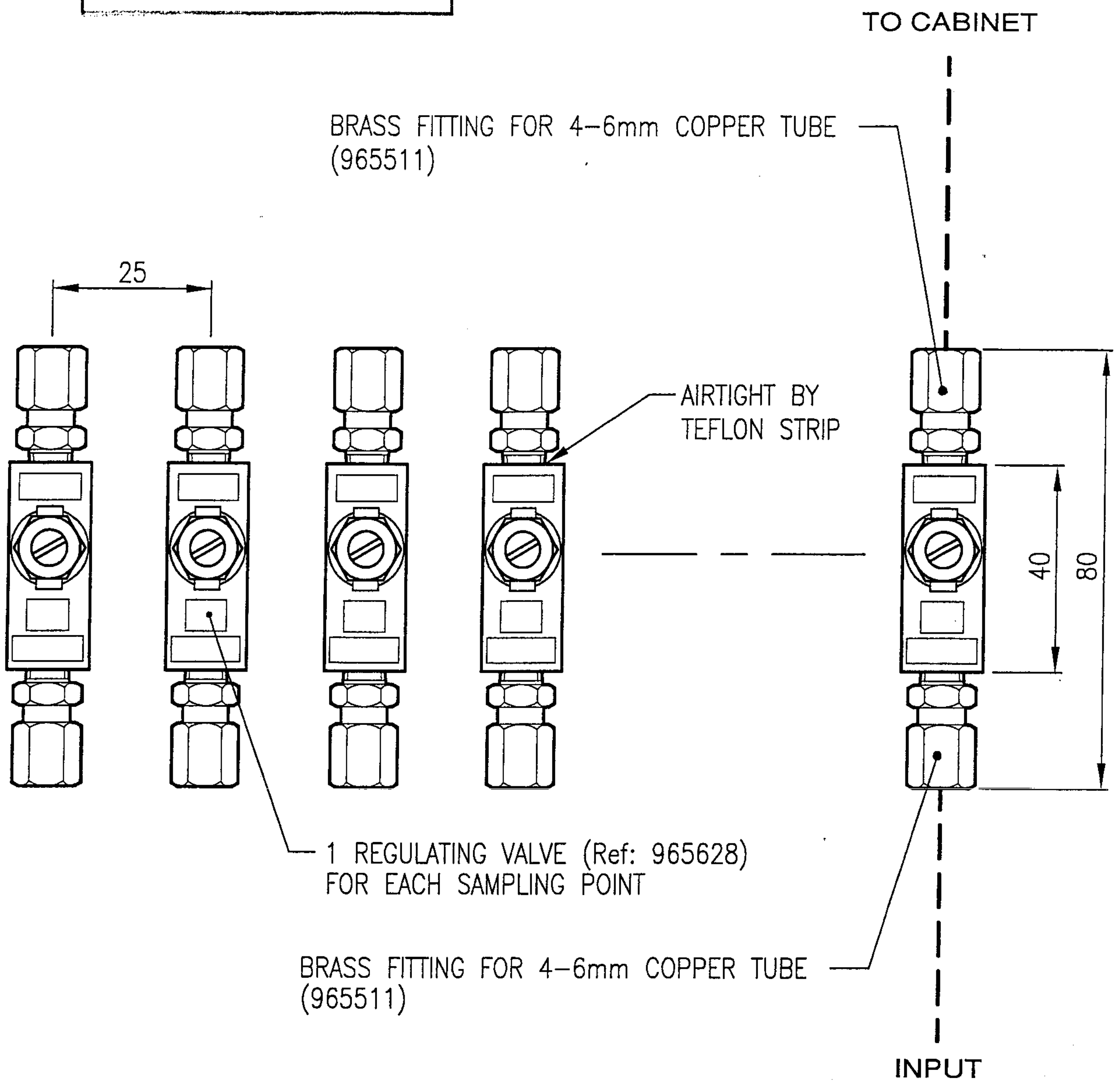
Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFI-CATEUR	APPRO-BATEUR
H	-	-	25-03-02	/	ADD ACCESSORIES SETS	M.D.	M.G.	L.M.
G	-	-	24-01-02	B1754	ADD CONNECTION FOR I.S. LOCAL SENSORS	M.D.	M.G.	L.M.
F	-	-	25-09-01	B1746	(see folio 1/2)	M.D.	M.G.	L.M.
E	-	-	29-06-01	/	(see folio 1/2), ADD FOLIO 2/2	M.D.	M.G.	L.M.
D	-	-	16-03-01	/	(see folio 1/2)	M.D.	M.G.	L.M.
C	-	-	02-03-01	/	(see folio 1/2)	M.D.	M.G.	L.M.
A			10-10-00		PREMIERE DIFFUSION	MD	POB	LM

Esterline AUXITROL

GAS CONTROLLER – GASBAL
Connecting drawing

ECHELLE: /
MASSE: –

M24442 folio 2/2



CODE: 28190

Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFI–CATEUR	APPRO–BATEUR
A	–	–	05-12-03	–	PREMIERE DIFFUSION	M.D.	M.C.	L.M.

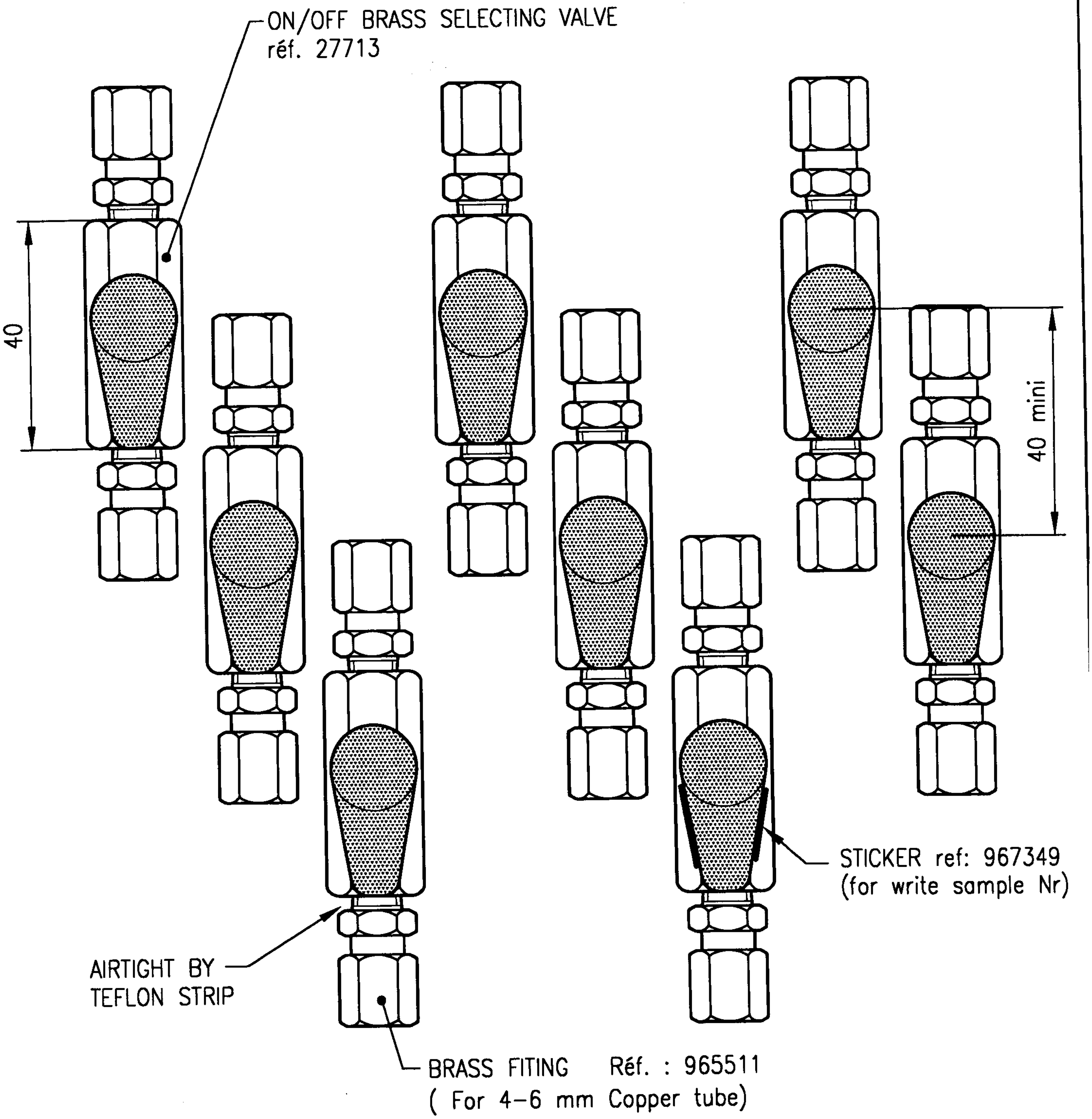
ENRAF AUXITROL MARINE

REGULATING VALVE
For copper tube O.D. 6mm

ECHELLE: /

MASSE: –

M28190



CODE : 27710 (1 VALVE and 2 Fittings)

DIFFUSION CONTROLEE
N° 76

Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFICATEUR	APPROBATEUR
A	-	a	13-09-01	B1746	MODIF DISPOSITION, AJOUTER STICKER.	MD	MG	-
A	-	-	05/07/01	-	PREMIERE DIFFUSION	LG	MG	-

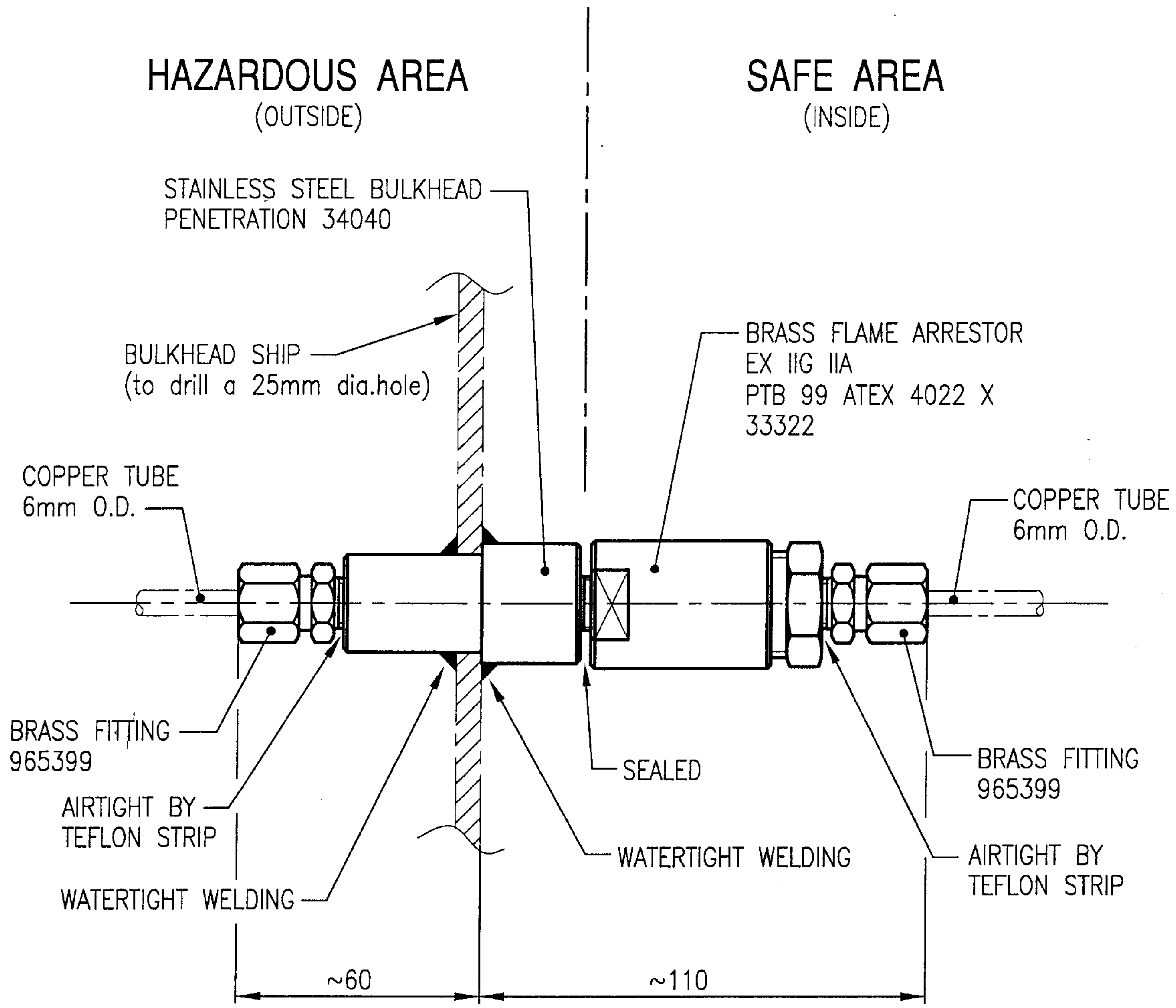
Esterline AUXITROL

CE PLAN EST LA PROPRIETE D'AUXITROL ET NE PEUT ETRE REPRODUIT OU COMMUNIQUE SANS SON AUTORISATION.

ON/OFF SELECTING VALVE
GASBAL gas detection

ECHELLE: 1/1
MASSE: -

M27852



DIFFUSION CONTROLEE
N° ....76....................

CODE: 34051

Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFICATEUR	APPROBATEUR
A	–	–	08-10-03	–	PREMIERE DIFFUSION	M.D.	M.G.	L.M.

ENRAF AUXITROL MARINE

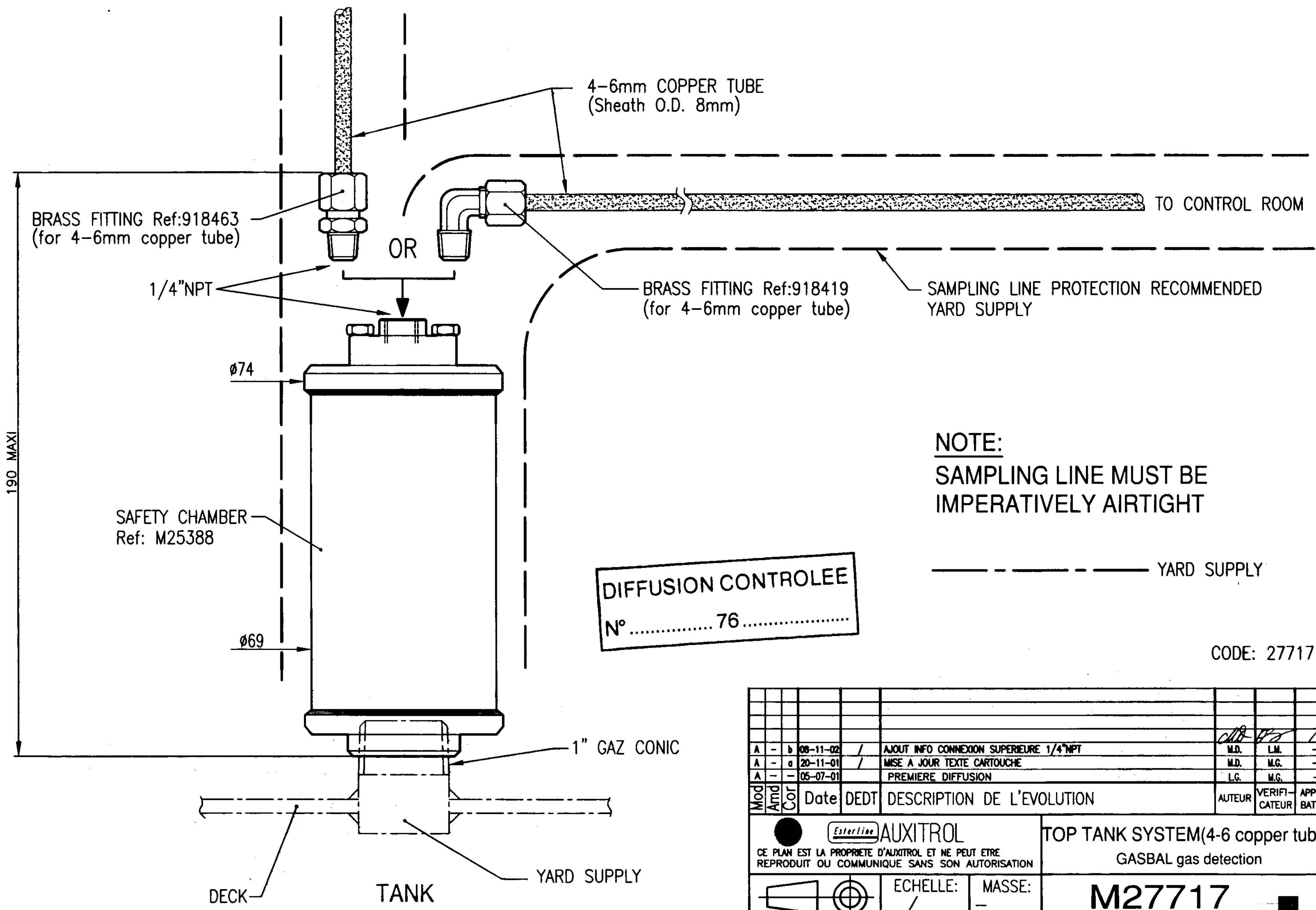
CE PLAN EST LA PROPRIETE D'E.A.M. ET NE PEUT ETRE REPRODUIT OU COMMUNIQUE SANS SON AUTORISATION

FLAME ARRESTOR AND BULKHEAD PENETRATION - copper tube Ø6

ECHELLE: /

MASSE: 0.4Kg

M34051



NOTE:
SAMPLING LINE MUST BE
IMPERATIVELY AIRTIGHT

YARD SUPPLY

DIFFUSION CONTROLEE
N° .... 76 ....

CODE: 27717

Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFI-CATEUR	APPRO-BATEUR
A	–	b	08-11-02	/	AJOUT INFO CONNEXION SUPERIEURE 1/4"NPT	M.D.	L.M.	–
A	–	a	20-11-01	/	MISE A JOUR TEXTE CARTOUCHE	M.D.	M.G.	–
A	–	–	05-07-01		PREMIERE DIFFUSION	L.G.	M.G.	–

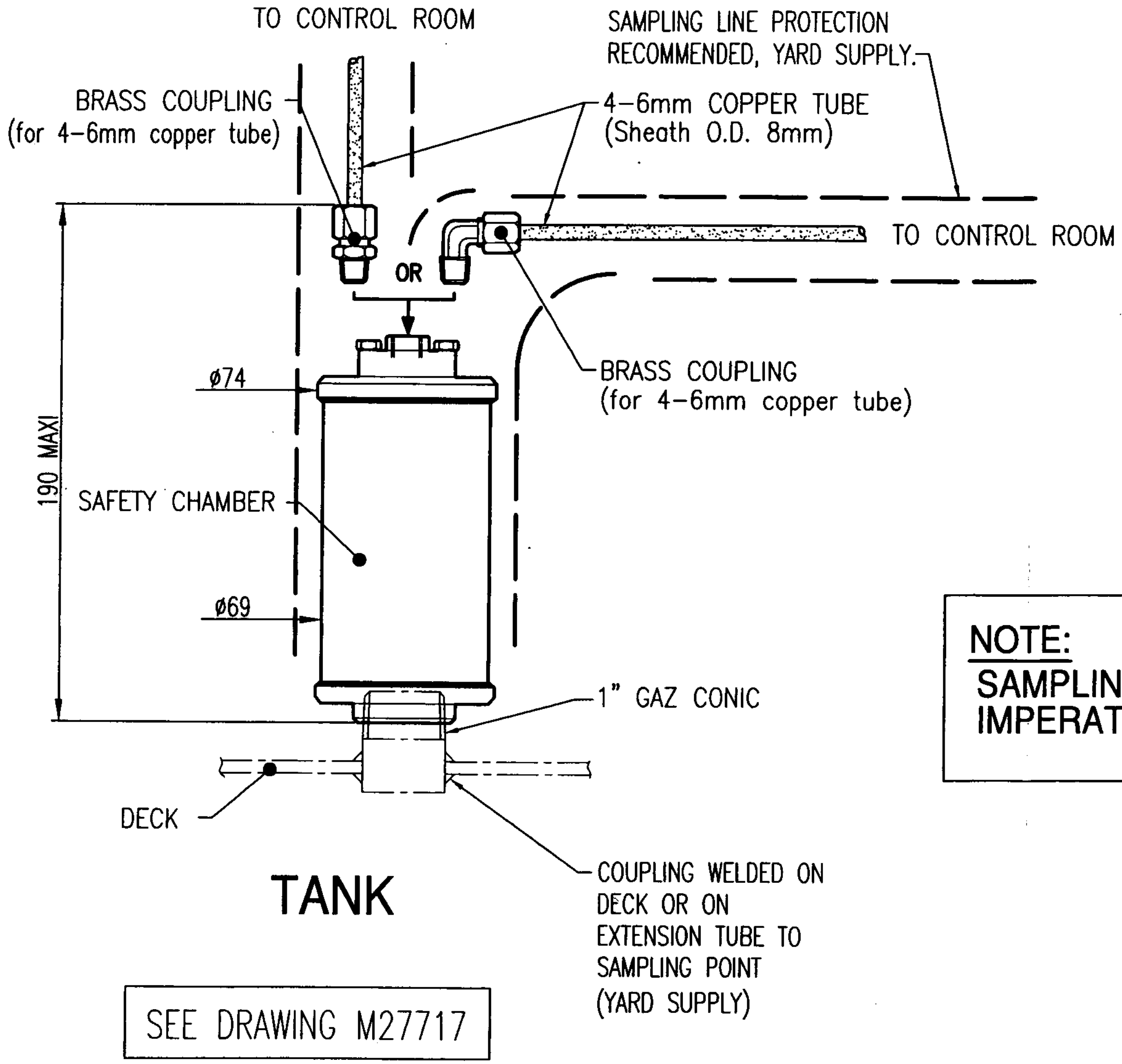
Esterline AUXITROL

TOP TANK SYSTEM(4-6 copper tube)
GASBAL gas detection

ECHELLE: /
MASSE: –

M27717

A) WET SAMPLING POINT :

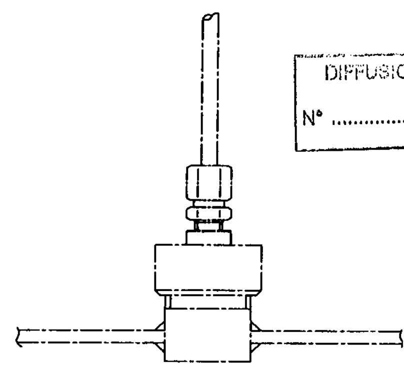


SEE DRAWING M27717

B) DRY SAMPLING POINT :

DIRECT CONNECTION
INITIATED BY YARD.

DIFFUSION CONTROLEE
N° 76



NOTE:
SAMPLING LINE MUST BE IMPERATIVELY AIRTIGHT

Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFICATEUR	APPROBATEUR
A	1	-	20/07/01	-	MISE A JOUR	LG	MG	/
A			19/07/01		PREMIERE DIFFUSION	LG	MG	-

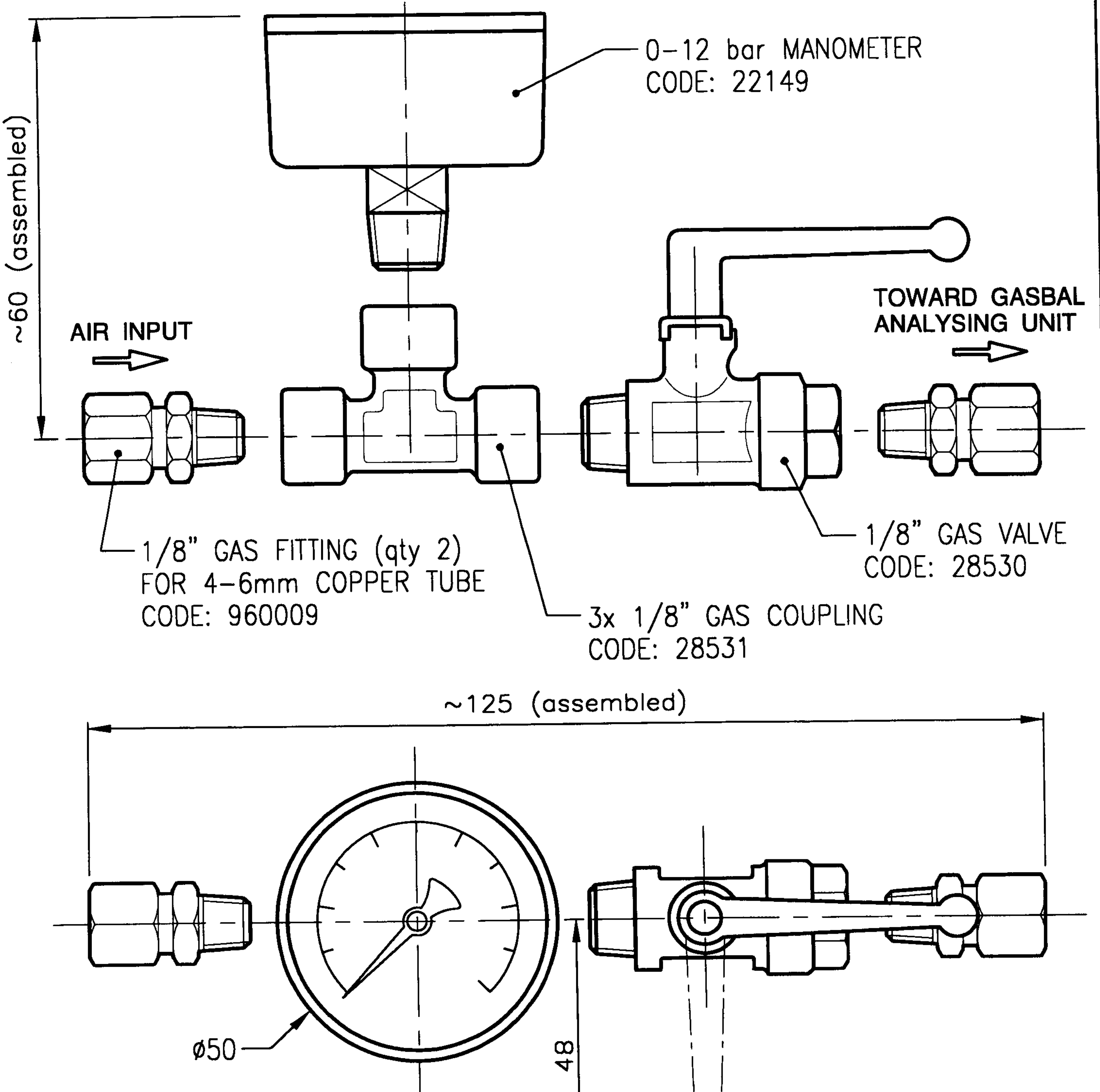
Esterline AUXITROL

CE PLAN EST LA PROPRIETE D'AUXITROL ET NE PEUT ETRE REPRODUIT OU COMMUNIQUE SANS SON AUTORISATION

GAS SAMPLING POINT INST.
GASBAL gas detection

ECHELLE: /
MASSE: –

M28194



NOTA:
ALL PARTS, ASSEMBLED WITH TEFLON RIB

CODE: 28524

DIFFUSION CONTROLEE
N° 76

Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFI-CATEUR	APPRO-BATEUR
A	–	–	12-09-01	–	PREMIERE DIFFUSION	MD	MG	–

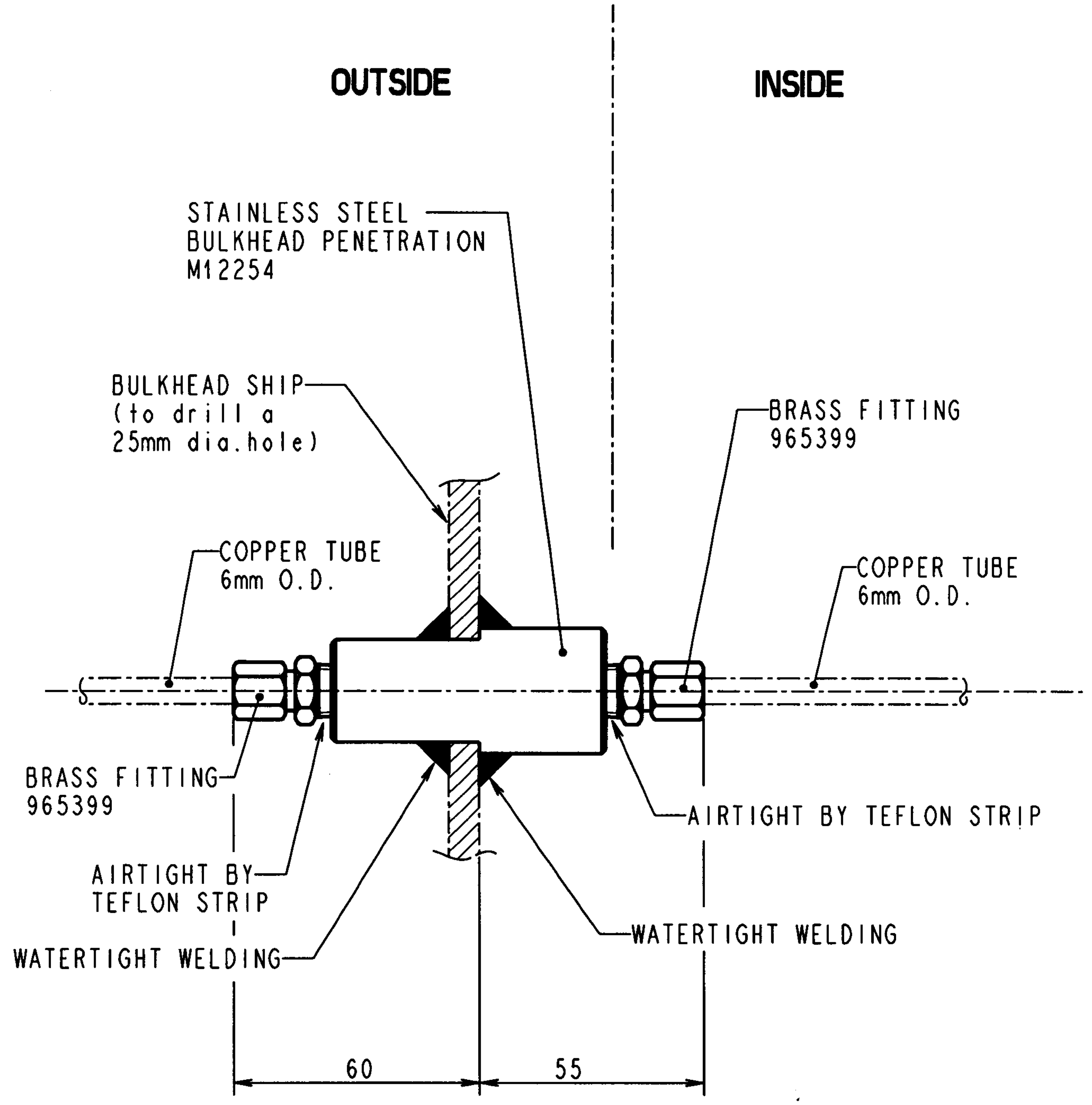
Esterline AUXITROL

KIT ON/OFF VALVE WITH MANOMETER (0 to 12 bar)

ECHELLE: /

MASSE: –

M28524



DIFFUSION CONTROLEE
N° 76

CODE: M12306

Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFICATEUR	APPROBATEUR
A	1	a	12-07-01	/	PRECISER RACCORDS POUR TUBE CUIVRE.	MD	MG	-
A	1	-	10-03-00	/	BRASS FITTING REF 918463 --> 965399	MD	JPL	-
A			04-03-97		PREMIERE DIFFUSION	JPG	GB	/

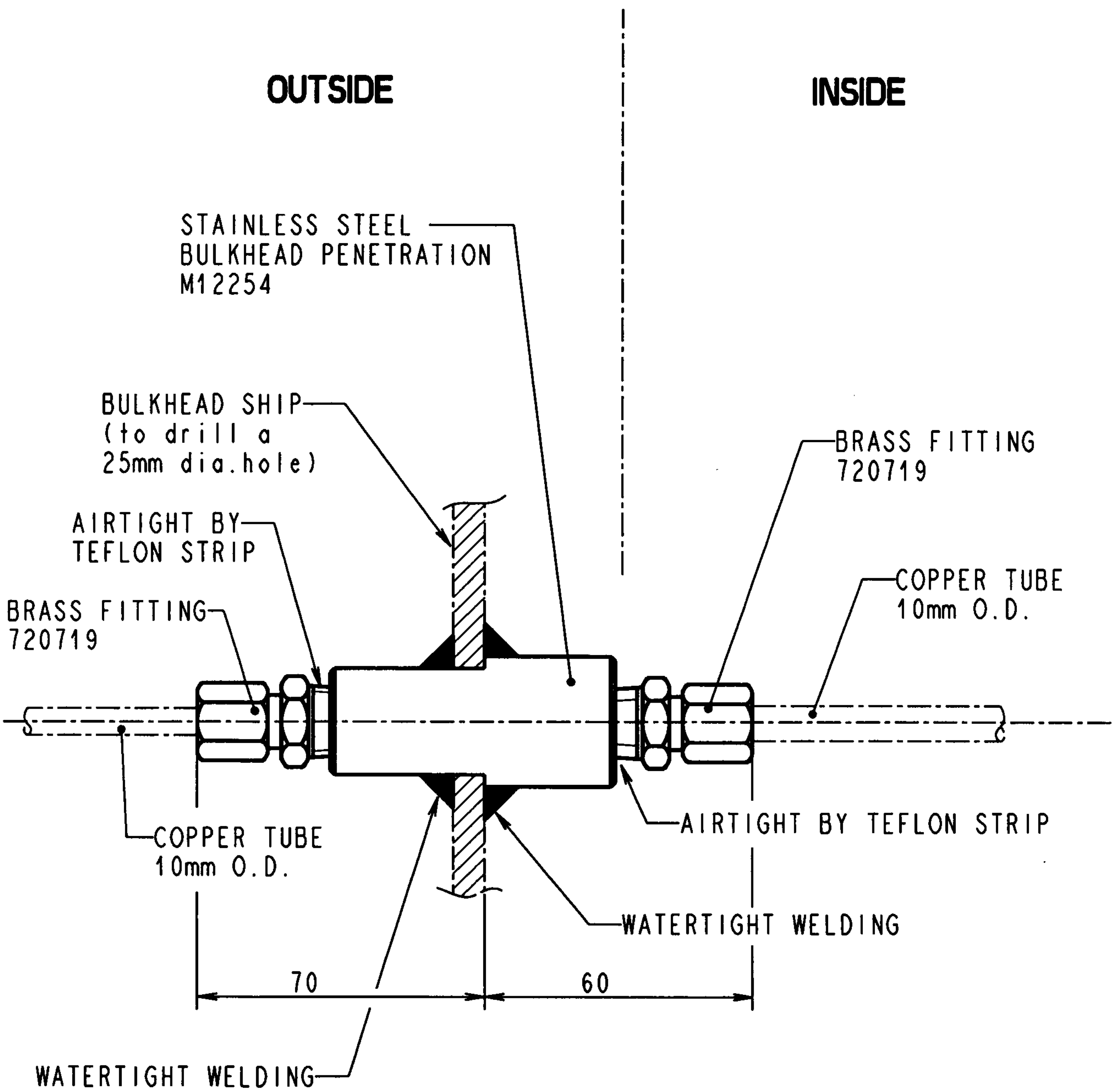
Esterline AUXITROL

BULKHEAD PENETRATION
for copper tube Ø6

ECHELLE: 1

MASSE: 0.2Kg

M12306



DIFFUSION CONTROLEE

N° 76

CODE: M15005

Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFICATEUR	APPROBATEUR
B	-	a	16-11-01	/	CORRECTION: LIRE "BRASS" ET NON "SS" FITTING POUR CODE 720719	MD	MG	-
B	-	-	12-07-01	B1743	RACCORDS INOX DEVIENNENT LAITON. CONNEXIONS NPT DEVIENNENT GAZ.	MD	MG	-
A			20-05-99		PREMIERE DIFFUSION	MD	MG	/

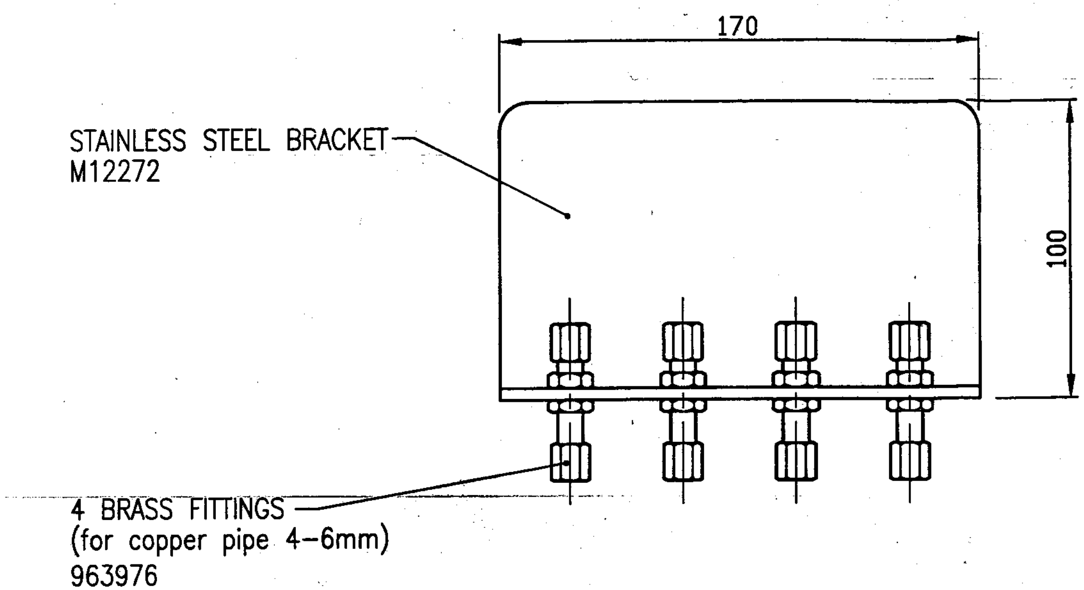
Esterline AUXITROL

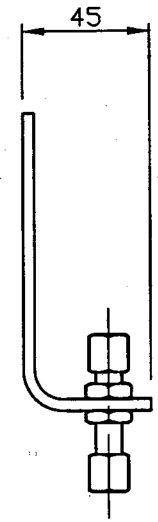
BULKHEAD PENETRATION
for copper tube Ø10

ECHELLE: 1

MASSE: 0.2Kg

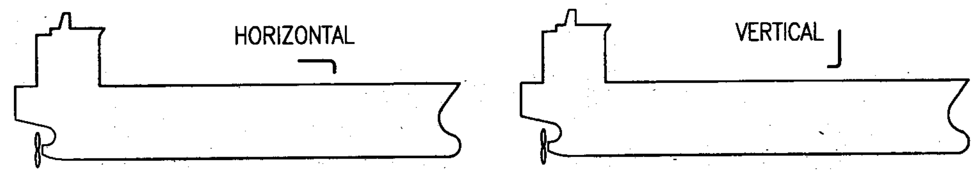
M15005



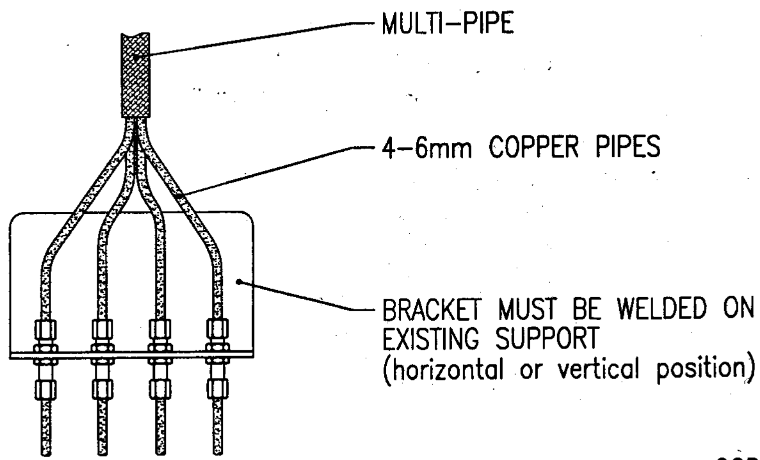


DIFFUSION CONTROLEE
N° ....76..........

DECK INSTALLATION



DETAIL CONNECTING



CODE: M12259

Mod	Amd	Cor	Date	DEDT	DESCRIPTION DE L'EVOLUTION	AUTEUR	VERIFI–CATEUR	APPRO–BATEUR
B	1	–	16-12-02	B1778	REMOVED CLAMPING, LENGTH 200 --> 100, THICKNESS 4 --> 3mm	M.D.	M.G.	L.M.
B	–	a	20-11-01	/	NEW ISSUE	M.D.	L.M.	–
B	–	–	01-02-99	/	NEW ISSUE	M.D.	L.M.	–
A	2	–	01-12-99	/	4 HOLES ⌀9 --> 6 HOLES ⌀10.5. PG29 --> PG21	M.D.	F.P.	–
A	1	–	12-07-99	/	NEW POSITION FOR CABLE GLAND. ADD 2 FIXING HOLES.	M.D.	M.G.	–
A	–	a	25-05-99	/	PART NAME "CONNECTION BOX" --> "CONNECTION BRACKET"	M.D.	M.G.	–
A			14-01-97		PREMIERE DIFFUSION	M.D.	G.B.	–

Esterline **Auxitrol**

CE PLAN EST LA PROPRIETE D'AUXITROL ET NE PEUT ETRE REPRODUIT OU COMMUNIQUE SANS SON AUTORISATION

CONNECTION BRACKET

(4-6 copper tube) Gas detection system

ECHELLE: 1/2

MASSE: 0.6Kg

M12259

2-2

Enraf Auxitrol Marine
158, Bureaux de la Colline
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France
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GASBAL™ SYSTEM

Gas Detection System for
ballast tanks, void spaces, pump room, accommodations

ISGOTT CHAP. 7.8 AND 8.2

INSTALLATION MANUAL

CONTENT

1 INTRODUCTION

The **GASBAL™ System** is dedicated to analyse Gases in Tanks (ballast, cargo,…), Void Spaces or any possible dangerous area, in order to detect any Gas Concentration level over safety limits and to monitor actions as visible and audible alarms. That System complies with **ISGOTT regulation chap. 7.8 and 8.2.**

The GASBAL™ System is a Gas Monitoring based on suction process sampling toward one or more common detector(s), combined with analog inputs from local gas detectors.

The Operation, Configuration, Commissioning, Calibration of the System are described in the **MT5094 Technical Manual**.

The Maintenance of the System is described in the **MM5094 Maintenance Manual**.

2 SYSTEM DESCRIPTION

The typical system is composed of :

- a cabinet named "Analysing Unit" (Type drawing **M24442**) including the suction process sampling gas detector(s) and pneumatic components;
- a panel named "Control Unit" including a backlighted LCD screen, LED indicators and keyboard. This panel can be mounted inside the Analysing Unit (Type drawing **M24442**) or in remote box (Type drawing **M27816**) or in existing console (Type drawing **M28193**);
- a set of suction sampling accessories : non-return valves for wet sampling points, pipe connection plates, bulkhead penetrations, flame arrestors, shut-off valves, regulating valves, shut-off valve / manometer for air supply;
- a set of local gas detectors if required, with safety class equipments if necessary (Zener barriers located in Analysing Unit, …);
- a gas calibration kit.

The typical system is shown by synopsis on following page. For complete description and specifications, refer to the Technical Manual.

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92213 Saint-Cloud cedex
France
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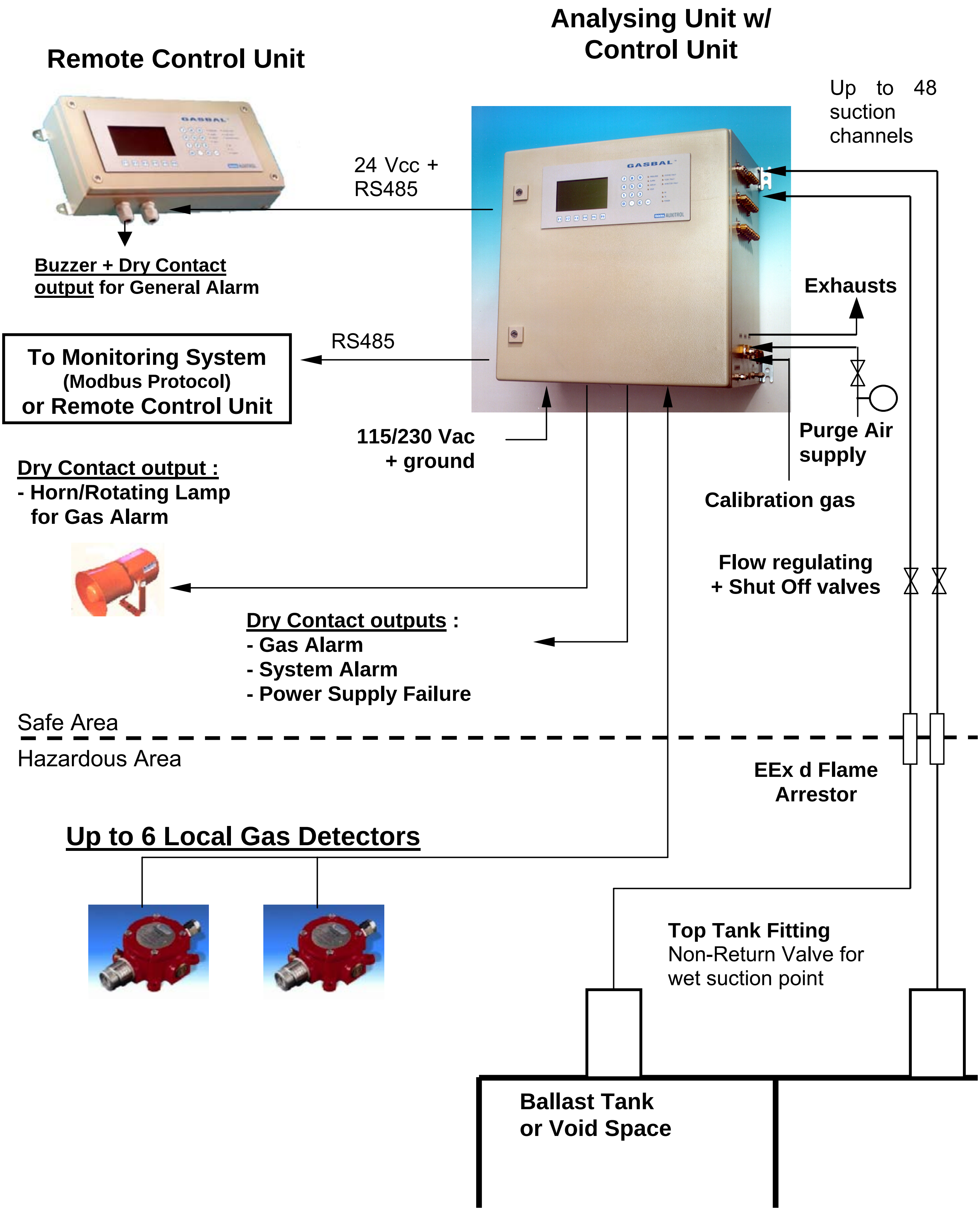

Enraf Auxitrol Marine
158, Bureaux de la Colline
92213 Saint-Cloud cedex
France
Tel + 33 (0) 1 49 11 65 75
Fax + 33 (0) 1 49 11 65 76
Email contact@enrafmarine.fr

3 UNPACKING AND INSPECTION.

3.1 Unpacking

Proceed carefully to the unpacking. If some damage is observed, mail immediately necessary claim to the carrier, in local legal conditions.

3.2 Inspection

Proceed to the inspection of all components in compliance with order, delivery bill and Manufacturer File.

3.3 Handling

- Analysing Unit cabinet : take care not to hurt the face of Control unit with LCD screen and keyboard, if present;
- Control Unit remote box (if present) : take care not to hurt the face of Control unit with LCD screen and keyboard;
- Local gas detectors : keep them in their original package until final installation (see 4.8);
- Other pneumatical components : take care not to damage them, nor to crush or to obstruct the pipes if present.

3.4 Storage

There is no particular requirement. The storage temperature of the Analysing and Control Units and local gas detectors is 0°C to +70°C.

4 MECHANICAL INSTALLATION

The normal procedure is successively as follow :

1. installation of sampling points, with non-return valves for wet points;
2. installation of bulkhead penetrations; model with flame arrestor on hazardous area limit if any, and standard model on other bulkheads to pass through if any;
3. continuation of sampling pipes from sampling points to bulkhead penetrations;
4. installation of bulkhead penetrations for exhausts;
5. installation of the exhausts pipes;
6. installation of the Analysing Unit;
7. installation of shut-off valves and regulating valves in the area of the Analysing Unit;
8. installation of the Remote Control Unit if any;
9. installation of local gas detectors if any, as well as zener barriers if any;
10. pneumatical and electrical connections on Analysing and Control Unit, described in chapter 5 and 6.

For best installation performance, the above procedure must be carried out step by step. The Installation Inspection Report in APPENDIX A must be fulfilled after each step of the system installation and control (see 7).

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4.1 Sampling points

See drawing **M28194**.

4.1.1 General

The sampling pipe extremity are to be placed in each controlled space as close as possible to the gas emission source depending of its type : in lower part of space for heavy gas, in upper part of space for light gas. For best efficiency, they should be installed as far as possible from the venting outlets, to avoid dilution of controlled atmosphere with fresh air. The used material must be compatible with the media.

However regarding the ballast tanks, they can be placed on tank top even if the gas emission is in tank bottom, in order to preserve the longest efficient period when ballast tanks are filling up, to ensure the fire prevention efficiency of the system as fire can be caused by gas coming out on deck from tank top, and also because gases are pushed toward top by tank filling (IMO Sub-Committee on fire protection).

4.1.2 Wet points

For each wet point, a stainless steel 316L non-return valve is provided, to be installed as close as possible to the sampling extremity.

For best reliability, they should be located such as the probability of ingress of water due to the movement of the vessel or other are as low as possible, for instance toward the Forward.

4.2 Bulkhead penetrations

The Bulkhead penetrations are to be installed in accordance with synopsis drawing included in the Manufacturer File proper to the project. The provided bulkhead penetrations are to be welded on the bulkhead or on a plate fixed on the bulkhead, according to drawing **M12306** for standard models on intermediate bulkheads if any, and to drawing **M12276** for model with Flame arrestor class EEx d on bulkhead / limit with safe area.

4.3 Sampling pipes

The sampling pipes are to be installed in accordance with synopsis drawing included in the Manufacturer File proper to the project. When not provided, copper type pipe 4 x 6 mm is recommended.

The pipes and non-return valves on deck have to be installed with the adequate protection (see 7.1). Copper pipes or cable comprising 1 to 4 4x6 mm copper pipes have to be connected to the connection plates (Drawing **M12259**).

On certain chemical tankers, standard copper pipes cannot be used for material compatibility reasons. Other materials can be used for pipes, but using at the same time suitable couplings. For instance, stainless steel pipes can be used but only with stainless steel couplings on non-return valves and bulkhead penetrations. Carbon steel coupling must never be used.

<u>IMPORTANT :</u> make sure not to get any low points on the continuation of any pipe between the sampling point and the arrival in the area of the Analysing Unit, to avoid condensation to obstruct the sampling pipe (see 7.1). It is always possible to throw out that condensation using the manual purge (Refer to the Technical Manual, system operation), but this could occur frequent flow faults.

4.4 Bulkhead penetrations for exhaust pipes

Three bulkhead penetrations are provided for exhaust pipes :

- Sample gas exhaust, copper pipe 4x6 mm, to be driven to outside safe area;
- Circulation gas exhaust, copper pipe 8x10 mm, to be driven to outside safe area;
- Drain water, copper pipe 4x6 mm, to be driven to bilge with regular down slope.

The bulkhead penetrations are to be installed in accordance with synopsis drawing **M27848** and welded on the bulkhead or on a plate fixed on the bulkhead, according to drawing **M12306** for O.D. 6 mm pipe model, and to drawing **M15005** for O.D. 10 mm pipe model. The pipes have to be installed with the adequate protection.

On certain chemical tankers, standard copper pipes cannot be used for part of pipes on deck, for material compatibility reasons. Other materials can be used for pipes, but using at the same time suitable couplings. For instance, stainless steel pipes can be used but only with stainless steel couplings on bulkhead penetrations. Carbon steel coupling must never be used.

4.5 Analysing Unit

The Analysing Unit consists in a wall mounted cabinet, position and fixing dimensions are shown on drawing **M24442**. A free space must be reserved on left side for ventilation, by respect of the requirement on drawing. Sufficient free space must also be reserved on right side for copper 4 x 6 mm tubing of sampling pipes, exhaust pipes and instrument air supply. The Pneumatical and Electrical connections are described in **Chapter 5 and 6**.

4.6 Valves

The sampled channels shut-off valves and regulating valves (one set per sampling line) are to be installed in accordance with synopsis drawing **M27848** and in the area of the Analysing unit, for easy access to adjust the flow and to shut off the line when skipped (See Technical Manual for operation). It is recommended to install the Shut-off valve on the line toward the sampling point, and the regulating valve on the line toward the Analysing Unit. The Shut-off valves are shown on drawing **M27852**, the regulating valves are shown on drawing **M15134**.

The shut-off valves are provided with labels, where can be reported the channel name, in order to identify them when some channel line must be open or closed (see 7.1).

The instrument air supply shut-off valve and manometer is to be installed in the area of the Analysing unit, for easy access to open or shut off the line and to check the pressure on the manometer. The Shut-off valve and manometer are shown on drawing **M28524**.

IMPORTANT NOTE :

- **The air supply for purge MUST be instrument quality, dry and clean, in order to preserve the Analysing Unit internal solenoid valves. In case of new installation, it must be purged before connection, in order to avoid internal moisture or condensation to enter the Analysing Unit.**
- **After installation, the regulating valves must be left in open position (turn anti-clockwise), and the shut-off valves must be left in closed position.**

4.7 Remote Control Unit

The Remote Control Unit consists in a wall mounted box, position and fixing dimensions are shown on drawing **M27816**, or a panel to be mounted in an existing console, fixing dimensions are shown on drawing **M28193**. The Electrical connections are described in **Chapter 6**.

4.8 Local Gas Detectors

The local detectors can be placed in pump room, engine room, accommodations, venting ducts, air-conditioned ducts.

They are to be installed in accordance with the Manufacturer File proper to the project and their specific installation document, proper to each type.

5 PNEUMATICAL CONNECTION TO ANALYSING UNIT

Refer to the drawing **M24442** and typical synopsis **M27848**.

5.1 Sampling pipes

The sampling pipes from bulkhead penetrations to Analysing Unit through shut-off and regulating valves are to be installed according to synopsis on drawing **M27848**.
The Analysing Unit sampling lines inlets are equipped with coupling for copper pipe 6 mm O.D., except when specified. The sampling tubes are to be connected on these couplings, by respect of the line number specified on the synopsis drawing in the Manufacturer File.

IMPORTANT :

- **All the unused inlets must be blocked with provided plugs.**
- **After completion of pipes installation, proceed to an absolute leakproof test, for instance by performing a pressure test of each sampling line from nearness of Analysing Unit till sampling point (see 7.1).**

5.2 Exhaust pipes

The Analysing Unit exhausts lines outlets are equipped with coupling for copper pipes 6 mm and 10 mm O.D., except when specified. The exhaust tubes are to be connected on these couplings.

IMPORTANT : make sure to maintain the coupling body with relevant spanner, for leakproof preservation.

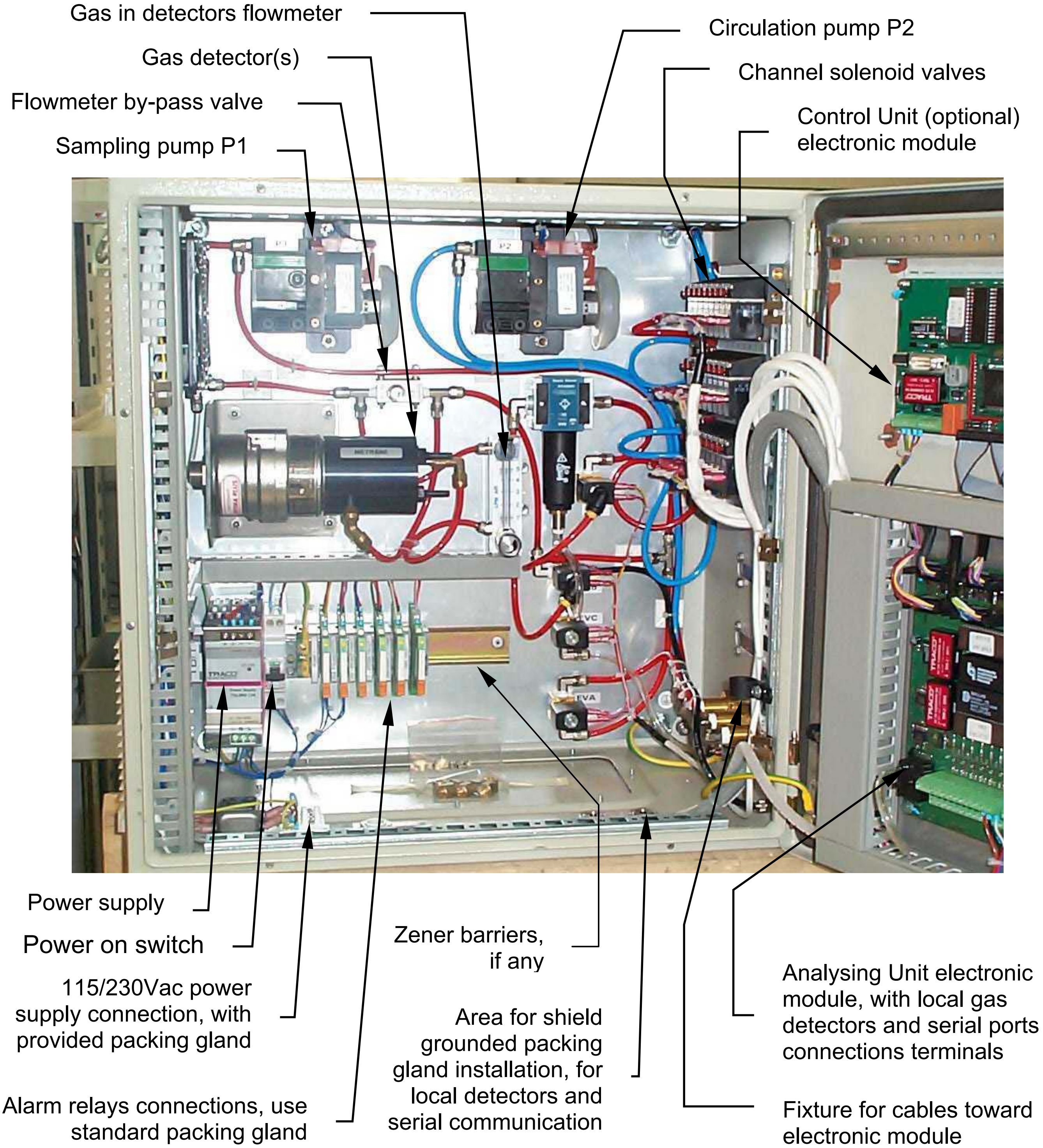
5.3 Air supply

The air supply inlet is equipped with coupling for copper pipe 6 mm.

IMPORTANT : make sure to maintain the coupling body with relevant spanner, for leakproof preservation.

6 ELECTRICAL CONNECTION OF THE SYSTEM

6.1 Analysing Unit



Refer to the drawing **M24442**.

IMPORTANT NOTE : The cables for communication port(s) and local gas detector(s) must be connected by shielded cable(s) passing through cable gland insuring the connection of shield with metallic body of cabinet, for electro-magnetic compatibility reasons. These packing gland are supplied by user and suitable for cable size, to be installed in area shown by above drawing (not on removable plate). Use wires max. 2 x 2.5 mm². The relevant cables are to be fixed on the dedicated fixture, in order to help to pass the cabinet hinge, then can enter the bottom cable path left side.

6.1.1 Power supply 115/230 Vac 50/60 Hz

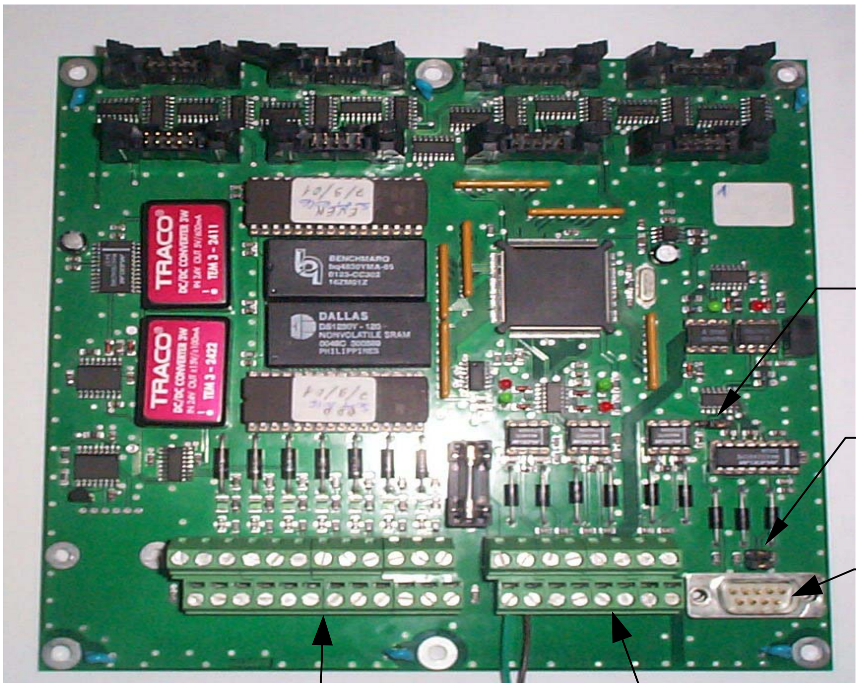
After checking that the power switch is in off position, install the delivered packing gland and connect the power supply including ground on relevant terminals, through the provided packing gland. Use wires max. 3 x 2.5 mm².

The power supply switch must be left off.

6.1.2 Dry contact outputs for Alarms

Four relays are provided for power supply fault, system fault, gas alarm, siren/rotating lamp for high gas level alarm, for connection to an extra-Alarm Management System if any. They can be connected to cable through standard packing gland(s) supplied by user and suitable for cable size, to be installed in area shown by above drawing. Use wires max. 2 x 2.5 mm².

Other connections are to be performed on the electronic board :



Jumper JP1 for Port 1 communication type selection

Jumpers JP2/3 for Port 1 RS232 wiring mode acc. to used cable

RS232 connector for port 1

J5 to J8 terminals for Local Gas Detectors

J1 to J3 for RS485 ports and Control Unit(s) supply

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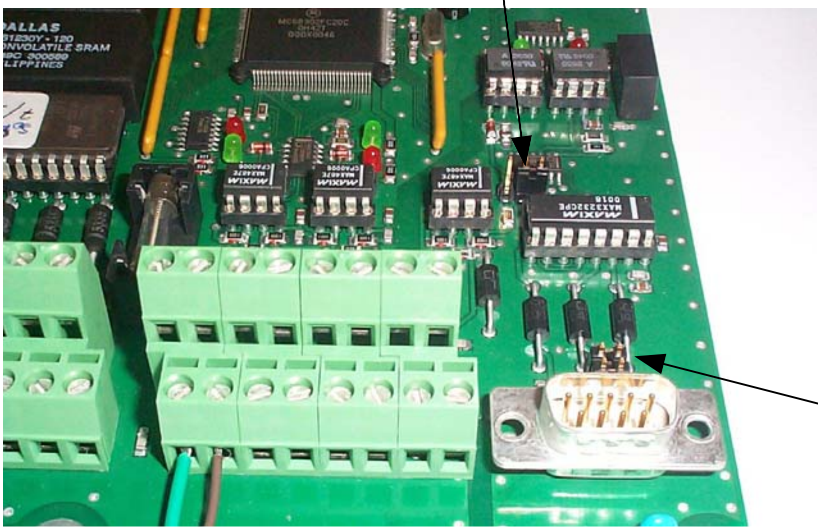
6.1.3 Serial communication ports

Three terminals sets J1 to J3 are provided for communication with remote Control Unit(s) and/or external Monitoring. Refer to the synopsis drawing included in the Manufacturer File proper to the project.

- Terminals J2 (port 2) and J3 (port 3) are dedicated for RS485 type communication and 24 Vcc power supply on 2 Control Units, one of them can be installed in Analysing cabinet.
- Terminals J1 (port 1) provides a galvanic insulation and is dedicated for communication toward an external Monitoring. RS485 type communication is available on terminal J1, RS232 type is also available using the DB9 connector :
  - jumper JP1allows to select RS485 or RS232 mode;
  - jumpers JP2/3 allows to cross emission/reception wires in RS232 mode, depending of used cable model

JP1 : left position for RS485
righ position for RS232



JP2/3 : vertical position for crossed emission/reception (Crossed cable)
horizontal position for uncrossed emission/reception (Pin to Pin cable)

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6.1.4 Local gas detectors inputs

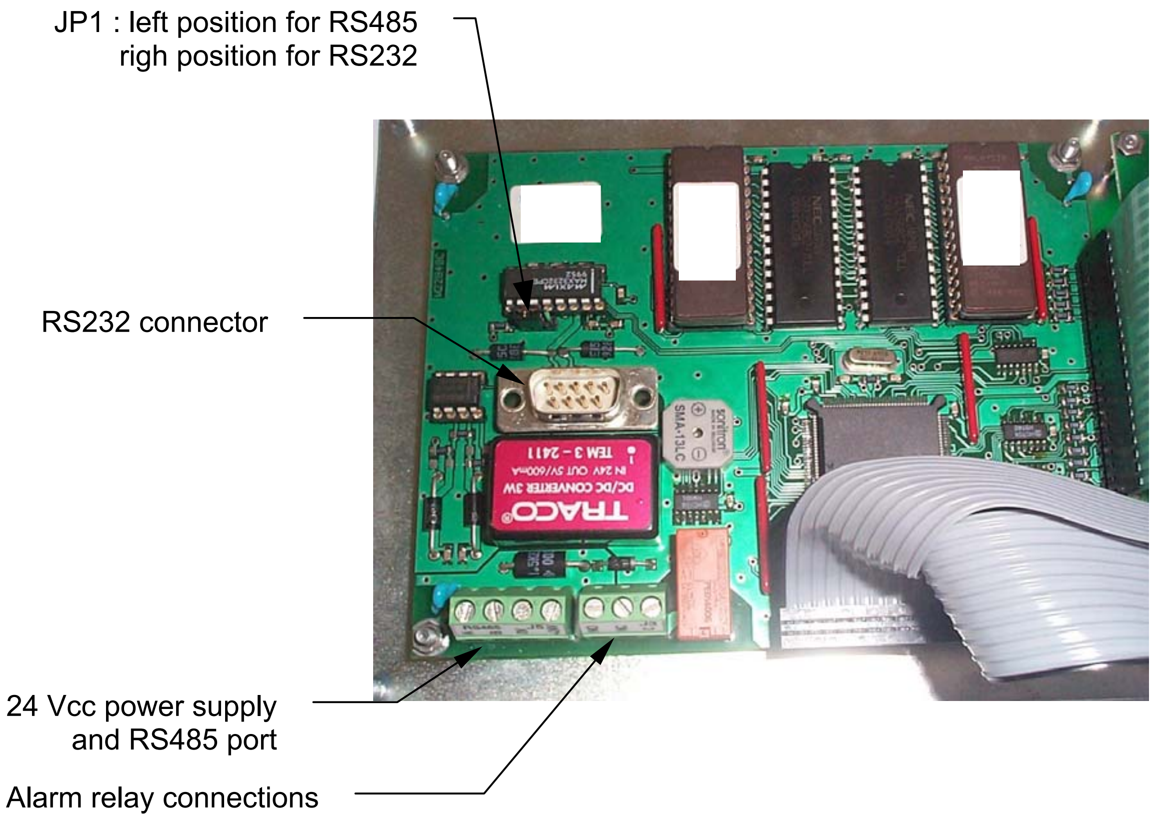
Terminals sets J5 to J8 are provided for connection of 7 Gas detectors. After factory connection of detectors installed in the Analysing Unit, the others are available for local Gas detectors. These are to be connected by respect of type and number with the Manufacturer File, in order to correspond with the factory-set System configuration.

Local gas detectors can be powered by 24 Vcc on pins 1 / 2 or 4 / 5; 4 – 20 mA inputs are to be connected on pins 3 or 6 (+) and pins 2 or 5 (-). 0V terminal is common with power supply and negative polarity of 4-20 mA input. When required, Zener barriers are inserted before terminals J5; so, the local gas detectors are to be connected to these barriers, according to the Manufacturer File.

6.2 Remote Control Unit

Refer to the drawing **M27816** (mounted in remote box) or **M28193** (mounted in console).

The connections are to be performed on the electronic board :



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6.2.1 Power supply, serial communication :

One terminal J5 is provided for 24 Vcc power supply and RS485 serial port for communication with Analysing Unit. RS232 type communication is also available, using the DB9 connector : jumper JP1allows to select RS485 or RS232 mode.

IMPORTANT NOTE : The power supply / communication cable must be connected by shielded cable passing through a cable gland insuring the connection of shield with metallic body of cabinet, for electro-magnetic compatibility reasons. This packing gland is supplied by user and suitable for cable size. Use wires max. 2 x 2.5 mm².

The other jumpers are factory pre-set, and must not be moved.

6.2.2 Dry contact output for Alarm

One relay is provided for general fault, for connection to an extra-Alarm Management System if any. It can be connected to cable through a standard packing gland supplied by user and suitable for cable size. Use wires max. 2 x 2.5 mm².

6.3 Local Gas Detectors

They are to be connected in accordance with their specific installation document, proper to each type.

7 INSTALLATION CONTROL – PREPARE TO RUN

The Installation Inspection Report in APPENDIX A must be fulfilled after each step of the system installation and control.

7.1 Sampling lines

- Check the integrity of the pipes on deck : no shock, no crushing, no bend which could reduce the internal section. The pipes and non-return valves on deck have to be installed with an adequate protection.
- Check the pipes profile **:** no low points on the continuation of any pipe between the sampling point and the arrival in the area of the Analysing Unit.
- Check the conformity of line number in correspondence with tanks or spaces, according to the Manufacturer File. Copy the channel number on each shut-off valve label.
- Check the absolute leakproof of each pipe. The best method is to pressurise each pipe using at least 6 bar compressed air, from nearness of Analysing Unit to sampling points, after plugging them; no pressure drop should be observed after several minutes.
- Fully open the regulating valves (turn anti-clockwise).
- Close temporarily the channel shut-off valves.

7.2 Local gas detectors

- Check the conformity of the local gas detector connection number in correspondence with gas type detector and area to control, according to the Manufacturer File.

7.3 Analysing / Control Units

- Check the complete wiring of the Analysing Unit : 115/230 Vac power supply, communication lines, local gas detectors, dry contact outputs.
- Check the complete wiring of the remote Control Unit if any : 24 Vcc power supply, communication line, dry contact output.
- Check the quality of air supply : it must Instrument air, dry and clean. In case of new installation, it must be purged before connection, in order to avoid internal moisture or condensation to enter the Analysing Unit. Close temporarily the shut-off valve.
- Sampling flow in Analysing Unit : open the valves of flowmeter as well as by-pass valve close to flowmeter (see 6.1).
- Sampling solenoid valves in Analysing Unit : make sure that the red handles on sampling solenoid valves are left in full horizontal position (see 6.1).

7.4 Commissioning

Commissioning and Operation are described in the Technical Manual MT5094 :
- System configuration has been carried out in factory according to the Manufacturer File. For modification, refer to the Technical Manual.
- Calibration of detectors has been carried out in factory. For other calibration, refer to the Technical Manual.

8 INFORMATIONS

8.1 Reference of Customer Service

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Manager : Alain AGOSTINI

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8.2 Personal notes

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APPENDIX A

GASBAL™ SYSTEM INSTALLATION INSPECTION REPORT

Vessel : .... Location : ....
Owner or Shipyard : ....
Analysing Unit P/N : .... Control Unit(s) P/N : ....

1. Deck pipes integrity ....:
No shock, no crushing, no excessive bend, adequate protection.

2. Sampling pipes profile ....:
No low points

3. Sampling lines identity ....:
Check with Manufacturer File. Fill in shut-off valve labels.

4. Sampling lines leakproof test..:
Absolute leakproof of sampling lines. Unused channels properly plugged.

5. Sampling lines shut-off valves closing until start-up**..........:**

6. Local gas detectors identity ....:
Check with Manufacturer File.

7. Analysing Unit wiring..:
Power supply, communication, local gas detectors, dry contact outputs lines.

8. Remote Control Unit wiring (if any) ....:
Power supply, communication, dry contact output lines.

9. Quality of air supply ....:
Instrument air dry and clean, water purged. Close shut-off valve until start-up.

10. Analysing Unit sampling flow opening until start-up**..........:**

11. Sampling lines solenoid valves handle position ....:

NAME :	**DATE :**
QUALITY :	**SIGNATURE :**

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GASBAL™ SYSTEM

Gas Detection System for
ballast tanks, void spaces, pump room, accommodations

ISGOTT CHAP. 7.8 AND 8.2

MAINTENANCE MANUAL

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CONTENT

1 INTRODUCTION

The GASBAL™ System is dedicated to analyse Gases in Tanks (ballast, cargo…), Void Spaces or any possible dangerous area, in order to detect any Gas Concentration level over safety limits and to monitor actions as visible and audible alarms. That System complies with **ISGOTT regulation chap. 7.8 and 8.2.**

The GASBAL™ System is a Gas Monitoring based on suction process sampling toward one or more common detector(s), combined with analog inputs from local gas detectors.

The Installation of the System is described in the **MI5094 Installation Manual**.

The Operation, Configuration, Commissioning, Calibration of the System are described in the **MT5094 Technical Manual**.

2 SYSTEM DESCRIPTION

The typical system is composed of :

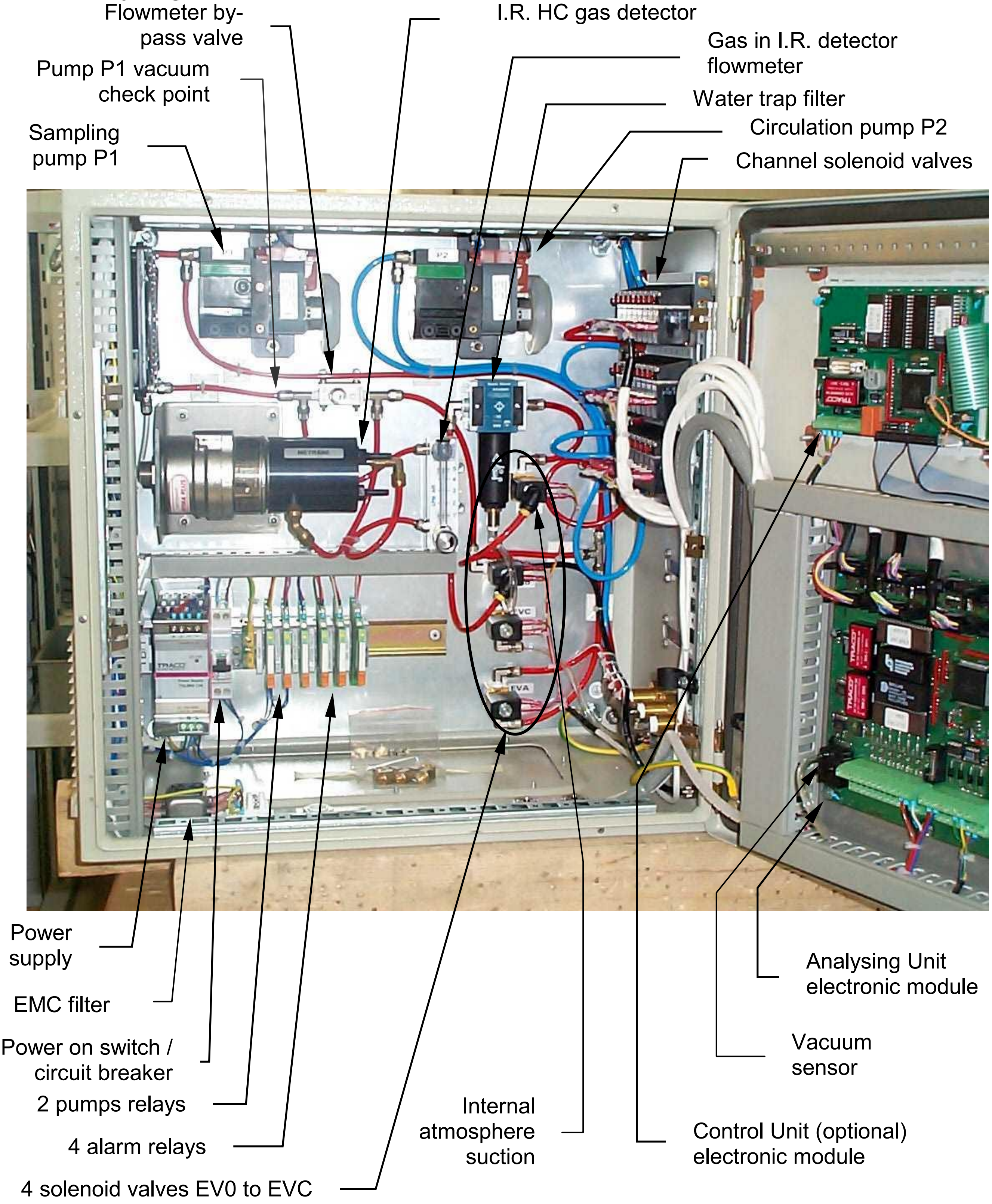
- a cabinet named "Analysing Unit" (Type drawing **M24442**) including the suction process sampling gas detector(s) and pneumatic components;
- a panel named "Control Unit" including a backlighted LCD screen, LED indicators and keyboard. This panel can be mounted inside the Analysing Unit (Type drawing **M24442**) or in remote box (Type drawing **M27816**) or in existing console (Type drawing **M28193**);
- a set of suction sampling accessories : non-return valves for wet sampling points, pipe connection plates, bulkhead penetrations, flame arrestors, shut-off valves, regulating valves, shut-off valve / manometer for air supply;
- a set of local gas detectors if required, with safety class according to their location;
- gas calibration kit(s).

Refer to drawing M24442 for Analysing Unit terminals detail, or M27816 or M28193 for Control Unit terminals detail.

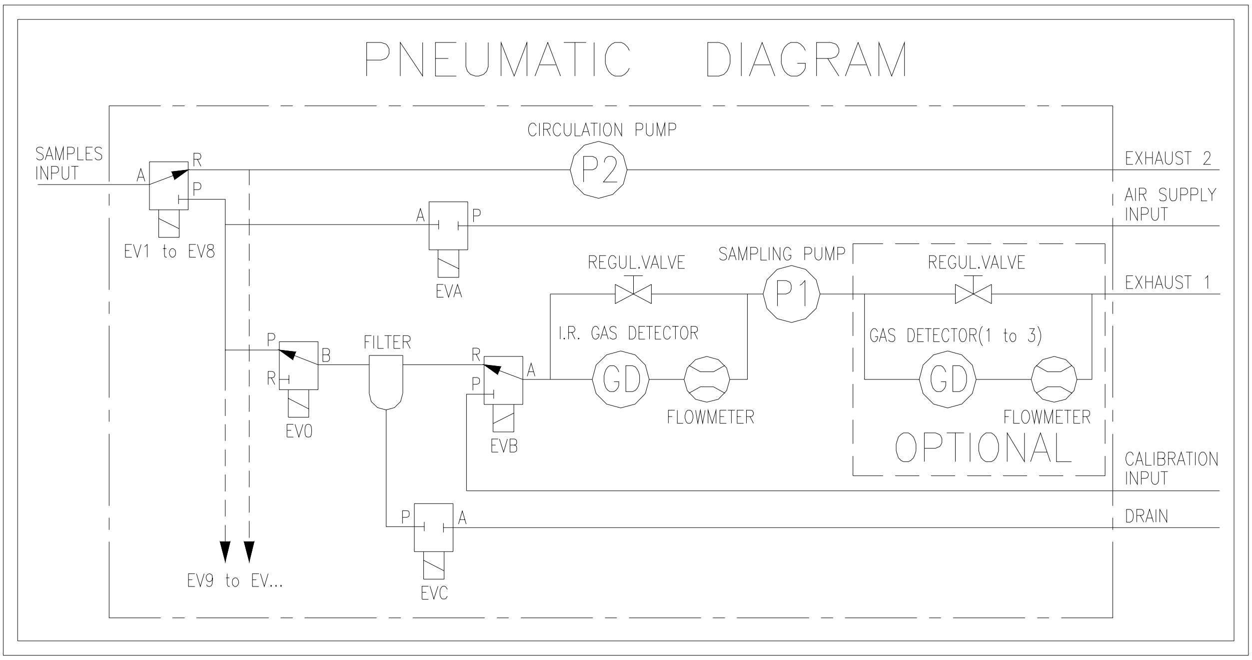
The complete typical system description and specifications are shown by synopsis with comments on **MT5094 Technical Manual**.

The parts detail overviews are shown in following paragraphs.

2.1 Analysing Unit internal overview



2.2 Pneumatic Diagram



- P1 is the sampling pump;
- P2 is the circulating pump providing parallel gas circulation of all unselected lines in order to reduce the response time by getting permanently fresh gas samples from all lines;
- EV0 allows the cabinet internal atmosphere control, or the P1 pump discharge;
- EVA allows to proceed to a purge of selected channel, automatically at each sampling cycle, or manually, in order to correct or prevent ingress of water, moisture, condensation, dust or foreign material;
- EVB allows to drive the Calibration Gas to the detector(s);
- EVC allows to purge the water trap.
- Gas Detectors (GD) : the Hydrocarbon gas Infrared detector is placed upstream P1 pump; other detector(s) is(are) placed downstream P1.

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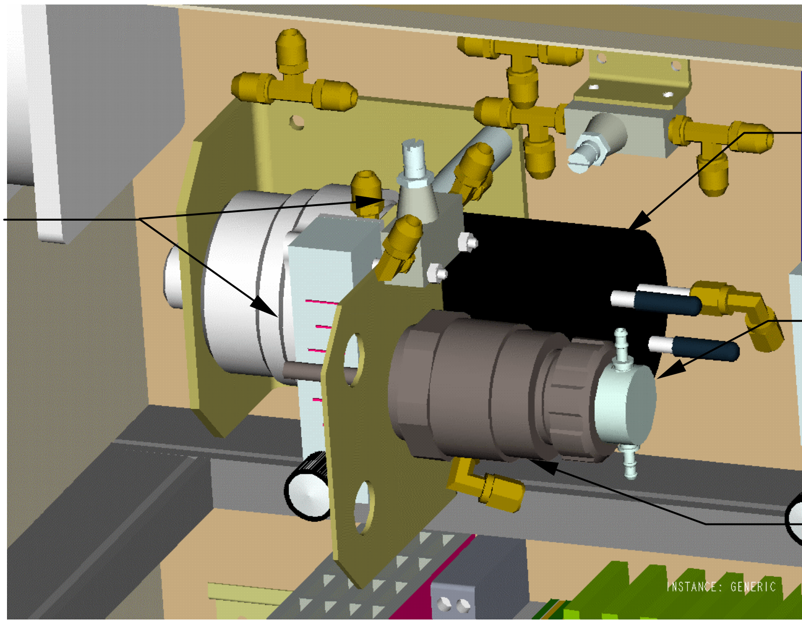
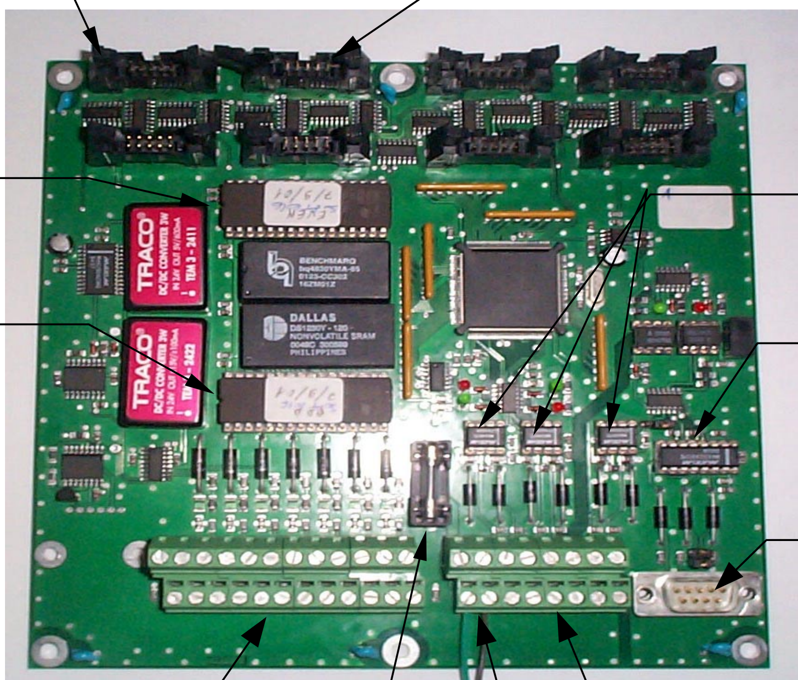
2.3 Additional detector(s) kit



2.4 Analysing unit PCB



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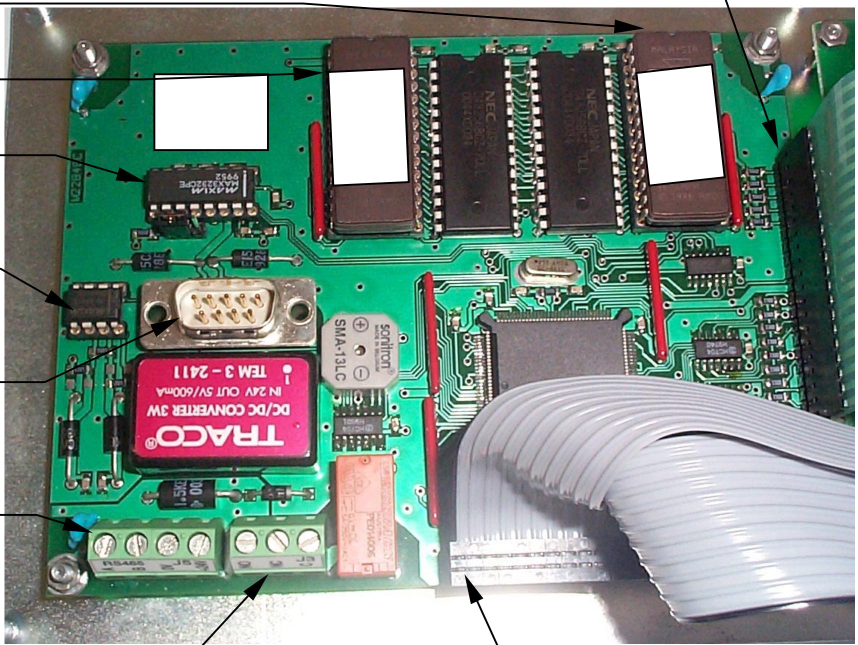
2.5 Control Unit PCB



3 PREVENTIVE MAINTENANCE

3.1 Gas calibration

3.1.1 General

The GASBAL™ Analysing Unit and local detectors if any are delivered with factory calibration, covered by the Conformity Certificate.

Further, the Gas Detectors should be periodically calibrated using reference concentration gas cylinder(s) (contact Enraf Auxitrol MArine for the supply of calibration gas). ENRAF AUXITROL MARINE recommends to proceed to such calibration at least each 6 months. In addition, gas detector's calibration can be required at any moment by Surveyors.

NOTE : the gas detectors must be left to stabilise for 1-2 hours after power on, before any calibration check.

3.1.2 Procedure

Refer to MT5094 Technical Manual ch. 7.4.1 for calibration gas concentration setup, and ch. 7.6 for use of automatic calibration sequence of the Analysing Unit.

a) Gas detectors in the Analysing Unit :
   - make sure that the calibration gas concentration(s) is(are) properly setup in the parameters data base;
   - engage the calibration procedure using relevant menu; when required connect the relevant calibration gas cylinder to the calibration inlet.

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b) Local detectors :

WARNING : for calibration of Flamgard detector, the calibration gas must contain air for proper operation of the pellistor sensor.

- make sure that the calibration gas concentration(s) is(are) properly setup in the parameters data base;
- install the calibration chamber on the detector if not present;
- engage the calibration procedure using relevant menu; when required connect the relevant calibration gas cylinder to the calibration chamber.

3.1.3 Calibration failure

a) Message "Calibration flow fault" on Control Unit : make sure to open the gas cylinder, then refer to the troubleshooting guide, calibration operation (See 4);
b) If the gas concentration is over ± 10% FS of the expected value for the Hydrocarbon infrared detector, check for the adequate test gas and the 24 Vcc power supply. If correct, the mirror must be cleaned or the detector should be replaced (See 5.2.1);
c) If the concentration is over ± 10% FS of the expected value for the catalytic or electrochemical detector, check for the adequate test gas, the 24 Vcc power supply if any, the detector insulation and the Zener barrier if any (See 5.2.16). If correct, the filter must be cleaned and/or the sensor cell must be replaced (See 5.2.2 or 5.2.3).

IMPORTANT NOTE : The Toxic Gas local detectors, including Oxygen, are equipped with electro-chemical cell. Their shelf life is limited depending of the type and operating conditions. The replacement of such cells is not covered by system warrantee.

3.2 Monthly controls

3.2.1 Visual inspection

a) Venting fan, pump P2 : check that the fan and pump P2 rotate; if not, refer to the troubleshooting guide, venting / pump P2 operation (See 4);
b) Venting filter : check that the venting filter on bottom left side of the Analysing Unit keeps clean from any dust; remove it and clean it when necessary;
c) Red handles on sampled channels solenoid valves : check that they are left in full horizontal position;
d) Flow in detector(s) : check that the flowmeter(s) either for hydrocarbon detector as for additional detectors are fully opened and show 1.5 l/min (except for channel 0). If necessary, adjust the flow(s) using **only** the by-pass valve close to the relevant flowmeter.

3.2.2 Display, Buzzer, Indicators and Alarms test

a) Proceed to a test function (See MT5094 Technical Manual ch. 6.8.1) from each Control Unit(s);
b) Check that the display LCD screen properly darken, that the indicators light on, that the buzzer and alarm relay on Control Unit are activated, and that the LED indicators on the 3 alarm relays (all relays except power supply alarm) of the Analysing Unit light on;
c) In case of negative result for one or more items, refer to the troubleshooting guide for relevant failure (See 4).

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3.2.3 Filter purge

a) Proceed to a filter purge (See MT5094 Technical Manual ch. 6.8.2);
b) Ear that the solenoid valve EVC opens and the possible water evacuates;
c) If not, refer to the troubleshooting guide, filter purge operation (See 4).

3.2.4 Pressure supply, purge efficiency

a) Check the presence of air supply on manometer;
b) Lightly loosen the tube connection on one sampled unskipped channel on the right side of the Analysing Unit;
c) Proceed to a sampled channel purge on that channel (See MT5094 Technical Manual ch. 6.8.3); check that the air blows through the inlet of that channel; stop the purge by starting a filter purge from same menu; carefully tighten the tube;
d) If the air doesn't blow, refer to the troubleshooting guide, channel purge operation (See 4).

3.2.5 Sampled channel Purge

Proceed to a sampled channel purge successively for each sampled channel (See MT5094 Technical Manual ch. 6.8.3), in order to contribute to clean the inside of the sampling lines.

3.2.6 Channel flows, comparison to these after commissioning

a) For each sampled channel using the display function (See MT5094 Technical Manual ch. 6.5), compare the current flow to this one registered after the commissioning (See Installation report);
b) If the flow differs more than 50 l/min for some or all channels, refer to the troubleshooting guide, flow fault for some or many channels (See 4).

3.3 Soft updating

The Analysing Unit soft is named SOFT 1016; this one of the Control Unit is named SOFT 1017. In case of updating, ENRAF AUXITROL MARINE can provides a set of 2 EPROM's per SOFT, one "EVEN" and one "ODD".

The EPROM's have to be inserted on relevant PCB (see 2.4 or 2.5) at the same location than the old one's, taking care with the notch of each EPROM.

3.4 Manufacturer visit

ENRAF AUXITROL MARINE strongly recommends to proceed to a general review of the GASBAL™ System by a Service Engineer each 2 years, in order to preserve the efficiency of the System as a safety system.

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4 TROUBLESHOOTING GUIDE

General table for common failures, causes and remedies. Refer to overviews in chapter 2 for details. Refer to chapter 5 for remedial maintenance.

Observation	Cause	Remedies
Complete system stop	- Circuit breaker off - Main PCB's fuse blown out - 24 Vcc Power supply LED off	- Set circuit breaker on; search for the reason of short circuit - Replace the fuse 1.8A Temp.; search for the reason of short circuit - Check the EMC filter - Check the power supply
Venting and/or pump P2 stop	- Connections of fan or P2 pump - Relay P2, LED off - Relay P2 out of service - Fan or pump out of service	- Tighten the connections - Check the relay connections, then connector J21 on main PCB - Replace the relay P2 - Replace fan or pump P2 (see 5.2.11)
Flow fault on some channels except channel 0 (Flow fault indicator lights on)	- Too high flow	- Check that solenoid valves red handles are in horizontal position - Search for leakage on sampling line
	- No flow	- Check the opening of the shut-off valve - Check non-return valve on wet tanks
	- No flow : wet tank full	- Skip that channel and close its shut-off valve
	- Flow just lower than low limit	- Purge that channel, several time if necessary, to clean the line - It is admitted to lower the limit by 50 l/h (see MT5094 ch. 7.3.3)
	- Too low flow : obstructed sampling line	- Purge that channel, several time if necessary, check the presence of air supply, valve open, pressure on manometer - Search for obstruction
	- Too low flow : obstructed local valve	- Regulating valve obstructed : clean it by counter flow or replace the valve (see 5.2.4) - Shut off valve obstructed : clean it by counter flow or replace the valve (see 5.2.4)
	- Too low flow : obstructed flame arrestor	- Clean it by counter flow - Clean the sintered steel part (see 5.2.5) - Replace the flame arrestor
	- No flow : solenoid valve for that channel keeps closed	- Check the connection of relevant valve and the integrity of wires - Replace the solenoid valve (see 5.2.6)

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Continuation :

Observation	Cause	Remedies
Flow fault on channel 0 (Flow fault indicator lights on)	- No flow : EV0 solenoid valve keeps closed	- Check the connection of EV0 and the integrity of wires - Check the connector J20 on main PCB - Replace EV0 (see 5.2.7)
Flow fault for many channels (Flow fault indicator lights on)	- Too low flow on gas detectors	- Fully open the flowmeter(s) and adjust the flow to 1.5 l/min (except for channel 0) using the by-pass valve(s)
	- Too low flow : water trap filter obstructed	- Purge the filter, several time when necessary - Remove the filter, clean the cartridge by blowing air (see 5.2.9) - Replace the cartridge (see 5.2.9)
	- Too low flow : exhaust 1 obstructed	- Search for possible pipe obstruction - Clean it by blowing air
	- No flow : closed solenoid valve for several channels	- Check the connection of relevant valves - Check the relevant connector J1x on main PCB
	- Too high flow : pump P1 internal leakage	- Check pump by holding on channel 0 and opening the check point (see 2.1), and plug it : if flow don't fall to 0, clean or replace the diaphragm and valves (see 5.2.10)
	- Too high flow : EV0 or EVA or EVB or EVC leakage	- Identify the leakage by searching for the input with additional suction - Replace the solenoid valve in cause (see 5.2.8)
	- Flow sensor	- Check vacuum sensor : see 5.2.12
No flow (Flow fault indicator lights on), pump P1 stops	- Connections of pump P1 - Relay P1, LED off - Relay P1 out of service - Pump out of service	- Tighten the connections - Check the relay connections, then connector J21 on main PCB - Replace the relay P1 - Replace pump P1 (see 5.2.11)
Sample leak (System fault indicator lights on)	- Couplings	- Tighten the internal couplings
	- EV0 or EVA or EVB or EVC leakage	- Identify the leakage by searching for the input with additional suction - Replace the solenoid valve in cause (see 5.2.8)
	- Channel EVn solenoid valve keeps open by foreign body	- Hold channels one by one to search for channel with too high flow; purge that channel
	- Pump P1 internal leakage	- Check pump by holding on channel 0 and opening the check point (see 2.1), and plug it : if flow don't fall to 0, clean or replace the diaphragm and valves (see 5.2.10)

Continuation :

Observation	Cause	Remedies
Shut-down (System fault indicator lights on)	- Coupling leakage inside Analysing Unit	- Tighten the internal couplings - Check the integrity of pipes
No gas concentration display from a detector	- Detector configuration	- Check the correct detector configuration and validation (See MT5094 ch.7.4.1)
Detector fault (Detector fault indicator lights on), check detector number message	- Detector power supply if any - Hydrocarbon gas infrared - Remote detector insulation - Remote detector Zener barrier - Other detector	- Check the 24 Vcc power supply and connections - Replace the detector - Check the insulation and line - Check / replace the Zener barrier (See 5.2.16) - Replace the sensor (See 5.2.2 or 5.2.3)
	- Analog Input	- Check the analog input (see 5.2.13)
Filter purge operation	- EVC solenoid valve keeps closed	- Check the connection of EVC - Check the connector J20 on main PCB - Replace EVC (see 5.2.7)
Channel purge operation	- EVA solenoid valve keeps closed	- Check the connection of EVA - Check the connector J20 on main PCB - Replace EVA (see 5.2.7)
Calibration operation	- EVB solenoid valve keeps closed	- Check the connection of EVB - Check the connector J20 on main PCB - Replace EVB (see 5.2.7)
No alarm output	- Connections of relay - Relay LED off - Relay out of service	- Tighten the connections - Check the relay connections, connector J21 on main PCB - Replace the relevant relay
Blank lines or columns on Control Unit LCD	- Connectors - LCD board	- Check the LCD connectors - Replace the LCD board (see 5.2.17)
Control Unit LED Indicator off or key out of service on keypad	- Front panel connector - Front panel board	- Check the front panel connectors - Replace the front panel (see 5.2.18)
Control Unit buzzer, alarm relay	- Control Unit board	- Replace the Control Unit board (see 5.2.19)
No communication from Control Unit RS232 or RS485 (LED Tx off)	- Line - Driver	- For RS485, check the polarity A and B on both Units - Replace the relevant driver on Control Unit (see 5.2.14)
No communication from Analysing Unit RS232 or RS485 (LED Tx flashing, LED Rx off)	- Line - Driver	- Check the line connection - For RS485, check the polarity A and B on both Units - For RS485 communication, change the port if another port is available - Replace the relevant driver on Analysing Unit (see 5.2.14)

5 REMEDIAL MAINTENANCE

5.1 Recommended spare parts

The standard Spare Parts kit (ENRAF AUXITROL MARINE code **28219**), for one year includes :

- 1 set diaphragm and valves for sampling or circulation pump (see 5.2.10)
- 2 channel solenoid valves (see 5.2.6)
- 1 EV0 solenoid valve (see 5.2.7)
- 1 filter cartridge (see 5.2.9 - 5.2.8)
- 2 flame arrestor (see 5.2.5)
- 1 regulating valve (see 5.2.4)
- 1 shut-off valve (see 5.2.4)
- 1 set of fuses

In case of other spare parts request, see 6, Spare Part list with ordering code.

5.2 Repair procedures

Refer to overviews in chapter 2 for details.

5.2.1 Hydrocarbon gas Infra-red detector : mirror cleaning or replacement

a) Switch off the power supply;
b) Remove the two red pipes, remembering their position;
c) Remove the plastic cover of the wires guide, in order to make free the cabling;
d) Remove the nut fixing the detector on its bracket;
e) Remove two nuts fixing the black cover of the detector, pull it carefully out of the detector;
f) Clean the mirror using a soft lint-free rag, with mild detergent when necessary;
g) Reassemble the detector in reverse order;
h) Re-install the detector and check the calibration;
i) If the detector is always in failure, order it to ENRAF AUXITROL MARINE (see 6) and replace it.

5.2.2 Hydrocarbon gas catalytic remote detector :

The expected operating life is 3 to 5 years. Over that period and when the calibration fails, the sensitive part needs to be changed. Refer also to the Instruction sheet attached to the detector.

Cleaning the sinter part :

a) Switch off the power supply, by respect of the hazardous area environment;
b) Open the junction box unscrewing the four M6 Allen head screws;
c) Disconnect the 3 sensor wires from the terminal block;
d) Unscrew the complete sensor housing from the junction box;
e) Open the housing by removing the four screws from the top cap with a 3 mm Allen key;
f) Check the cleanness of the sinter assembly; if necessary clean this part by blowing air or better by soaking in a grease thinner or hot water;
g) Reassemble in reverse order; make sure that the screws are well secured;
h) Switch the power on; leave the detector for stabilisation during 1-2 hours and proceed to a calibration check.

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Replacing the sensor :

a) Proceed as above a) to e);
b) Remove the sensor from the top cap PCB. An extra black sleeve may be separate from the sensor; the sleeve may be re-used;
c) Fit the replacement sensor; observe the pin alignment with PCB;
d) Re-fit the gasket;
e) Reassemble in reverse order; make sure that the screws are well secured;
f) Switch the power on; leave the detector for stabilisation during 1-2 hours and proceed to a calibration check.

5.2.3 Oxygen and toxic gas electrochemical detector, installed in the Analysing Unit or in remote location

The expected operating life is 1 to 3 years depending of the gas type. Over that period and when the calibration fails, the sensitive part needs to be replaced.

Cleaning the filter on detector installed in the Analysing Unit :

a) Switch off the power supply;
b) Pull off the red pipes and unscrew the flow chamber;
c) In remote location, unscrew the flow chamber if any;
d) Unscrew by hand the sensor cap;
e) Clean the filter by blowing air or better by soaking in a grease thinner;
f) Reassemble in reverse order handtight;
g) Switch the power on; leave the detector for stabilisation during 1-2 hours and proceed to a calibration check.

Replacing the sensor cell of detector installed in the Analysing Unit :

a) Proceed as above a) to c);
b) Unscrew the M20 nut to loosen the detector for better access;
c) Unscrew by hand the sensor cap;
d) Gently pull the electrochemical cell from the PCB;
e) Put the new cell into the PCB;
f) Reassemble in reverse order handtight;
g) Switch the power on; leave the detector for stabilisation during 1-2 hours and proceed to a calibration check.

For remote detector (refer also to the Instruction sheet attached to the detector) :

a) Switch off the power supply;
b) Remove the plastic bayonet housing;
c) Gently pull the electrochemical cell from the PCB;
d) Put the new cell into the PCB; reassemble in reverse order.

5.2.4 Shut-off or regulating valves : cleaning or replacement

a) Attempt to clean by blowing counter-flow air;
b) Reassemble the regulating valve by respect of the way : refer to the flow arrow on the valve body;
c) After reassembling the regulating valve of a channel, proceed to re-adjustment of channel's flow to the common value by holding that channel (see MT5094 Technical Manual, ch. 7.1 and Configuration Sheet);
d) If the cleaning is unsuccessful, replace the relevant valve with spare part.

5.2.5 Flame arrestor : cleaning or replacement

a) Remove the flame arrestor from the bulkhead penetration;
b) Remove the cap from the hexagonal body, to pull out the sintered part;
c) Clean this part by blowing air or better by soaking in a grease thinner or hot water;
d) Reassemble the flame arrestor for check;
e) If the cleaning is unsuccessful, replace the relevant flame arrestor with spare part.

5.2.6 Channel n solenoid valve EVn

a) Switch off the power supply;
b) For solenoid valve replacement, put off carefully the two terminals;
c) Remove the two screws fixing the solenoid valve on the manifold;
d) Replace the solenoid valve by a spare part; make sure to keep in place the three holes gasket; tighten the two screws;
e) Reconnect the two terminals on the two shortest pins;
f) Make sure that the red handle is in horizontal position.

5.2.7 System solenoid valves EV0, EVA, EVB, EVC : replacement of coil

a) Switch off the power supply;
b) For coil replacement, put off carefully the two terminals;
c) For EV0 or EVB, remove the top pipe by pulling the pipe while pushing the yellow ring, then the top coupling using a spanner 11 mm, taking care with the o'ring gasket;
d) Remove the nut fixing the solenoid using a spanner 11 mm;
e) Replace the solenoid by a spare part; tighten the nut;
f) For EV0 or EVB, reassemble the top coupling, taking care with the o'ring gasket, then push the top pipe home;
g) Reconnect the two terminals on the two shortest pins.

5.2.8 System solenoid valves EV0, EVA, EVB, EVC : replacement of valve

a) Switch off the power supply;
b) For solenoid valve replacement, put off carefully the two terminals;
c) For EV0 or EVB, remove the top pipe by pulling the pipe while pushing the yellow ring, then the top coupling using a spanner 11 mm, taking care with the o'ring gasket;
d) Remove the nut fixing the solenoid using a spanner 11 mm;
e) Remove the coil; remove the four screws fixing the valve;
f) Replace the valve by a spare part; EV0's valve is provided in spare part kit; for other valve, order it to ENRAF AUXITROL MARINE (see 6); tighten the four screws;
g) Reassemble the coil, then for EV0 or EVB, reassemble the top coupling, taking care with the o'ring gasket, then push the top pipe home;
h) Reconnect the two terminals on the two shortest pins;

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5.2.9 Filter : cleaning or replacement of cartridge

a) Switch off the power supply;
b) Remove the two plastic black plates fixing the filter on its support;
c) Pull out firmly the filter, taking care with the two o'ring gaskets;
d) Rotate anticlockwise the filter tank from the filter body;
e) Unscrew the black disc fixing the cartridge;
f) Replace the cartridge; reassemble the filter tank;
g) Engage the filter by respect of the way : refer to the flow arrow on the filter body, which must be toward left; take care with the two o'ring gaskets;
h) Reassemble the two plastic black plates.

5.2.10 Pump P1 or P2 : cleaning and/or replacement of diaphragm and valves

a) Switch off the power supply;
b) Remove the electrical terminals, the two pipes remembering their position;
c) Remove the two nuts fixing the pump on the back plate;
d) Holding the pump vertically, remove the four screws on the top of the pump;
e) Pull up carefully the top black part;
f) Check for cleanness of the valves and absence of foreign part; when necessary, replace the gaskets and valves;
g) To replace the diaphragm, pull up the green round part supporting the gaskets and valves;
h) Unscrew the diaphragm from the conrod; replace the diaphragm, screw it with thread brake and reassemble in reverse order; all parts have non-symmetric items in order to avoid any mistake.

5.2.11 Pump P1 or P2: replacement

a) Order a pump to Enraf Auxitrol Marine (see 6);
b) Switch off the power supply;
c) Remove the electrical terminals, the two pipes remembering their position;
d) Remove the two nuts fixing the pump on the back plate;
e) Replace the pump and reassemble in reverse order.

5.2.12 Vacuum sensor test

This test can be performed using a pressure generator :

a) Hold on any channel, in order to directly read the flow;
b) Disconnect carefully the plastic pipe from the pressure sensor on the main PCB (see 2.1);
c) Connect the pressure generator on the center inlet of the vacuum sensor;
d) Generate 130 ± 5 mbar, the displayed flow should be 350 ± 20 l/h;
e) Generate 470 ± 5 mbar, the displayed flow should be 185 ± 20 l/h;
f) If these values are much different, the vacuum sensor needs to be replaced together with the complete PCB (see 5.2.15);
g) Reconnect the plastic pipe to the upper inlet of the vacuum sensor.

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5.2.13 Analog input test

This test can be performed using a milliampere generator. Refer to type drawing **M24442** for terminals numbering.

a) Hold on any channel, in order to directly read the gas concentration;
b) Disconnect the gas detector from the analog input to be tested (4-20 mA and 0V);
c) Connect the current generator to the analog input, "+" to 4-20 mA, "-" to 0V;
d) Generate 4 mA, the displayed gas concentration should be 0 ± 0.2 % of full range;
e) Generate 20 mA, the displayed gas concentration should be 100 ± 0.2% of full range;
f) If these values are much different, the vacuum sensor needs to be replaced together with the complete PCB (see 5.2.15);
g) Reconnect the gas detector to its analog input (4-20 mA and 0V).

5.2.14 Communication driver replacement

In case of communication failure of one port, RS485 or RS232, the driver may be in cause. Order the relevant driver to ENRAF AUXITROL MARINE.

The driver has to be inserted on relevant PCB (see 2.4 or 2.5) at the same location than the old one's, taking care with the notch.

5.2.15 Analysing Unit PCB replacement

Order to Enraf Auxitrol Marine an Analysing Unit PCB for replacement (see 6). Don't forget to mention the name of vessel, the serial number of the Analysing Unit by fulfilling the report in Annex A, and provide a copy of the configuration sheet (see MT5094 Technical Manual) in order to help Enraf Auxitrol Marine to deliver a PCB with correct downloaded parameters and configuration. Refer to 2.4.

a) Switch off the power supply;
b) Disconnect the solenoid valves and relays connectors J20, J21, J10,…;
c) Disconnect the detectors' wires, remembering the terminals for each detector;
d) Disconnect the power supply 24 Vcc, the communication port(s) remembering the terminals for each port;
e) Disconnect the vacuum sensor pipe;
f) Remove the six screws and washers fixing the PCB;
g) Install the new PCB and reconnect in reverse order;
h) Switch on the power supply;
i) Check the configuration by comparison with the origin configuration sheet (see MT5094 Technical Manual).

5.2.16 Zener barrier test and replacement

a) By inserting a milliamperemeter in input and output current loops, compare the current values; if different, the Zener barrier needs to be changed;
b) To change a Zener barrier, order it to Enraf Auxitrol Marine, disconnect the wires from the old barrier, remembering the terminals, and reconnect the new one accordingly.

5.2.17 Control Unit LCD display PCB replacement

Order to ENRAF AUXITROL MARINE a LCD display PCB for replacement (see 6). Refer to 2.5.

a) Switch off the power supply;
b) Disconnect the two 10 pins connectors from the Control Unit PCB and the backlight 2 pins connector;
c) Remove the four nuts and washers fixing the PCB, using a 5.5 mm spanner;
d) Install the new PCB and reconnect in reverse order.

5.2.18 Control Unit front panel

Order to ENRAF AUXITROL MARINE a Control Unit front panel for replacement (see 6). Refer to 2.5.

a) Switch off the power supply;
b) Disconnect the connector from the front panel to the Control Unit PCB;
c) Disconnect the two 10 pins connectors from the Control Unit PCB and the backlight 2 pins connector;
d) Remove the four nuts and washers fixing each Control Unit PCB and LCD display PCB using a 5.5 mm spanner;
e) Install the two PCB's on the new front panel; reassemble in reverse order.

5.2.19 Control Unit PCB replacement

Order to ENRAF AUXITROL MARINE a Control Unit PCB for replacement (see 6). Refer to 2.5.

a) Switch off the power supply;
b) Disconnect the connector from the front panel;
c) Disconnect the two 10 pins connectors from the LCD display PCB and the backlight 2 pins connector;
d) Disconnect the power supply 24 Vcc and the communication port;
e) Remove the four nuts and washers fixing the PCB using a 5.5 mm spanner;
f) Install the new PCB and reconnect in reverse order.

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6 SPARE PARTS LIST

Analysing and Control Units components, with ordering code :

CODE	DESIGNATION
721302	Fan 230 Vac
721303	Venting filter
27711	Sampling solenoid valve
25734	Solenoid valve EVA or EVC
25780	Solenoid valve EV0
25735	Solenoid valve EVB
28267	Spare valves for pump P1 or P2
28269	Spare gaskets for pump P1 or P2
28270	Spare diaphragm for pump P1 or P2
25726	Pump P1 or P2
27712	Filter cartridge
25727	Filter
25729	Flowmeter
965628	By-pass valve
M20510	Analysing Unit PCB
28273	Fuse for Analysing Unit PCB, 1.8 A temporised
25730	24 Vcc power supply
25731	Circuit breaker
27168	Relay
27725	EMC electric filter
22819	Control Unit front panel
25720	LCD display PCB
22833	Control Unit PCB
24408	Hydrocarbon gas I.R. OPTIMA+ detector LEL Methane
28585	Hydrogen sulphide detector 50 ppm
29325	Hydrogen sulphide spare cell 50 ppm
29004	Sulphur dioxide detector15 ppm
29327	Sulphur dioxide spare cell 15 ppm
29005	Carbon monoxide detector 100 ppm
29328	Carbon monoxide spare cell 100 ppm
29006	Oxygen detector 25% V/V
29329	Oxygen spare cell 25% V/V
29007	Ammoniac detector 50 ppm
29330	Ammoniac spare cell 50 ppm
29008	Nitrogen oxide detector 100 ppm
29331	Nitrogen oxide spare cell 100 ppm
29009	Nitrogen dioxide detector 10 ppm
29332	Nitrogen dioxide spare cell 10 ppm
29562	Zener barrier 7787+

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Remote detectors, with ordering code :

CODE	DESIGNATION
29547	Hydrocarbon gas detector LEL Methane
29630	Hydrocarbon gas spare cell LEL Methane
29628	Calibration chamber for Hydrocarbon gas detector
29559	Hydrogen sulphide detector 25 ppm
29555	Hydrogen sulphide spare cell 25 ppm
29560	Sulphur dioxide detector 10 ppm
29556	Sulphur dioxide spare cell 10 ppm
29558	Carbon monoxide detector 250 ppm
29553	Carbon monoxide spare cell 250 ppm
29631	Oxygen detector 25% V/V
29626	Oxygen spare cell 25% V/V
29632	Ammoniac detector 50 ppm
29627	Ammoniac spare cell 50 ppm
29561	Nitrogen dioxide detector 10 ppm
29557	Nitrogen dioxide spare cell 10 ppm
29629	Calibration chamber for toxic gas detector

Suction accessories components, with ordering code :

CODE	DESIGNATION
27710	Shut-off valve w/ brass coupling
28190	Regulating valve w/ brass coupling
M12306	Bulkhead penetration for copper deck pipe ∅ 6 mm
29027	Bulkhead penetration for S.S. deck pipe ∅ 6 mm
M15005	Bulkhead penetration for copper deck pipe ∅ 10 mm
29028	Bulkhead penetration for S.S. deck pipe ∅ 10 mm
965394	Flame arrestor
27717	Non-return valve kit for copper deck pipe ∅ 6 mm
29066	Non-return valve kit for S.S. deck pipe ∅ 6 mm

7 INFORMATIONS

7.1 Claim report

The target is : help us to help you !
Despite the Troubleshooting guide and the Repair Procedures, in case of spare part need or of request for Service, the Claim Report in annex A needs to be fulfilled and transmitted by fax to Enraf Auxitrol Marine. This will help us to confirm the nature of failure and remedies, for better service.

This report will be requested before any other intervention.

7.2 Return for repair

The Return for Repair form in annex B needs to be fulfilled and transmitted to Enraf Auxitrol Marine together with the defective equipment in purpose. This will help us to identify the defect and the action to carry out, for better service.

7.3 Reference of Customer Service

Address :

Enraf Auxitrol Marine
Customer Servic
Bât 59, rue Isaac Newton
ZA Port Sec Nord
Esprit 1
18941 BOURGES Cedex 9

Manager : Alain AGOSTINI

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7.4 Personal notes

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APPENDIX A

GASBAL™ SYSTEM - CLAIM REPORT

Vessel :...Hull number :...
Owner or Shipyard :..

Analysing Unit	P/N :	S/N :.....................	Soft version : ....
Control Unit 1	P/N :	S/N :.....................	Soft version : ....
Control Unit 2	P/N :	S/N :.....................	Soft version : ....
Control Unit 3	P/N :	S/N :.....................	Soft version : ....
Remote detector(s) type :	....	S/N :.....................	

1) Description of trouble, with read values, messages, indicators status, alarms status, etc…:

2) Result of troubleshooting item applied, osbervations :

3) Carried out remedies :

4) Requested spare parts :

NAME :	**DATE :**
QUALITY :	**SIGNATURE :**

APPENDIX B

GASBAL™ SYSTEM - RETURN FOR REPAIR FORM

Vessel :.. Hull number :..
Owner or Shipyard :..

Analysing Unit	P/N :	S/N :.....................	Soft version : ....
Control Unit 1	P/N :	S/N :.....................	Soft version : ....
Control Unit 2	P/N :	S/N :.....................	Soft version : ....
Control Unit 3	P/N :	S/N :.....................	Soft version : ....
Remote detector(s) type :	....	S/N :.....................	

1) Name of returned defective equipment :

2) Description of defect, reason for suspecting the returned equipment (Result of troubleshooting item applied, osbervations, read values, messages, indicators status, alarms status, etc…) :

3) Requested spare parts :

NAME :	**DATE :**
QUALITY :	**SIGNATURE :**

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GASBAL™ SYSTEM

Gas Detection System for ballast tanks, void spaces, pump room, accommodations

ISGOTT CHAP. 7.8 AND 8.2

TECHNICAL MANUAL

CONTENT

1 INTRODUCTION

The **GASBAL™ System** is dedicated to analyse Gases in Tanks (ballast, cargo,…), Void Spaces or any possible dangerous area, in order to detect any Gas Concentration level over or under safety limits and to monitor actions as visible and audible alarms. That System complies with **ISGOTT regulation chap. 7.8 and 8.2.**

The GASBAL™ System is a Gas Monitoring based on suction process sampling toward one or more common detector(s), combined with analog inputs from local gas detectors.

Controlled Gases are :

- Hydrocarbon gases
- Oxygen
- Hydrogen Sulphide
- Carbon monoxyde
- Sulphur dioxyde
- Nitrogen monoxyde
- Nitrogen dioxyde
- Ammonia
- Chlorine
- Hydrides
- Other on request

The sampling points for suction process are to be placed as close as possible to the gas emission source. However regarding the ballast tanks, they can be placed on tank top even if the gas emission is in tank bottom, in order to preserve the longest efficient period when ballast tanks are filling up, to ensure the fire prevention efficiency of the system as fire can be caused by gas coming out on deck from tank top, and also because gases are pushed toward top by tank filling (IMO Sub-Committee on fire protection).

The local detectors can be placed in pump room, engine room, accommodations, venting ducts, air-conditioned ducts.

The Installation of the System is described in the **MI5094 Installation Manual**.
The Maintenance of the System is described in the **MM5094 Maintenance Manual**.

2 SYSTEM DESCRIPTION

The typical GASBAL™ System is described by the synopsis on following page.

The GASBAL™ System is delivered with a calibration gas kit designed for calibration of all gas detectors, installed in analysing Unit as well as local ones.

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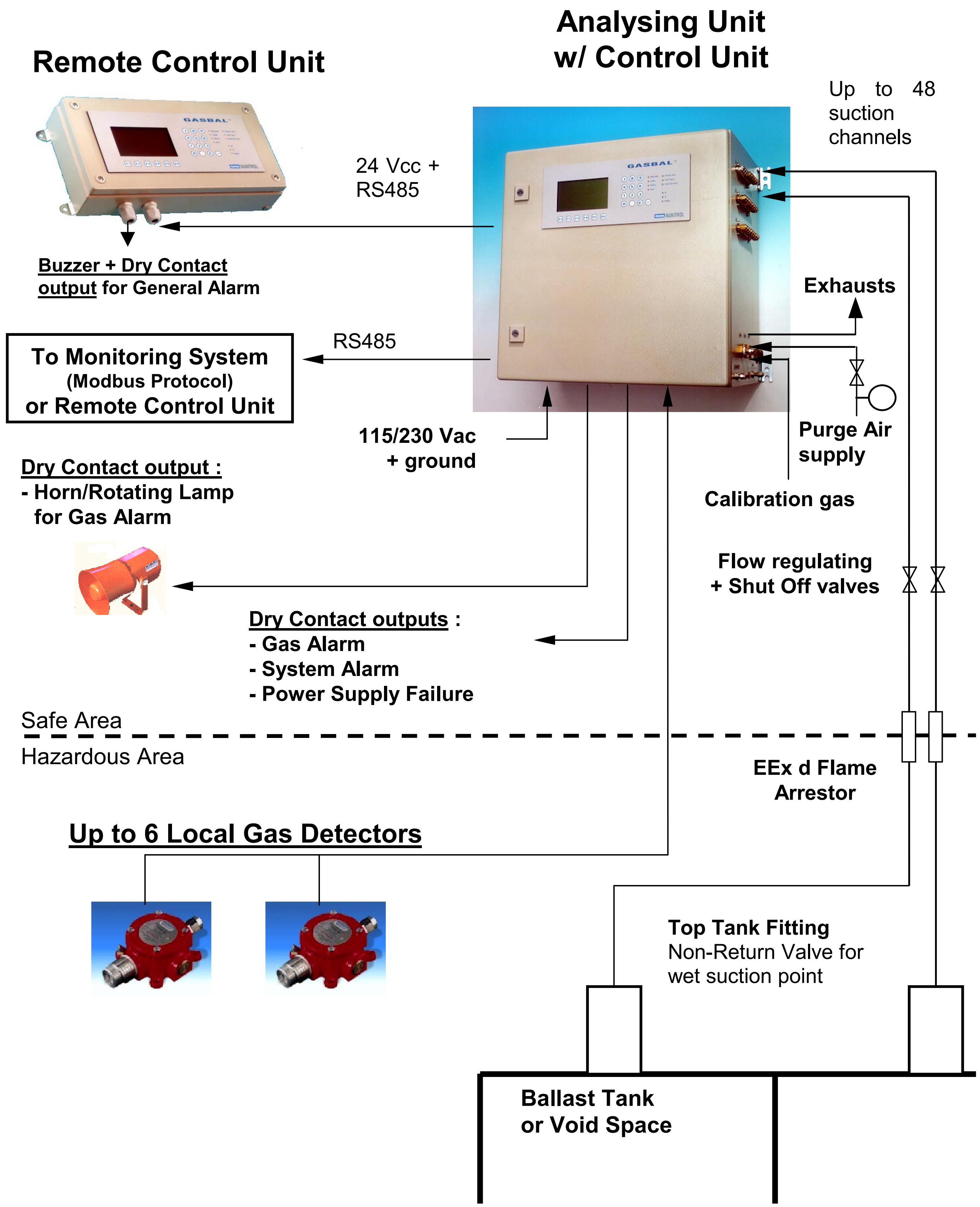

2.1 Analysing Unit

A cabinet named "Analysing Unit" (Type Drawing **M24442**) includes the suction process sampling gas detector(s) (see 2.4) and pneumatic components for up to 48 channels, from multiple(s) of 8 channels (pumps, solenoids valves, water trap filter, pressure sensor). The unused channels are plugged. A modular arrangement of solenoid valves allows easy sizing of channels number, easy extension as well as easy maintenance. An 8 channels extension module can easily be installed further.

After suction sampling of a channel, an instrument air purge is started, in order to correct or prevent ingress of water, moisture, condensation, dust or foreign bodies.

The Unit provides a parallel gas circulating pump in all unselected channels, in order to ensure fresh gas samples in Unit in any circumstances.

In addition, a solenoid valve EV0 (See hereafter) allows to control the internal atmosphere of the Analysing Unit, named channel 0.

In case of several sampling gas detectors, the sampled gas passes through each detector one after the other, so that all gas concentrations are measured at the same time. An additional Gas detector can easily be installed further. Each Gas detector can be enabled or disabled at will (see 7.4.1).

An electronic module supervises over the suction process, provides up to 7 analog inputs for gas detectors placed either in suction process sampling or in remote conditions. The local gas detectors are classed as additional individual channels as well as additional gas detectors. That module periodically performs and saves all gas measurements. It provides a non-volatile memory for saving all data, alarm status and operating parameters.

The Analysing Unit provides two high or low gas concentration alarm levels on each channel and each gas (see 7.4.2), monitoring a permanent "Gas alarm" dry contact output (see 6.2) :
- pre-alarm limit, to advise that gas concentration approaches the alarm limit;
- alarm limit, with activation of another dry contact output dedicated to a horn/rotating lamp, until alarm acknowledgement.

When some gas are without interest for some channels, the relevant alarm limits can be set at the extremities of the gas range, in order to avoid gas concentration alarm monitoring for that gas on these channels.

The Analysing Unit provides self-tests : sample flow, internal gas concentration, internal leakproof, detector's current output, calibration gas flow, communication between Analysing and Control Units, power supply. These tests monitor a "System fault" and "Power supply" dry contact outputs.

This module also manages 2 digital RS485 communication ports and one RS485/RS232 isolated communication port for connection to one or more Control Unit(s) as described in synopsis, and/or an external Monitoring System, using MODBUS RTU protocol (see Specification ST1086 GASBAL Communication Protocol). The Analysing Unit operates as slave.

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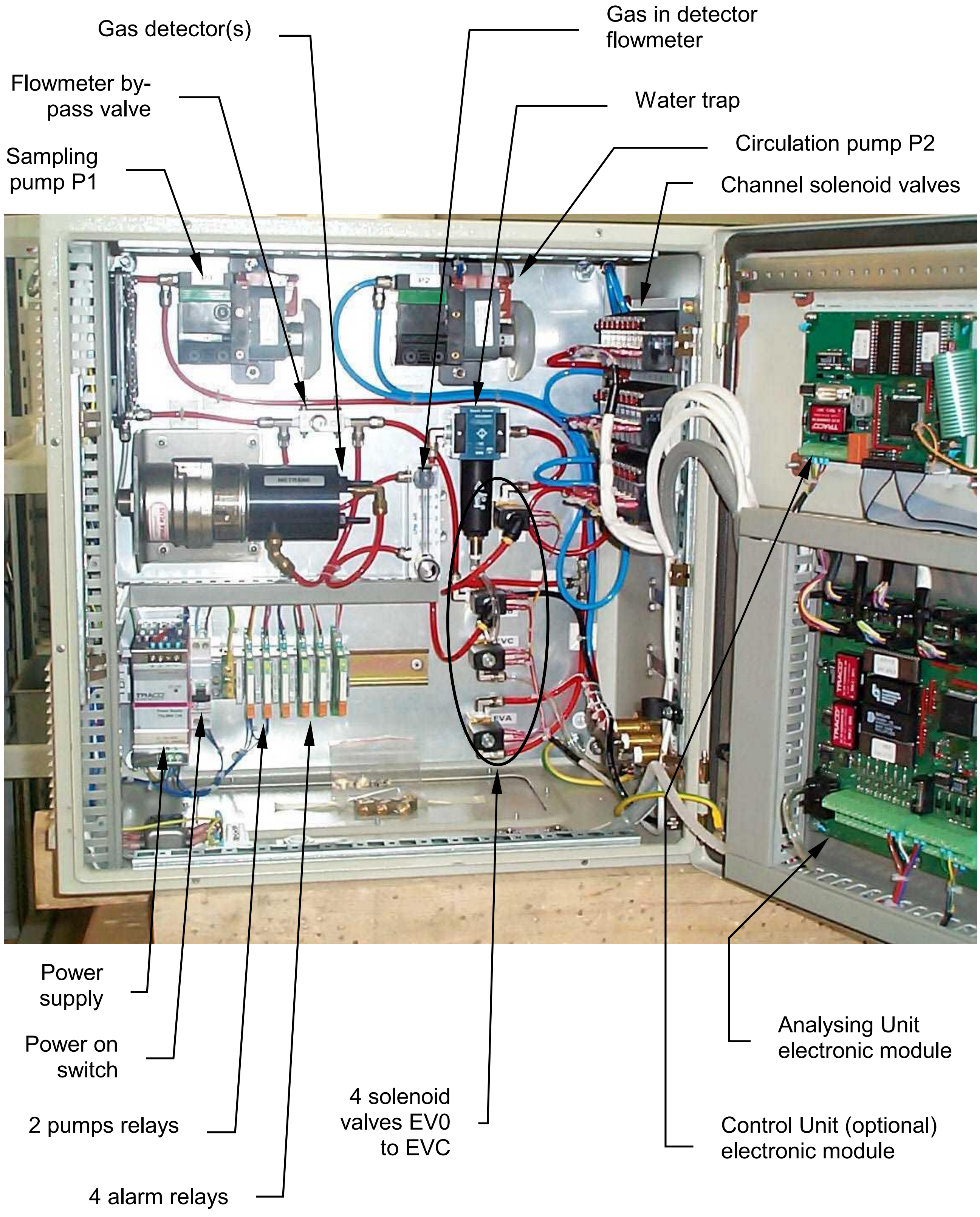
Analysing Unit internal overview :



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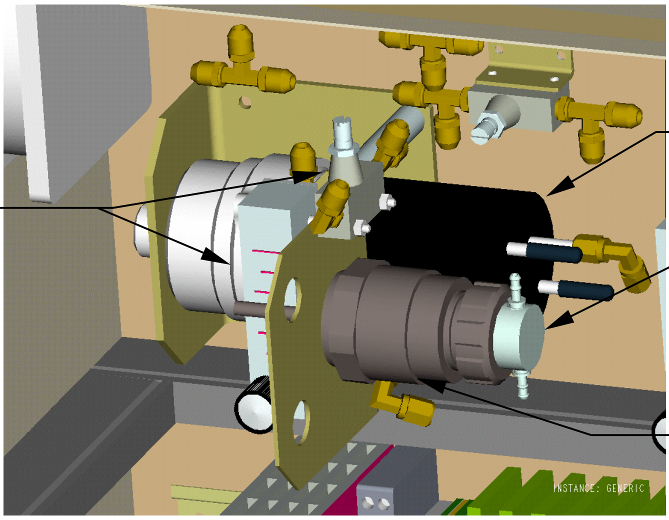
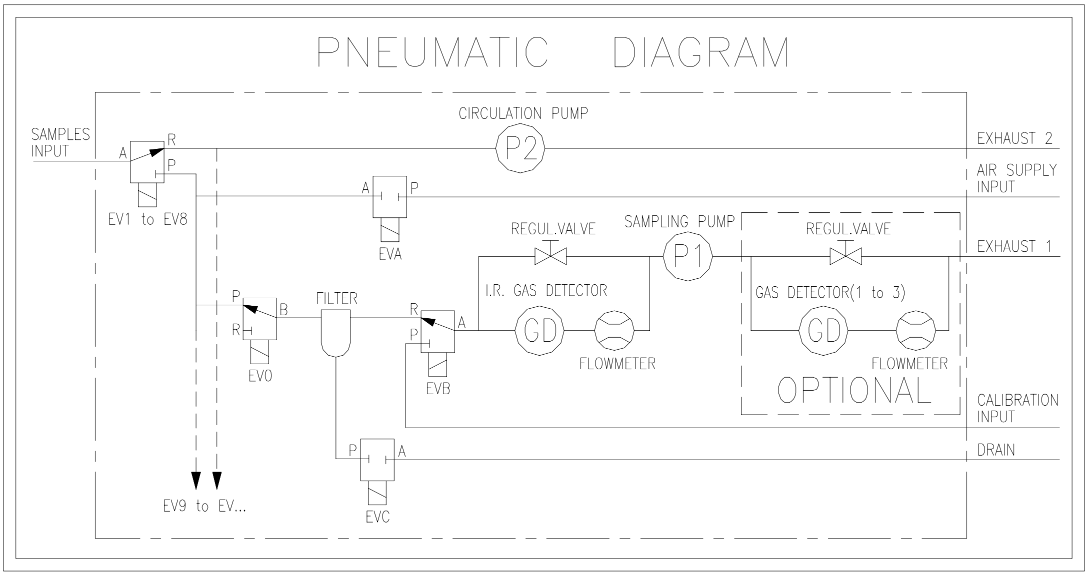
Additional detector(s) kit :



Pneumatic Diagram :



- P1 is the sampling pump;
- P2 is the circulating pump providing parallel gas circulation of all unselected lines;
- EV0 allows the cabinet internal atmosphere control, and the P1 pump discharge;
- EVA allows to proceed to an air purge of selected channel;
- EVB allows to drive the Calibration Gas to the detector(s);
- EVC allows to purge the water trap;
- Gas Detectors (GD) : the Hydrocarbon gas Infrared detector is placed upstream P1 pump; other detector(s) is(are) placed downstream P1.

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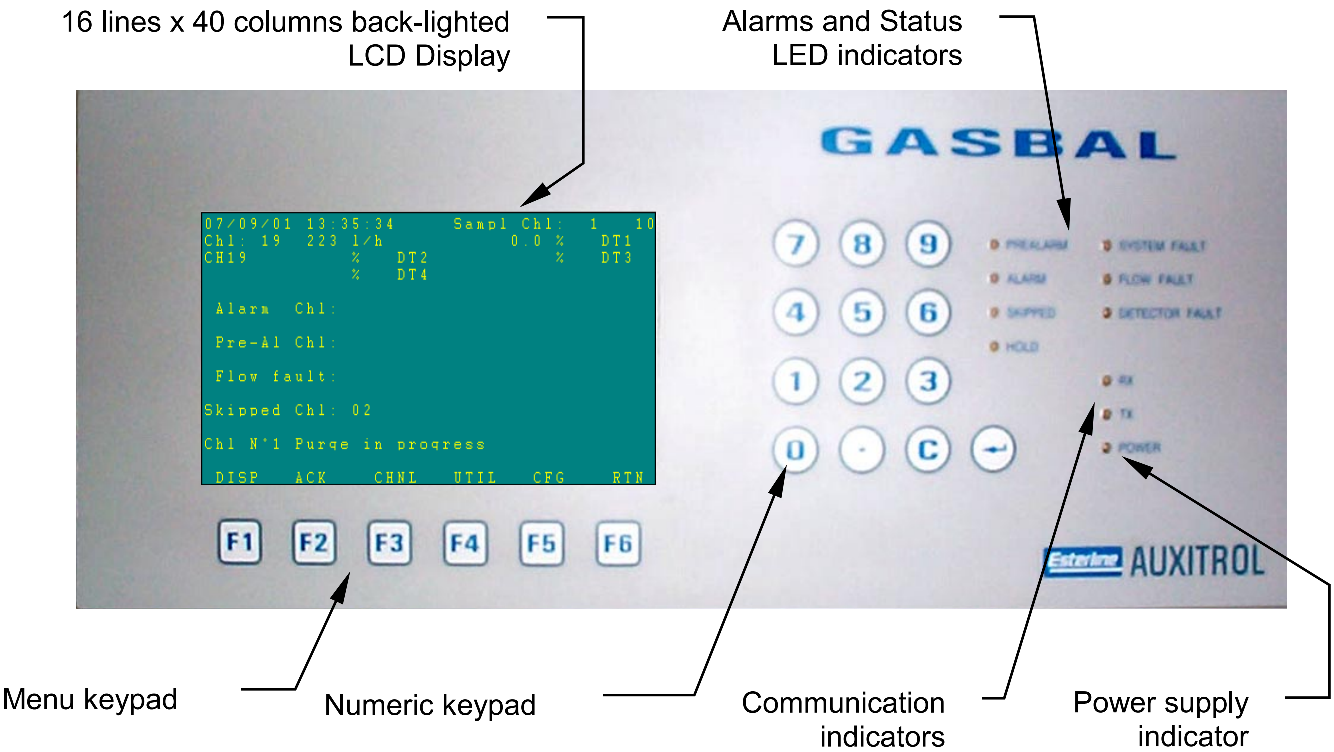
2.2 Control Unit

A panel named "Control Unit" includes a backlighted LCD screen for measurements and alarm display, LED indicators, buzzer/dry contact output for alarm and keyboard for System operation and configuration. The large LCD screen increases the data availability and the man-machine interface is simplified for easy access to functions and configuration using scrolling menus. It communicates with the Analysing Unit using a Modbus protocol serial link RS485, operating as master in order to get all data and status from the Analysing Unit, and to set any configuration change. The Control Unit is powered by 24 Vdc from the Analysing Unit.

One Control Unit can be incorporated in the Analysing Unit (Type Drawing **M24442**) or located in a remote box (Type Drawing **M27816**) or in existing console(Type Drawing **M28193**). One or two others can be used as repeater(s), but all Control Units will operate identically, the last order from any Control Unit is considered. As the common database (gas measurements, fault status, operating parameters) is saved in the Analysing Unit, all Control Units display the same data.

Control Unit front overview :



- LCD display is described in 6.1.2;
- Menu keypad are function keys described in 6.4;
- Numeric keypad allows to key-in channel's number and all numeric values; "↵" is "Enter" key; "C" cancels the last keyed-in or displayed digit.

IMPORTANT NOTE : the Analysing Unit and Control Unit operate independently. So, in case of failure of communication with the Control Unit, the Gas Concentration alarms and System faults remain under control by the way of the Analysing Unit Gas Alarm and System fault relays.

2.3 Suction sampling accessories

The accessories are described on drawing M28194 for deck equipments and drawing M27848 for equipments adjacent to the Analysing Unit.

2.3.1 Deck equipments

These equipments consist in non-return valves for top wet tank fitting.

2.3.2 Analysing Unit adjacent equipments :

- Bulkhead penetrations for suction process channels : the number of bulkhead penetration set is defined by the number of bulkhead to pass through. For each channel a flame–arrestor EEx d class is placed at the limit between safe and hazardous area, on relevant bulkhead penetration;
- Shut-off valves for each channel;
- Flow-regulating valve for each channel allows to equilibrate the flow whatever is the length of the line, for best efficiency of the system;
- Bulkhead penetrations for gas exhaust and filter drain;
- Shut-off valve and manometer for instrument air supply for channel purge.

2.4 Gas detectors

The GASBAL™ System uses two locations of Gas detectors : inside the Analysing Unit, in suction process sampled Gas, and in local areas to be controlled. Any gas detectors of Infrared, catalytic, electro-chemical or any other type, 4-20 mA output, can be connected to the Analysing Unit.

2.4.1 Gas detectors located inside the Analysing Unit, in suction process sampled Gas :

- detector dedicated to Hydrocarbon gases is Infrared type, for best reliability and maintenance free.
- other detectors for :
  - Oxygen
  - Hydrogen Sulphide
  - Carbon monoxyde
  - Sulphur dioxyde
  - Nitrogen monoxyde
  - Nitrogen dioxyde
  - Ammonia
  - Chlorine
  - Hydrides

2.4.2 Local Gas detectors

Same detectors can be used, but installed with connection box. The Safety class and the installation process comply with hazardous location; when required, Zener barriers are installed in the Analysing Unit.

3 SYSTEM SELECTION

See definition synthesis on following page, where are mentioned the Part Numbers and Order Codes.

1. Select model of Analysing Unit, with or without integrated Control Unit;
2. Select number of suction channels : the Analysing Unit includes as minimum a kit of 8 channels; select the number of additional 8 valves kits up to 5, in order to get 48 channels maximum;
3. Select number of plug kits by difference from the number of additional 8 valve kits;
4. Select type of Gas Detector(s) to be included in the Analysing Unit;
5. Select the number and type of local Gas Detector(s), together with Zener barrier(s) when required;
6. Select the calibration gas : the System is delivered with an empty cylinder with pressure/flow regulator, to be filled with selected calibration gas;
7. Select the optional Remote Control Unit(s) model, mounted in box or in existing console. If the Analysing Unit doesn't include a Control Unit, at least a Remote Control Unit is necessary;
8. Select the number of regulating valve kits and shut off valves according to the number of suction channels. Select the number of fixtures for regulating valves, each one is available for 10 valves maximum;
9. Select the number and type of bulkhead penetrations for exhausts, multiplied by the number of bulkheads to pass through;
10. Select the number of bulkhead penetrations with flame-arrestor according to number of suction channels. If the lines need to pass through additional bulkheads, select the number of simple bulkhead penetrations, multiplied by the number of bulkheads to pass through;
11. Select the number of non-return valves according to the number of wet suction points;
12. Select the number of deck pipes fixing kits, according to the pipes arrangement on deck;
13. Select the number of Spare Parts kits; one kit is available for one year.

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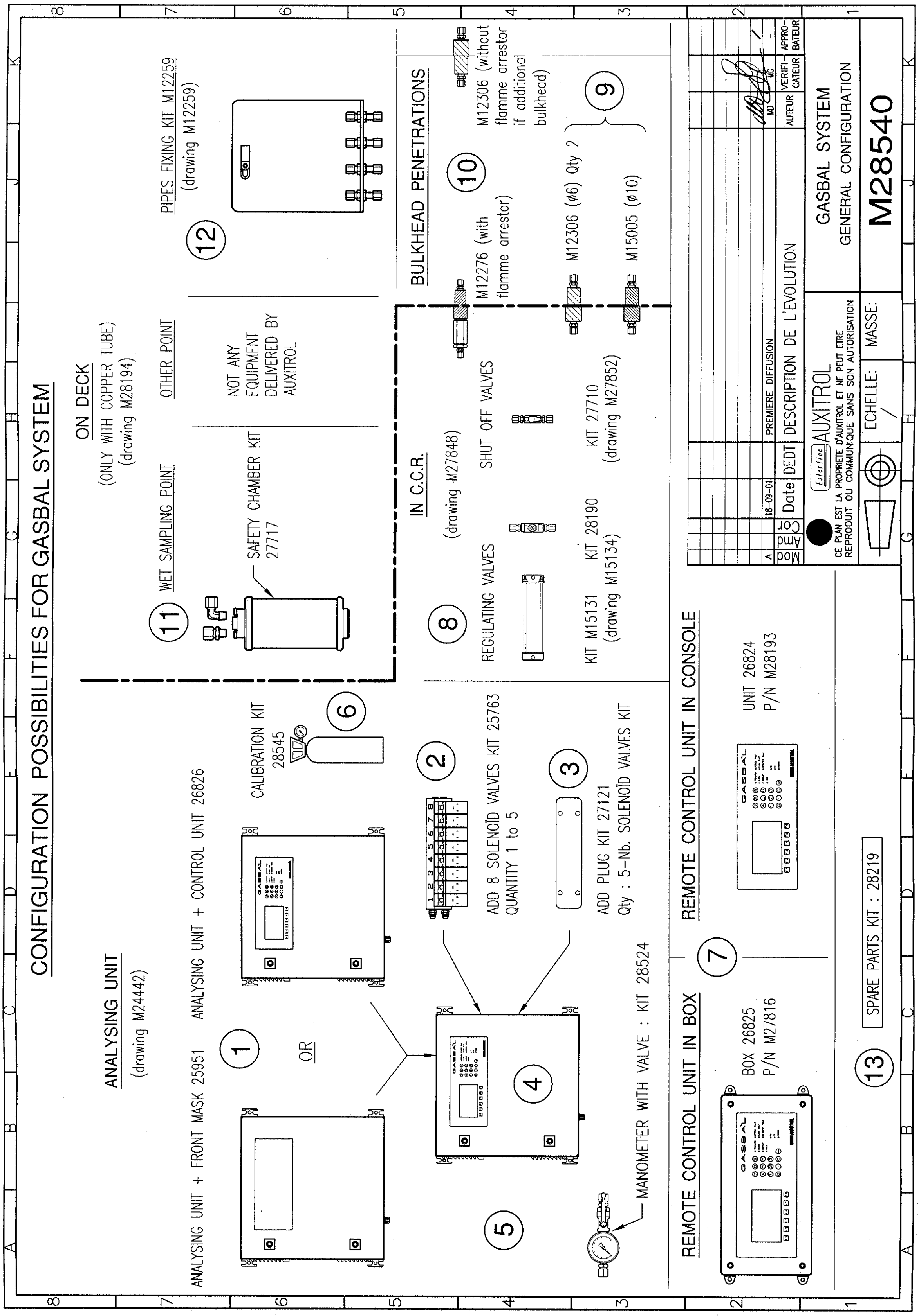

4 SPECIFICATIONS

4.1 Analysing Unit

• Suction channels number	: up to 48, connections for O.D. 6 mm copper pipes
• Suction capacity	: up to 400 meters w/ I.D. 4 mm pipes
• Sampling exhaust output	: connection for O.D. 6 mm copper pipe to safe area
• Circulation exhaust output	: connection for O.D. 10 mm copper pipe to safe area
• Drain water trap output	: connection for O.D. 6 mm copper pipe to safe area
• Calibration Gas input	: connection for O.D. 6 mm plastic pipe
• Instrument air supply for purge	: 4 / 7 bars, connection for O.D. 6 mm copper pipe
• 4-20 mA Analog calibrated inputs for sampling or local detectors	: up to 7
• 4-20 mA inputs accuracy	: 0.1 %
• Output power supply for detectors	: 24 Vcc
• Internal Detector for Hydrocarbon Gas	: Infra-Red type, 4 – 20 mA output
• Other Detectors	: miscellaneous type according to gas type, 4 – 20 mA 2 wires output or 3 wires with negative polarity common with 0 Vcc power supply (source); safety class according to their location
• Gas concentration alarm levels	: 2 adjustable, high or low Pre-Alarm and Alarm
• Alarm from Analysing Unit, dry contact outputs	: 1 for Gas Concentration Alarm 1 for System Alarm 1 for remote Horn / Rotating Lamp 1 for Power Supply failure
contact rating	: 24 Vdc – 140 W 250 Vac – 1500 VA
• Communication ports	: 2 RS485, 1 isolated RS232 or RS485 Modbus RTU protocol Slave status (acc. ST1086) Baudrate 1200, 2400, 4800, 9600, 19200, 38400 Default address 1; default baudrate 9600
• Output power supply for Control Units	: 24 Vcc
• Operating/storage temperature for Analysing and Control Units	: 0°C to +70°C
• Location for Analysing and Control Units	: Safe area in enclosed space (Control Room, accommodations, bridge,…)
• Enclosure protection	: IP22 minimum
• Electromagnetic compatibility	: acc. standard EN50081 – EN50082 European directive 89/336/CEE (CE mark)
• Electrical, climatical, mechanical environment tests	: acc. IACS test procedure E10
• Power Supply	: 115 / 230 Vac 50/60 Hz
• Power consumption	: 260 VA with one Control Unit 270 VA with three Control Units

4.2 Control Unit

- LCD display : 16 lines, 40 columns, back-lighted
- Keypad : tactile type, numeric + functions keys
- Indicators : front integrated color LED's
- Alarm : 1 internal buzzer + 1 dry contact output
- Communication port : RS232/485, Modbus RTU protocol Master status (acc. ST1086)
- Front face : Polyester continuous film
- Power supply : 24 Vcc from Analysing Unit

4.3 Gas detectors

The Gas detector(s) are selected according to the project. The Gas Detector specifications are in the Manufacturer File.

5 INSTALLATION

5.1 Mechanical connections

Refer to the Installation Manual MI5094E.

- ➢ The Analysing Unit is wall-mounting type; preserve sufficient free space for ventilation and tubing. The Remote Control Unit if any can be wall mounting type or installed in an existing console;
- ➢ The sampling pipes are connected to the channels' inlets through the shut-off valves and regulating valves, by respect to their identification and taking care to the perfect leakproof;
- ➢ The exhaust pipes are connected to the relevant exhaust connections;
- ➢ The clean and dry instrument air supply pipe is connected to the relevant inlet, through a shut-off valve with witness manometer;
- ➢ The calibration gas cylinder is connected to the relevant inlet.

5.2 Electrical connections

Refer to the Installation Manual MI5094E.

- ➢ The 115/230 Vac power supply is connected to the relevant terminals;
- ➢ The local Gas Detector(s) is(are) connected to the relevant terminals on the PCB or to Zener barrier, through shield grounded packing gland(s);
- ➢ The Remote Control Unit if any is connected to the relevant terminals, 24 Vcc power supply and RS485 communication line through shield grounded packing gland(s);
- ➢ The Monitoring System RS232 or RS485 communication line is connected to the relevant connector or terminals through shield grounded packing gland;
- ➢ Alarm on/off dry contact outputs are connected to the relevant terminals.

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6 OPERATION

6.1 Normal operating mode

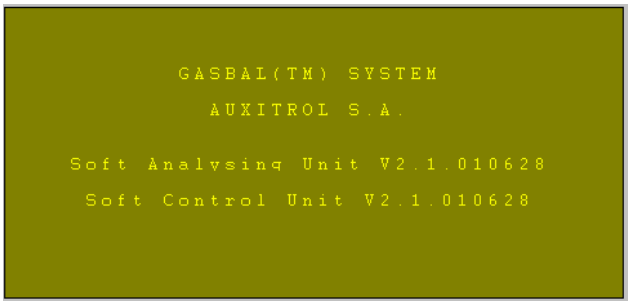
6.1.1 Start-up

The System is configured in factory with channel number, channel names (4 characters), gas detectors number and dispatching, ranges, default operating periods and alarm limits. So the System can be started up without further operations, after being properly installed and connected according to Installation Manual MI5094.

IMPORTANT NOTE before switching power on :

- the regulating valves must be in full open position (turn anti-clockwise);
- make sure that the red handles on channel solenoid valves are in full horizontal position;
- open the shut-off valves;
- open the instrument air supply shut-off valve;
- make sure that the valves of flowmeter(s) as well as by-pass valve(s) close to flowmeter(s) are open (see 2.1).

The power is switched on using the switch inside the Analysing Unit (see 2.1); the green LED indicator lights on. As soon as the power is on, the Control Unit communicates with the Analysing Unit : the Rx / Tx indicators flash one time. In case of communication failure (Rx / Tx indicators don't flash), the Control Unit display "!!! Communication Fault !!!", and an alarm starts (see 6.3.7 Communication fault). When communication is correct, the screen shows the versions of both Analysing and Control Units during 5 seconds :

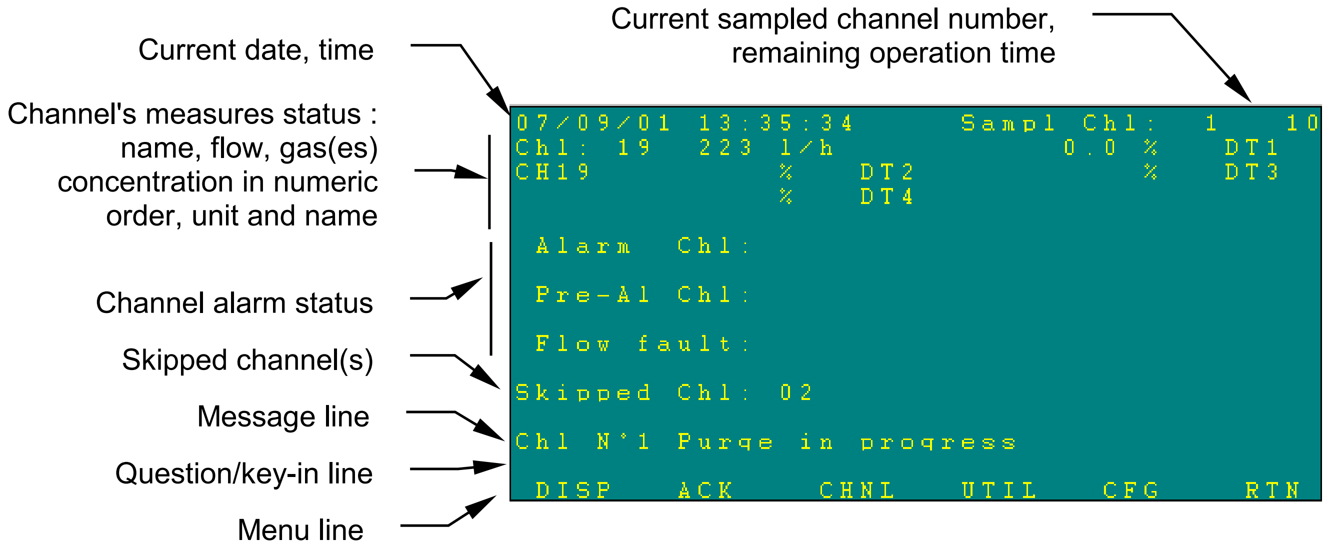


Then, the main screen is displayed (see 6.1.2), the Rx / Tx indicators flash. The message "Please wait…" is displayed during default time 90 seconds, and the Analysing Unit operation doesn't start before the end of waiting period, in order to stabilise the Gas Detectors before starting the measurements.

No function key is active, except "ACK" = alarm acknowledgement (see 6.6).

IMPORTANT NOTE : Just after starting sampling on channel 0, adjust the flow in the detector(s) on the flowmeter(s) to **2.8 liter/minute**, using **only** the by-pass valve(s) close to the flowmeter(s) (see 2.1).
The flowmeter(s) valve must stay open.

6.1.2 Control Unit's LCD Display arrangement.



- Line 1 : current date, time, then current sampled channel number and remaining operation time on that channel. Channel 99 means no sampled channel.
- Lines 2 to 5 : last saved measurements of all channels successively each 5 seconds, with channel number, name, flow for sampled channel in liter per hour, gas concentration from all enabled detectors in suction process sampling for sampled channels, or relevant detector for individual channels. First displayed are x Gas detectors from suction process sampling, next till 7th max. are for individual channels.
- Lines 6 to 11 : number of channel(s) in gas concentration alarm, gas concentration pre-alarm, flow fault, in numeric order. Two lines per alarm allows to display 22 channels (see 6.2, 6.3.4 for alarm procedure and display). Further channels make disappear the first ones.
- Lines 12, 13 : number(s) of skipped channels in numeric order (See 6.7.4). Two lines allows to display 22 channels. Further channels make disappear the first ones.
- Line 14 : used for message in regards with current operation.
- Line 15 : used for questions and key-in copy.
- Line 16 : menu tittles above relevant keys.

The backlight lights on for 15 seconds when pushing any key, or when an alarm occurs until acknowledged.

As soon as power is on, and despite the waiting period, the main screen starts displaying :

- the Channels' saved measurements on lines 2 to 5. The display begins with channel 0 (Internal atmosphere of Analysing Unit, named SYST), then all sampled channels from channel 1, then all individual channels which numbers follows those of sampled channels. The detectors' numbers for individual channels also follow those of the suction process sampling.
- as the alarm status are saved in the Analysing Unit, all present and/or not acknowledged alarms are recovered and displayed, together with buzzer monitoring.

After the waiting period, the main screen shows on line 1 current sampled channel, and remaining operation time on that channel.

6.1.3 Sampling sequence

The Analysing Unit operates independently of the Control Unit operation, except when some command is started from the Control Unit : Channel HOLD (see 6.7.3), Filter Purge (see 6.8.2), Channel Purge (see 6.8.3), Calibration (see 7.6). After the waiting period, the typical sequence is as follow :

1. Starting with channel 0 (Internal atmosphere of Analysing Unit);
2. Water trap purge during 15 sec. : the message "Filter purge in progress" is displayed (if no alarm message is present), the time is scrolling down on line 1;
3. Sampling line leakproof test during 5 sec. : the message "Leak test in progress" is displayed (if no alarm message is present), the time is scrolling down on line 1;
4. Channel 0 sampling during configured time, the time is scrolling down on line 1, no message is displayed;
5. At the end of the sampling period on channel 0, saving of gas concentration(s) and flow, and comparison with the alarm limits :
   - if the flow is over the limits, starting a flow fault (see 6.3.4, Flow alarm);
   - if one gas concentration is over the limits, starting a shutdown (see 6.3.3, System Shutdown);
6. Channel 1 sampling during configured time, the time is scrolling down on line 1, no message is displayed;
7. At the end of the sampling period, saving of gas concentration(s) and flow, and comparison with the alarm limits :
   - if the flow is under or over the limits, starting a channel purge during the configured time in order to attempt to restore the correct flow : the message "Chl N x Purge in progress" is displayed (if no alarm message is present), the time is scrolling down on line 1. Then starting a sampling period for the double configured time, the time is scrolling down on line 1. If after the second sampling period, the flow is always uncorrect, starting a flow alarm (see 6.3.4, Flow alarm);
   - if one gas concentration is over the limits, starting a pre-alarm or pre-alarm/alarm according to the concentration value (see 6.2, Gas Concentration alarms);
8. Channel 1 purge during the configured time : the message "Chl N x Purge in progress" is displayed (if no alarm message is present), the time is scrolling down on line 1;
9. Continuation up to the last sampled channel, except for skipped channels (see 6.7.4) and coming back to channel 0;
10. All the individual channels and detectors are measured and saved continuously, independently of the sampled channels. Pre-alarm or alarm sequence starts according to the concentration value (see 6.2, Gas Concentration alarms).

6.2 *Gas Concentration Alarms*

Typical alarm sequence on any channel except channel 0 (Unit internal atmosphere) :

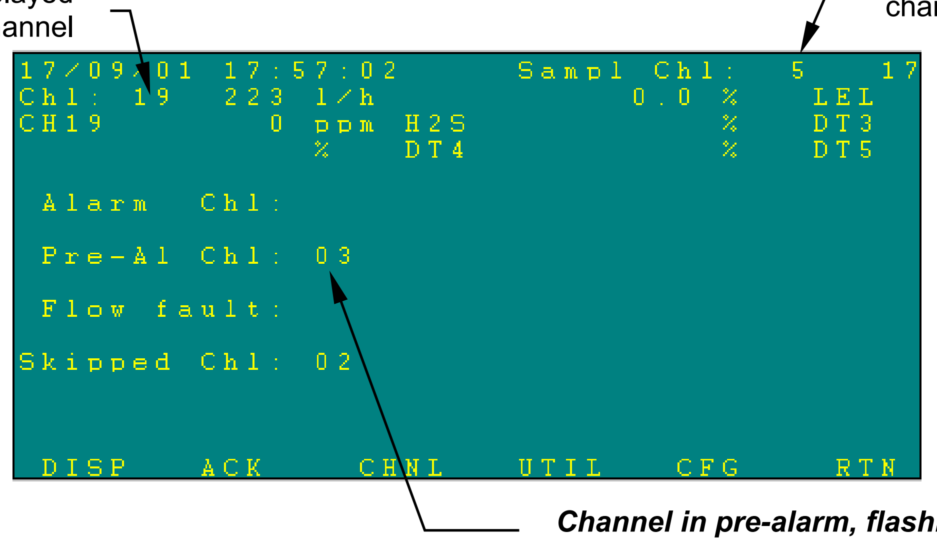
When a Gas Concentration goes over or under the pre-alarm configured limit, the Gas alarm sequence is as follow :

- the number of the relevant channel appears flashing on the main screen pre-alarm line (8-9) of the Control Unit screen, and the screen backlight lights on;
- the Pre-alarm red LED indicator lights on;
- the Gas Alarm relay is activated;
- the buzzer of the Control Unit is activated in continuous mode;
- the alarm relay of the Control Unit is activated.

Example of display :

Current displayed channel

Current sampled channel



Channel in pre-alarm, flashing

When a Gas Concentration goes over or under the alarm configured limit, the alarm sequence is as follow in addition to the pre-alarm sequence :

- the number of the relevant channel appears flashing on the main screen alarm line (6-7) of the Control Unit screen, and the screen backlight lights on;
- the Alarm red LED indicator lights on;
- the Horn/Rotating lamp relay is activated.

When the Operator pushes the ACK = Acknowledgement key :

- the display of channel number stops flashing and becomes fix, the Gas alarm acknowledgement in numeric order has priority against Gas pre-alarm acknowledgement in numeric order;
- the buzzer of the Control Unit is activated during 2 seconds each 15 seconds, except if another unacknowledged alarm is present;
- in case of alarm, the Horn/Rotating lamp relay falls in normal position except if another unacknowledged gas concentration alarm is present;
- the ACK = Acknowledgement key needs to be pushed for <u>each</u> Pre-alarm or Alarm to be acknowledged; Alarm is acknowledged in priority before Pre-alarm.

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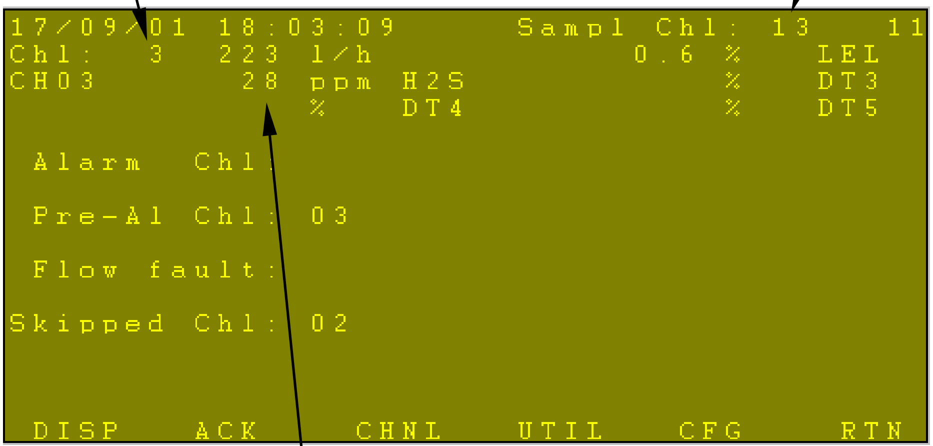
In order to know which gas concentration causes the alarm, use the "Display" function (see 6.5) for the channel in alarm. The "Display" request for channel 3 allows to read on screen :

Channel 3 display requested

Current sampled channel



H2S gas concentration causes the pre-alarm

When and only when the relevant Gas Concentration on relevant channel comes under or over the relevant limit after a further sampling :

- the channel number disappears from the relevant line;
- the Pre-alarm or Alarm indicator switches off, except if another Gas concentration Pre-alarm or Alarm is present;
- the Gas Alarm relay falls in normal status, except if another Gas concentration Pre-alarm or Alarm is present;
- the buzzer and the alarm relay of the Control Unit stops, except if another Gas Pre-alarm or Alarm or System alarm is present.

6.3 System alarms

6.3.1 Presentation

Following self-tests are performed and cause System alarm in case of failure :

- Analysing Unit internal gas concentration
- sampled channel flow
- gas detector(s)' output current
- Analysing Unit internal leakproof
- communication checked from Analysing Unit
- communication checked from Control Unit

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6.3.2 General notes

The typical System alarm sequence is as follow :
- the relevant LED indicator lights on;
- in addition when necessary, an explanation message is displayed on message line on Control Unit screen (line 14), and the screen backlight lights on. In case of several alarms situation, the priority for message display in decreasing importance order is : Communication fault, Shut-down, Sample leak, Detector fault in numeric order; alarm messages have priority against action messages (Filter purge, …);
- the System fault relay is activated;
- the buzzer of the Control Unit is activated in continuous mode;
- the alarm relay of the Control Unit is activated.

When the Operator pushes the ACK = Acknowledgement key :
- the buzzer of the Control Unit is activated during 2 seconds each 15 seconds, except if another unacknowledged alarm is present;
- the ACK = Acknowledgement key needs to be pushed for each fault to be acknowledged. The priority in decreasing importance order is : Shut-down, Detector fault, Sample leak, Flow fault in numeric order, before gas alarm, gas pre-alarm in numeric order.

When and only when the fault disappears after a further sampling for fault regarding a channel :
- the message disappears on the Control Unit screen;
- the relevant LED indicator (if any) lights off, except if another similar system fault is present;
- the System fault relay falls in normal position, except if another system fault is present;
- the buzzer of the Control Unit stops, except if another alarm is present;
- the alarm relay of the Control Unit falls in normal status, except if another alarm is present.

6.3.3 System Shutdown

When a Gas Concentration overpasses the Pre-alarm/Alarm configured limit on channel 0 (Analysing Unit internal atmosphere) , the fault is considered as a Gas concentration alarm as well as a System alarm :
- the System fault LED indicator lights on;
- the Analysing Unit stops completely : pumps, solenoid valves, venting.

For re-starting the System after check, the ACK = Acknowledgement key needs to be pushed a second time :
- the Analysing Unit starts a complete sequence from channel 0;
- the System fault relay falls in normal position, except if another System fault is present.

If the Gas Concentration overpasses again the pre-alarm or alarm limit on channel 0, the shutdown alarm sequence re-starts as above. If the Gas Concentration on channel 0 comes under the relevant limit, alarm sequence is finished and the System continues.

6.3.4 Flow alarm

When the flow on a sampled channel is under or over the configured limits, the typical System alarm sequence starts with particularities as follow :

- the number of the relevant channel appears flashing on the main screen Flow fault line (10-11) of the Control Unit screen.

When the Operator pushes the ACK = Acknowledgement key :

- the display of channel number stops flashing and becomes fix; till the flow on relevant channel comes back to normal.

In order to know the value of flow which causes the alarm, use the "Display" function (see 6.5) for the channel in alarm.

6.3.5 Detector (Gas detector) fault

When a Gas detector output current is under 3.6 mA or over 20.4 mA, the typical system alarm sequence starts, with display of a message "Detector x fault".

NOTE : if the relevant Gas detector cannot be repaired in a short time, it is possible to disable it, in order to cancel the alarm (see 7.4.1). As soon as the Gas detector is repaired or replaced, take care to enable that detector.

6.3.6 Sample leak

The leak test consists to close all solenoid valves when the sampling pump keeps running. If the flow remains over the high limit of 50 l/h, the typical system alarm sequence starts, with the System fault LED indicator lightning on and display of a message "Sample leak".

6.3.7 Communication fault

6.3.7.1 From Analysing Unit

When the Analysing Unit becomes not periodically questioned during operation, the System fault relay is activated, until the communication is restored.

6.3.7.2 From Control unit

When the communication with the Analysing Unit is not available at power on (the Rx / Tx indicators don't flash), the Control Unit display "!!! Communication Fault !!!"

When the communication with the Analysing Unit becomes unavailable during operation (the Rx / Tx indicators don't flash), the typical system alarm sequence starts, with the System fault LED indicator lightning on and display of a message "Communication Fault".

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6.4 Main menu

6.4.1 Presentation

After the waiting period, following functions are available from the main menu; they are summarised in APPENDIX B, which can be separated from present Manual to be held by operating personnel :

- DISP key F1 : selected channel "Display" function
- ACK key F2 : alarm "Acknowledgement"
- CHNL key F3 : menu for operations in regard with "Channel" command
- UTIL key F4 : menu for "Utilities"
- CFG key F5 : menu for system "Configuration", using a engineer password
- RTN key F6 : "Return" back to the normal operating mode, or going back to the previous menu, till main menu

6.4.2 General notes

- RTN key F6 = "Return" function allows also in any situation to cancel current action;
- Numeric keypad allows to key-in channel's number and all numeric values; "↵" is "Enter" key; "C" cancels the last keyed-in or displayed digit;
- Pushing a forbidden key causes a short beep from the buzzer.

6.5 Display function F1

That function allows to display a selected channel on lines 2 to 5, instead of channels scrolling each 5 seconds. The displayed values are the last one saved in memory, except for individual channels, where the values are shown in real time (see 6.1.3). That function is mainly used when an alarm occurs on a channel, in order to read immediately the cause of the alarm (see 6.2 and 6.3.4).

- Push DISP key F1 : the system asks for the channel number;
- Key-in the channel number, then "Enter";
- The "Display" indicator lights on, in order to remind the stop scrolling of channels on lines 2 to 5;
- Push RTN key F6 to go back to normal display mode, the "Display" indicator switches off.

IMPORTANT NOTE : during the Display mode, the Analysing Unit continues to sample and to proceed to Gas measurements and to alarms monitoring.

6.6 Alarm Acknowledgement F2

That function allows to acknowledge an alarm. See 6.2 to 6.3 for Alarm management.
The ACK = Acknowledgement key needs to be pushed for each Alarm to be acknowledged.

6.7 Channel menu F3

Pushing CHNL key F3 enters the Channel menu.

6.7.1 Normal mode

Pushing NORM key F1 allows to come back to the normal mode with only manually skipped channels (see 6.7.4), after using Handling mode (see hereunder).

IMPORTANT NOTE : make sure to open relevant shut-off valves, when coming back from Handling mode, otherwise flow faults would start on those channels.

6.7.2 Handling mode

Pushing HNDL key F2 allows to skip together all channels configured to be skipped during cargo or handling operation (see 7.3.2), i.e. channels corresponding to tanks likely to be filled with water (Ballasts), in order to avoid some flow faults when the non-return valves would be closed, or the risk to draw up water in the sampling lines and to damage the gas detectors despite the non-return valves and the water trap. The list of such skipped channels is displayed on lines 12,13.

IMPORTANT NOTE : make sure to close relevant shut-off valves, when starting Handling mode.

6.7.3 Channel Hold function

That function allows to sample only one particular channel.

- Push HOLD key F3 : the system asks for the channel number;
- Key-in the channel number, then "Enter" (only a sampled not skipped channel can be held);
- The "Hold" indicator lights on, in order to remind the sampling stop;
- Push RTN key F6 to go back to normal sampling operating mode, the "Hold" indicator switches off.

IMPORTANT NOTE : during the Hold mode, the Analysing Unit stops sampling, so **stops the Gas concentration alarm monitoring for other sampled channels.** So make sure to come back to normal sampling mode as soon as possible. Anyway, the Analysing Unit comes back to normal sampling mode automatically after 10 minutes, this time is scrolling down on line 1, current operation.

6.7.4 Channel skip function

That function allows to skip from sampling cycle a particular channel corresponding to tank likely to be filled with water (Ballasts), in order to avoid some flow faults when the non-return valves would be closed, or the risk to draw up water in the sampling lines and to damage the gas detectors despite the non-return valves and the water trap.

That function is also used to skip a channel in alarm during the repair, in order to avoid the alarm sequence starting again, or when carrying out some works generating dusts, in order to avoid to obstruct the Flame-arrestor or the filter.

- Push SKIP key F4 : the system asks for the channel number;
- Key-in the channel number, then "Enter" (only a sampled not held channel can be skipped, except channel 0);
- The number of the skipped channel is displayed on lines 12,13;
- Proceed same way to unskip that channel, the number of the skipped channel disappears from lines 12,13.

IMPORTANT NOTE : make sure to close relevant shut-off valve when skipping a channel, and to re-open it when going back to normal sampling.

6.7.5 Acknowledgement function

Pushing ACK key F5 allows to acknowledge an alarm on a held channel or a System alarm, as for acknowledgement in main menu. That key avoids to go out Channel menu, and so to stop the Hold function.

6.8 Utilities menu F4

Pushing UTIL key F4 enters the Utilities menu.

6.8.1 Test function

When pushing TEST key F1, the following items are checked during 3 seconds :

- the screen of the Control Unit becomes completely dark;
- all indicators except Rx / Tx light on;
- the buzzer of the Control Unit is activated;
- The Gas concentration alarm, System fault and Horn / Rotating lamp relays of the Analysing Unit, the alarm relay of the Control Unit are activated.

6.8.2 Filter Purge function

That function allows to purge the water trap manually, in case of doubt of water presence, and in addition to the automatic purge at each channels sampling cycle.

- Push PURG.FLT key F3 : the System stops sampling the current channel, which is kept displayed on line 1;
- A filter purge sequence starts for the configured time; the message "Filter purge in progress" is displayed, the time is scrolling down on line 1;
- The System starts again the sampling of the current channel before purge.

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6.8.3 Channel Purge function

That function allows to purge manually a selected channel by instrument air counter-flow, in case of obstruction or unusual flow value, or to help to open the non-return valve after use, in addition to the automatic purge of each channels just after sampling.

- Push PURG.CHL key F5 : the system asks for the channel number;
- Key-in the channel number, then "Enter" (only a sampled not skipped channel can be purged, except channel 0);
- The System stops sampling the current channel, which is kept displayed on line 1;
- A channel purge sequence starts for the configured time; the message "Channel N x purge in progress" is displayed, the time is scrolling down on line 1;
- The System starts again the sampling of the current channel before purge.

7 COMMISSIONING AND CONFIGURATION

7.1 Commissioning

The main commissioning operation for preparing the GASBAL™ System configuration, is the equalisation of the suction flow for all channels. Proceed as follow :

- Hold on all channels one by one, and note the displayed flow value in l/h, after waiting for stabilisation.
- Check these values : the lower must correspond to the channel with the longer sampling line, and the values must increase according to the reduction of the sampling line length. If not, check the relevant line to find the discrepancy (flow in excess : possible leak; low flow : possible obstruction).
- Hold on all the channel one by one, and adjust the relevant regulating valve in order to get the lower flow ± 20 l/h. So, the suction flows for all channels will be identical at ± 20 l/h.
- Correct the adjustment of the flow in the detector(s), in order to get 1.5 l/min for the common flow on all channels as set above (except channel 0), using **only** the by-pass valve(s) close to the flowmeter(s) (see 2.1).
  NOTE : the flowmeter(s) valve must stay open.

Take care to the presence of instrument air supply : open the shut-off valve and check the pressure on the manometer, in order to get efficient channel purge.

7.2 Configuration menu F5

- When pushing CFG key F5 from the main menu, the system asks for the Engineer Password : type *6854* then "Enter". If the password was properly entered, the Configuration menu is displayed. If not, the message "Invalid password" is displayed, and the system asks again for the Engineer Password.
- If the Configuration menu is entered by another Control Unit, the display shows "Configuration in progress".

7.2.1 Presentation

Following operations are available from the configuration menu; they are summarised in APPENDIX B, which can be separated from present Manual to be held by operating personnel :

- CHL key F1 : configuration of parameters in regards with the channels
- GAS key F2 : configuration of parameters in regards with the Gas detectors
- TIME key F3 : setting date and time
- CAL key F5 : detectors' calibration
- RTN key F6 : "Return" back to the previous configuration menu, or to the main menu

7.2.2 General notes

- During most configuration operation, the communication with Analysing Unit is broken (Rx/Tx indicators don't flash), so the display is not refreshed as long as the configuration menu is not exited, but the Analysing Unit keeps working.
- When the System asks for a channel number in order to set a parameter regarding some channel, it is possible to set that parameter available for all channels (except channel 0) : enter 99 as channel number. So, if only a few channels has a different value for such parameter; first enter the main value for all channels, then go back with that setting to the few channels for the different value of the parameter.
- When required, the system displays the limits of the parameter to be set, as guideline. And also the system proceeds to a control of coherency of the entered values, for instance a high alarm limit cannot be lower that the relevant low alarm limit, or a gas concentration must be inside the detector's range.
- When possible, the system reminds the current saved value of parameters; before entering a new value, cancel the uncorrect digits using "C" key. It is also possible to type "Enter" to let that value unchanged.
- Each parameter change is followed by the question "Exit / Save ?" : type 1 for exit and save, 2 for cancellation. Remind that RTN key F6 allows in any situation to cancel current action.
- RTN key F6 : "Return" back to the previous configuration menu or to the main menu with the question "Exit configuration ?". Answer Yes or No by typing F2 or F4.
- Don't forget to fulfil the Configuration Sheet in APPENDIX A, for archive use, and for detector's and channel's number reference.

7.3 Channel menu F1

Pushing CHL key F1 enters the configuration of channels' parameters menu.

7.3.1 Total menu

That menu allows to enter the channel manual purge duration. The recommended value is the double of automatic purge duration (see 7.3.2).

Pushing TOT key F1 :

- The system asks for manual purge duration;
- Key-in the value, then "Enter";

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7.3.2 Sample menu

That menu allows to enter the sampled channel sampling and purge durations. These values can be defined according to the following guide :

- sampling and purge duration setting :

Longest line length (m)	100	150	200	250	300	350	400
(Flow for information l/h)	(230)	(200)		(180)	(170)		
Sampling period (s)	25	35	50	70	95	125	155
Purge period (s)	2	3	4	5	7	9	11

- It is recommended to use the same sampling and purge periods for all channels (except channel 0) according to the longest line.
- The sampling period is set so as longest line is filled during one sampling cycle.
- The purge period is set so as the half volume of longest pipe is pushed with fresh air.
- The recommended value for channel 0 sampling is 10 seconds.

Pushing SAMP key F3 :

- The system asks for the channel number;
- Key-in the channel number or 99, then "Enter";
- Enter the sampling (dwell) time between the displayed limits, according the guide hereabove, then "Enter";
- Enter the Purge time between the displayed limits, according the guide hereabove, then "Enter";
- For a particular channel, enter 1 to declare that channel to be skipped in Handling mode; otherwise enter 0, then "Enter".

7.3.3 Flow menu

That menu allows to enter the low and high limits of a sampled channel flow. The recommended values are ±100 l/h from the average flow value for all channels set during the commissioning (see 7.1).

Pushing FLOW key F4 :

- The system asks for the channel number;
- Key-in the channel number or 99, then "Enter";
- Enter the low limit between the displayed limits, then "Enter";
- Enter the high limit between the displayed limits t, then "Enter".

The recommended values for channel 0 are low = 200 l/h, high = 400 l/h.

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7.4 Gas menu F2

Pushing GAS key F2 enters the configuration of Gas detectors' parameters menu.

7.4.1 Gas menu

That menu allows to enter for each gas detector, the calibration gas concentration and gas detector's enabling.

Pushing GAS key F2 :

- The system asks for the Gas detector number;
- Key-in the Gas detector number, then "Enter";
- Enter the calibration gas concentration in tenth % or in ppm, then "Enter";
- Enter 0 for gas detector disabling, 1 for enabling, then "Enter".

IMPORTANT NOTE : when one Gas detector is in failure and cannot be repaired immediately, it is possible to disable it to avoid false alarms, and to enable it when repaired.

7.4.2 Channel Alarm menu

That menu allows to enter the pre-alarm and alarm limits for each channel and each Gas detector.

Pushing CHL key F3 :

- The system asks for the channel number;
- Key-in the channel number or 99, then "Enter";
- The channel number or all channels is kept displayed.

The Alarm menu appears. Pushing ALRM key F4 :

- The system asks for the Gas detector number;
- Key-in the Gas detector number, then "Enter"; for an individual channel, key-in the number of detector associated to the channel number (this number is suggested from current configuration);
- Enter 1 for high limit alarms, 2 for low limit alarms, then "Enter";
- Enter the pre-alarm limit in tenth % or in ppm, then "Enter";
- Enter the alarm limit in tenth % or in ppm, then "Enter";
- Push again CHL for another channel or ALRM for another Gas detector if any and proceed same.

NOTE :

- when some Gas are without interest for some channels, the relevant pre-alarm / alarm limits can be set to the relevant Gas detector range extremities, in order to avoid gas concentration alarms monitoring for that gas on these channels;
- the factory pre-set values for Hydrocarbon gas are pre-alarm 20.0 %, alarm 50.0 %;
- the factory pre-set values for Hydrocarbon gas on channel 0 are 30.0 % for pre-alarm and alarm, according to the IMO rules;
- the factory pre-set values for Hydrocarbon gas on pump room channels are 10.0 % for pre-alarm and alarm, according to the IMO rules.

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7.5 Time menu F3

Pushing TIME key F3 allows to key-in successively the year, month, date, hour, minute, second; type "Enter" between each keyed-in value.

7.6 Gas detector(s) check / calibration menu F5

The GASBAL™ Analysing Unit's and local detectors if any are delivered with factory calibration, covered by Enraf Auxitrol Marine's Conformity Certificate. However, it is a good practice to check the System during the commissioning. Due to restriction of Gas transportation, Vessel should have relevant calibration gases available onboard, supplied by Shipyard or local supplier. Any concentration inside the detector's range can be used; however Enraf Auxitrol Marine recommends to reach a concentration between half and full range.

IMPORTANT NOTE : before proceeding to any gas detector check or calibration by following automatic procedure, make sure that the used calibration gas concentration is properly setup in configuration, see 7.4.1 .

Pushing CAL key F5 enters the Gas detectors checking or calibration using proper calibration gas cylinder :

- For an Analysing Unit's detector calibration, keep the Gas calibration inlet free of any connection; for a local detector calibration, install its calibration chamber;
- Push GAS key F1 to select the Gas detector, key-in the Gas detector number, then "Enter"; for an Analysing Unit's detector calibration, the Unit begins to draw up on the calibration inlet;
- Push ZERO key F3 : the flow for an Analysing Unit's detector, the current and gas concentration are displayed;
- Make sure that fresh air without gas pollution enters the Analysing Unit or a local detector. For an Oxygen detector, connect a Nitrogen gas cylinder to the Gas calibration inlet for an Analysing Unit's detector calibration, or on the local detector's calibration chamber. See further for gas flow adjustment;
- When the concentration measure is stabilised close to the expected value ± 5% of the normal range, push ZERO key F3 a second time : a beep and the displayed message "Zero done" confirms that the Zero is recorded;
- Push SPAN key F5 : the flow for an Analysing Unit's detector calibration, the current and gas concentration keep displayed. For an Analysing Unit's detector calibration, the message "Cal.Flow Fault" is displayed when the flow is under the low limit for channel 0;

- Connect the Gas calibration cylinder on Gas calibration inlet for an Analysing Unit's detector calibration, or on the local detector's calibration chamber;
  - For an Analysing Unit's detector calibration, adjust the pressure/flow regulator from the cylinder so that the flow on the detector(s) is about 3 l/min, and the general displayed flow is about 400 l/h;
  - For a local detector calibration, adjust the pressure/flow regulator from the cylinder to about 2 l/min.
  - For an Oxygen detector, make sure that fresh air without gas pollution enters the Analysing Unit or a local detector.
- When the concentration measure is stabilised close to the expected value and less than ± 5% of the normal range, it is not necessary to correct the current calibration parameters; push RTN key F6 to exit the calibration procedure without calibration parameters modification;
- When the concentration measure is stabilised close to the expected value and less than ± 10% of the normal range, push SPAN key a second time : a beep and the displayed message "Span done" confirms that the Span is recorded; then the calibration parameters are changed accordingly. If the measure is over ± 10% of the normal range, the proper calibration of the Gas detector should be checked (See Maintenance Manual MM5094);
- Push RTN key F6 to exit the calibration menu.
- Close and disconnect the Gas calibration cylinder; uninstall the local detector's calibration chamber.

8 PERSONAL NOTES

APPENDIX A

CONFIGURATION SHEET

Analysing Unit P/N : .... Soft version : .... Date :..................
Control Unit P/N : .... Soft version : ....

CHANNEL NAMES (4 characters) :
Note :
- individual channel numbers follow sampled channel numbers
- tick box for sampled channel to be skipped in Handling Mode

1:		2:		3:		4:		5:		6:	
7:		8:		9:		10:		11:		12:	
13:		14:		15:		16:		17:		18:	
19:		20:		21:		22:		23:		24:	
25:		26:		27:		28:		29:		30:	
31:		32:		33:		34:		35:		36:	
37:		38:		39:		40:		41:		42:	
43:		44:		45:		46:		47:		48:	
49:		50:		51:		52:		53:		54:	

GAS DETECTORS :

N°, TEXT (3 c)	Unit (% or ppm)	Range (tenth)	Channel	Cal. concentration
1:		-	to	
2:		-	to	
3:		-	to	
4:		-	to	
5:		-	to	
6:		-	to	
7:		-	to	

MANUAL PURGE DURATION (seconds) : ....

SAMPLING PERIOD (seconds) :

0:	1:	2:	3:	4:	5:	6:	7:	8:
9:	10:	11:	12:	13:	14:	15:	16:	17:
18:	19:	20:	21:	22:	23:	24:	25:	26:
27:	28:	29:	30:	31:	32:	33:	34:	35:
36:	37:	38:	39:	40:	41:	42:	43:	44:
45:	46:	47:	48					

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PURGE PERIOD (seconds) :

1:	2:	3:	4:	5:	6:	7:	8:	9:
10:	11:	12:	13:	14:	15:	16:	17:	18:
19:	20:	21:	22:	23:	24:	25:	26:	27:
28:	29:	30:	31:	32:	33:	34:	35:	36:
37:	38:	39:	40:	41:	42:	43:	44:	45:
46:	47:	48:						

AVERAGE FLOW (l/h) after regulating valves adjustment :........................

FLOW LIMITS (low, high, l/h) :

0:	1:	2:	3:	4:	5:	6:
7:	8:	9:	10:	11:	12:	13:
14:	15:	16:	17:	18:	19:	20:
21:	22:	23:	24:	25:	26:	27:
28:	29:	30:	31:	32:	33:	34:
35:	36:	37:	38:	39:	40:	41:
42:	43:	44:	45:	46:	47:	48:

GAS ALARM LIMITS (pre-alarm, alarm, in tenth % or ppm) :

Channel N°	Gas Detector 1	Gas Detector 2	Gas Detector 3	Gas Detector 4	Gas Detector 5	Gas Detector 6	Gas Detector 7
0	-	-	-	-	-	-	-
1	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-
3	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-
5	-	-	-	-	-	-	-
6	-	-	-	-	-	-	-
7	-	-	-	-	-	-	-
8	-	-	-	-	-	-	-
9	-	-	-	-	-	-	-
10	-	-	-	-	-	-	-
11	-	-	-	-	-	-	-
12	-	-	-	-	-	-	-
13	-	-	-	-	-	-	-
14	-	-	-	-	-	-	-
15	-	-	-	-	-	-	-
16	-	-	-	-	-	-	-
17	-	-	-	-	-	-	-
18	-	-	-	-	-	-	-
19	-	-	-	-	-	-	-
20	-	-	-	-	-	-	-

Continuation :

21	-	-	-	-	-	-	-
22	-	-	-	-	-	-	-
23	-	-	-	-	-	-	-
24	-	-	-	-	-	-	-
25	-	-	-	-	-	-	-
26	-	-	-	-	-	-	-
27	-	-	-	-	-	-	-
28	-	-	-	-	-	-	-
29	-	-	-	-	-	-	-
30	-	-	-	-	-	-	-
31	-	-	-	-	-	-	-
32	-	-	-	-	-	-	-
33	-	-	-	-	-	-	-
34	-	-	-	-	-	-	-
35	-	-	-	-	-	-	-
36	-	-	-	-	-	-	-
37	-	-	-	-	-	-	-
38	-	-	-	-	-	-	-
39	-	-	-	-	-	-	-
40	-	-	-	-	-	-	-
41	-	-	-	-	-	-	-
42	-	-	-	-	-	-	-
43	-	-	-	-	-	-	-
44	-	-	-	-	-	-	-
45	-	-	-	-	-	-	-
46	-	-	-	-	-	-	-
47	-	-	-	-	-	-	-
48	-	-	-	-	-	-	-
49	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-
51	-	-	-	-	-	-	-
52	-	-	-	-	-	-	-
53	-	-	-	-	-	-	-
54	-	-	-	-	-	-	-

COMMUNICATION WITH EXTERNAL MONITORING :

Modbus Address : ....
Baudrate :..

APPENDIX B

COMMAND SUMMARY

1. **DISP, Display function,** then key-in number of channel : displays selected channel, the others are managed in normal mode. Mainly used for channel display after alarm. Push RTN to exit.
2. **ACK, Acknowledgement** :
   - stops flashing channel number in case of Gas concentration pre-alarm / alarm or flow alarm;
   - stops Horn / Rotating lamp relay in case of Gas concentration alarm;
   - activate the buzzer during 2 seconds each 15 seconds;
   - one key push for one fault to acknowledge.
3. **CHNL :** operation on channels menu, **RTN** to come back to main menu.

3.1. **NORM** : normal mode channel sampling, with manually skipped channels.

Caution : open sampled channel shut-off valves.

3.2. **HNDL** : cargo handling mode channel sampling, with skipped channels listed in configuration (see 7.3.2).

Caution : close relevant shut-off valves.

3.3. **HOLD,** then key-in number of unskipped channel : stops channel normal sampling, for holding on selected channel. Get out that mode as soon as possible, or automatically after 10 minutes.

Caution : the other sampled channels are not managed in that mode.

3.4. **SKIP,** then key-in number of unheld channel and except channel 0 : skip relevant channel. Mainly used for tank filled with water or channel in repair. Same sequence to cancel channel skipping.

Caution : close relevant shut-off valve. Open it after channel skipping.

4. **UTIL :** utilities menu, **RTN** to come back to main menu.

4.1. **TEST** : lights on screen, lamps, buzzer, relays during 3 seconds.

4.2. **PURG.FLT.** : starts a water trap (filter) purge during 15 seconds.

4.3. **PURG.CHL,** then key-in number of unskipped channel and except channel 0 : starts a channel purge for configured time.

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5. **CFG :** configuration menu, then key-in password.

   Note 1 : see message "Configuration in progress" when configuration menu entered by another Control Unit.

   Note 2 : each parameter change is followed by the question "Exit/save ?" : key-in 1 when sure with settings, or 2 to cancel.

   Note 3 : key-in **RTN** in any situation allows to cancel and exit current operation. Key-in **RTN** also for going back to previous or main menu.

5.1. **CHL** : configuration of parameters regarding the channels.

- **TOT** : for entering the channel manual purge duration.
- **SAMP,** then key-in number of channel or 99 for all channels (except channel 0) : for entering successively the channel sampling and purge durations.
- **FLOW,** then key-in number of channel or 99 for all channels (except channel 0) : for entering successively the low and high limits for flow alarm.

5.2. **GAS** : configuration of parameters regarding the Gas detectors.

- **GAS,** then key-in number of gas detector : for entering successively the calibration gas concentration, and the enabling.
- **CHL,** then key-in number of channel or 99 for all channels (except channel 0) : the menu ALRM appears.
- **ALRM** : for entering successively the gas detector number, the type of alarm high or low limits, the pre-alarm and alarm limits.

5.3. **TIME** : for entering successively the year, month, date, hour, minute, second.

5.4. **CAL** : calibration menu.

- **GAS,** then key-in number of gas detector to calibrate.
- **ZERO** first time : to display current flow for suction process gas detector, current, gas concentration. Keep free the calibration gas inlet, with proper air free of gas pollution.
- **ZERO** second time : to calibrate the low extremity of range.
- **SPAN** first time : to display current flow for suction process gas detector, current, gas concentration. Connect, open and adjust the flow of the calibration gas.
- **SPAN** second time : to calibrate the high extremity of range.

3

ENRAF AUXITROL MARINE

Enraf Auxitrol Marine SAS
158, Bureaux de la Colline
92213 Saint Cloud cedex
France

Tel. + 33 (0)1 49 11 65 75
Fax. + 33 (0)1 49 11 65 76
Email : contact@enrafmarine.fr

GASBAL SYSTEM
TECHNICAL DATA QUESTIONNAIRE

NUMBER OF CHANNELS : 19

Number of sampled channels (up to 48) :
Number of detectors (or detector types) for sampling (up to 7) :
Number of individuals channels (remote detectors) (up to 7 – above) :

MAIN POWER SUPPLY VOLTAGE / FREQUENCY : 220Volts, 60Hz

CHANNEL NAMES :

Comments

- individual channel (remote detector) numbers follows sampled channel numbers ;
- shortened name : allowed characters are alphabetic, numeric, " ", "-" ;
- tick check box for sampled channel to be skipped in Handling Mode.

ENRAF AUXITROL MARINE SAS
SUPPORT TECHNIQUE DES VENTES
09 JAN. 2004
VISA :

Channel	Full name	Shortened name (4 charac)				Chk Box
1	Ballast Tank Port 1	B	T	P	1	
2	Ballast Tank Stb. 1	B	T	S	1	
3	Ballast Tank Port 2	B	T	P	2	
4	Ballast Tank Stb. 2	B	T	S	2	
5	Ballast Tank Port 3	B	T	P	3	
6	Ballast Tank Stb. 3	B	T	S	3	
7	Ballast Tank Port 4	B	T	P	4	
8	Ballast Tank Stb. 4	B	T	S	4	
9	Ballast Tank Port 5	B	T	P	5	
10	Ballast Tank Stb. 5	B	T	S	5	
11	Ballast Tank Port 6	B	T	P	6	
12	Ballast Tank Stb. 6	B	T	S	6	
13	Ballast Tank Port 7	B	T	P	7	
14	Ballast Tank Stb. 7	B	T	S	7	
15	Ballast Dist. Channel	B	D	C		
16	Fresh Water Tank Port	F	W	T	P	
17	Fresh Water Tank Stb.	F	W	T	S	
18	Ballast Pump Room 1	B	P	R	1	
19	Ballast Pump Room 2	B	P	R	2	

GAS ALARM LIMITS (pre-alarm, alarm, in relevant detector unit % or ppm) :

Comments :

- channel 0 is Analysing Unit internal atmosphere;
- for LEL detection on general, the default rule is : pre-alarm limit 20%, alarm limit 50%;
- for LEL detection on channel 0, the default rule is : pre-alarm and alarm limit 30%;
- for LEL detection in pump room and equivalent, the default rule is : pre-alarm and alarm limit 10%;

4

1 EC-TYPE EXAMINATION CERTIFICATE

2 **Equipment or Protective System Intended for use**
in Potentially Explosive Atmospheres
Directive 94/9/EC

3 EC-Type Examination Certificate Number : **BAS99ATEX2259X**

4 Equipment or Protective System: **SEARCHPOINT OPTIMA 2 INFRARED POINT DETECTOR**

5 Manufacturer: **ZELLWEGER ANALYTICS LTD**

6 Addrcss: **Hatchpond House, 4 Stinsford Road, Nuffield Estate, Poole, Dorset, BH17 0RZ**

7 This equipment or protective system and any acceptable variation thereto is specified in the schedule to this certificate and the documents therein referred to.

8 The Electrical Equipment Certification Service, notified body number 600 in accordance with Article 9 of the Council Directive 94/9/EC of 23 March 1994, certifies that this equipment or protective system has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of equipment and protective systems intended for use in potentially explosive atmospheres given in Annex II to the Directive.

The examination and test results are recorded in confidential Report N°

99(C)0425 dated 15 November 1999

9 Compliance with the Essential Health and Safety Requirements has been assured by compliance with:

EN 50014: 1997 **EN 50018: 1994**

except in respect of those requirements listed at item 18 of the Schedule.

10 If the sign "X" is placed after the certificate number, it indicates that the equipment or protective system is subject to special conditions for safe use specified in the schedule to this certificate.

11 This EC-TYPE EXAMINATION CERTIFICATE relates only to the design and construction of the specified equipment or protective system. If applicable, further requirements of this Directive apply to the manufacture and supply of this equipment or protective system.

12 The marking of the equipment or protective system shall include the following:-

Ex II 2 G **EEx d IIC T5 (T_{amb} = -40°C to +55°C)**
or T4 (T_{amb} = -40°C to +65°C)

This certificate may only be reproduced in its entirety and without any change, schedule included.

File No: EECS 0981/01/028

I M CLEARE
DIRECTOR
19 November 1999

HSE
Health & Safety

Electrical Equipment Certification Service
Health and Safety Executive
Harpur Hill, Buxton, Derbyshire. SK17 9JN. United Kingdom
Tel: 01298 28000 Fax: 01298 28244

13 **Schedule**

14 **EC-TYPE EXAMINATION CERTIFICATE N° BAS99ATEX2259X**

15 **Description of Equipment or Protective System**

The Searchpoint Optima 2 Infrared Point Detector is rated up to 32V d.c. with a maximum power dissipation of 8 watts. The unit comprises a cylindrical enclosure, manufactured in stainless steel, with a central spigot joint secured by 2 off M5 by 16mm long stainless steel socket head cap screws of grade A4-80. At the rear of the unit is an M25 or ¾" NPT male thread with integral conductors passing through an epoxy cement barrier. At the front of the unit is a quartz window and two heated arms to support an external mirror assembly.

The interior of the unit comprises a series of internal PCB's and an optical assembly dissipating a maximum of 5.5W, together with heater resistors mounted within the support arms dissipating up to a further 2.5W.

Internal earthing is by means of the supply cable and external earth connection facilities are provided at the rear of the central joint fixings.

To obviate the risk of hotspots and capacitor energy storage associated with this unit the cover must not be opened, even when isolated, when a flammable atmosphere is present. Each enclosure is marked with this information

This apparatus is to be installed and used in accordance with the appropriate codes of practice and the manufacturer's instructions.

A range of weather protection covers and sensing heads may also be provided. The sensing heads are for the detection of sporadic explosive atmospheres or calibration only, and are not intended for continuous monitoring/measurement of explosive gas concentrations.

This certification is to European Directive 94/9/EC only. Functional performance and compliance with other European directives is the responsibility of the equipment manufacturer and/or the user as appropriate.

16 **Report Nos.**

99(C)0425

17 **SPECIAL CONDITIONS FOR SAFE USE**

1. The integral supply cables must be mechanically protected and terminated in a suitable terminal or junction facility.
2. For replacement purposes the cover fixing screws shall be grade A4-80 minimum.

13 Schedule

14 EC-TYPE EXAMINATION CERTIFICATE N° BAS99ATEX2259X

18 Essential Health and Safety Requirements

ESSENTIAL HEALTH & SAFETY REQUIREMENTS not covered by Standards listed at (9)

Clause	Subject	Compliance
1.0.2	Analysis of possible operating faults	BASEEFA Report No 99C0425
1.0.3	Special checking and maintenance conditions	Not applicable
1.0.6	Instructions	BASEEFA Report No 99C0425
1.2.2	Components for incorporation or replacement	Not applicable
1.2.4	Dust deposits	Not applicable
1.2.5	Additional means of protection	Not applicable
1.3.5	Hazards arising from pressure compensation	Not applicable
1.4.2	Withstanding attack by aggressive substances	BASEEFA Report No 99C0452
1.5	General requirements for safety devices	Not applicable
1.6.1	Manual override	Not applicable
1.6.2	Emergency shutdown	Not applicable
1.6.3	Hazards arising from power failure	Not applicable
1.6.5	Placing of warning devices as parts of equipment	Not applicable
2.0	Category M	Not applicable
2.1	Category 1	Not applicable
2.2.1	Category 2G	BASEEFA Report No 99C0452
2.2.2	Category 2D	Not applicable
2.3	Category 3	Not applicable
3	Requirements for protective Systems	Not applicable

19 DRAWINGS

Number	Issue	Date	Description
2108E0001	3	22.09.99	General Arrangement, Searchpoint Optima 2 Infrared Point Detector
2108E0002	2	16.07.99	Enclosure Detail, Rear
2108E0003	3	14.09.99	Enclosure Detail, Front
2108E0004	A	04.11.99	Accessories

BASEEFA List Keywords
2GASDETE

1 SUPPLEMENTARY EC-TYPE EXAMINATION CERTIFICATE

2 **Equipment or Protective System Intended for use in Potentially explosive atmospheres Directive 94/9/EC**

3 Supplementary EC-Type Examination Certificate Number **BAS99ATEX2259X/1**

4 Equipment or Protective System: **SEARCHPOINT OPTIMA 2 INFRARED POINT DETECTOR**

5 Manufacturer: **ZELLWEGER ANALYTICS LTD**

6 Address: **Hatchpond House, 4 Stinsford Road, Nuffield Estate, Poole, Dorset, BH17 0RZ**

7 This supplementary certificate extends EC-Type Examination Certificate No. BAS99ATEX2259X to apply to equipment or protective systems designed and constructed in accordance with the specification set out in the Schedule of the said Certificate but having any variations specified in the Schedule attached to this certificate and the documents therein referred to.

This Supplementary Certificate shall be held with the original Certificate

File No: EECS 0981/01/028

This certificate is granted subject to the general conditions of the Electrical Equipment Certification Service. It does not necessarily indicate that the apparatus may be used in particular industries or circumstances.

I M CLEARE
DIRECTOR
17 January 2000

HSE
Health & Safety Executive

Electrical Equipment Certification Service
Health and Safety Executive
Harpur Hill, Buxton, Derbyshire, SK17 9JN, United Kingdom
Tel: 01298 28000 Fax: 01298 28244

CERT/ATEX/EQUIP/CAT1-2 S, Issue 1, Dated September 1998

13 Schedule

14 EC-TYPE EXAMINATION CERTIFICATE N° BAS99ATEX2259X/1

Description of the Variation to the Equipment or Protective System

VARIATION 1.1

Alternative rear casing assembly incorporating RFI filters

In this form the unit is designated

A SEARCHPOINT OPTIMA PLUS

SPECIAL CONDITIONS FOR SAFE USE

See original certificate

Essential Health and Safety Requirements

See original certificate

DRAWINGS

Number	Issue	Date	Description
2108E0001	4	10.12.99	General Arrangement
2108E0002	3	10.12.99	Enclosure Detail, Rear
2108E0004	1	10.12.99	Accessories

Physikalisch-Technische Bundesanstalt

Braunschweig und Berlin

PTB

(1) EG-Baumusterprüfbescheinigung

(2) Geräte und Schutzsysteme zur bestimmungsgemäßen Verwendung in explosionsgefährdeten Bereichen - **Richtlinie 94/9/EG**

(3) EG-Baumusterprüfbescheinigungsnummer

PTB 99 ATEX 4022 X

(4) Schutzsystem: Detonationssicherung „DS 20/MS" und „DS 20/VA"

(5) Hersteller: Firma ASF THOMAS Industries GmbH

(6) Anschrift: Siemensstraße 4, D-82178 Puchheim

(7) Die Bauart dieses Schutzsystems sowie die verschiedenen zulässigen Ausführungen sind in der Anlage zu dieser Baumusterprüfbescheinigung festgelegt.

(8) Die Physikalisch-Technische Bundesanstalt bescheinigt als benannte Stelle Nr. 0102 nach Artikel 9 der Richtlinie des Rates der Europäischen Gemeinschaften vom 23. März 1994 (94/9/EG) die Erfüllung der grundlegenden Sicherheits- und Gesundheitsanforderungen für die Konzeption und den Bau von Geräten und Schutzsystemen zur bestimmungsgemäßen Verwendung in explosionsgefährdeten Bereichen gemäß Anhang II der Richtlinie.

Die Ergebnisse der Prüfung sind in dem vertraulichen Prüfbericht PTB Ex 99-40022 festgelegt.

(9) Die grundlegenden Sicherheits- und Gesundheitsanforderungen werden erfüllt durch Übereinstimmung mit

PTB - Prüfverfahren (Verfahrensanweisungen) in Anlehnung an prEN 12874 „Flammendurchschlagsicherungen"

(10) Falls das Zeichen „X" hinter der Bescheinigungsnummer steht, wird auf besondere Bedingungen für die sichere Anwendung des Schutzsystems in der Anlage zu dieser Bescheinigung hingewiesen.

(11) Diese EG-Baumusterprüfbescheinigung bezieht sich nur auf Konzeption und Bau des festgelegten Schutzsystems gemäß Richtlinie 94/9/EG. Weitere Anforderungen dieser Richtlinie gelten für die Herstellung und das Inverkehrbringen dieses Schutzsystems.

(12) Die Kennzeichnung des Schutzsystems muß die folgenden Angaben enthalten:

Ex II G IIA

Zertifizierungsstelle Explosionsschutz
Im Auftrag

Dr. H. Förster
Regierungsdirektor

Braunschweig, 1999-04-06

Physikalisch-Technische Bundesanstalt • Bundesallee 100 • D-38116 Braunschweig

Physikalisch-Technische Bundesanstalt
Braunschweig und Berlin

PTB

(13)

Anlage

(14) **EG-Baumusterprüfbescheinigung PTB 99 ATEX 4022 X**

(15) Beschreibung des Schutzsystems

Die Detonationssicherung Typ „DS 20/MS" und „DS 20/VA" soll einen Flammendurchschlag bei Deflagrationen und stabilen Detonationen von explosionsfähigen Dampf-Luft- bzw. Gas-Luft-Gemischen der Explosionsgruppe IIA, unter atmospärischem Druck in der vorgeschalteten Rohrleitung ≤ DN 10 verhindern.

Die Sicherung besteht aus einem zylindrischen Gehäuse und einer Flammensperre. Die Flammensperre besteht aus zwei Bandsicherungen. Die Bänder von je 10 mm Breite und 0,2 mm Dicke sind in dichten Lagen spiralförmig aufgerollt. Hierdurch werden dreieckförmige Kanäle von höchstens 0,5 mm gebildet, durch die Dampf-Luft- bzw. Gas-Luft-Gemische strömen können, ein Flammendurchschlag jedoch verhindert werden soll.

Die Bauart, Werkstoffe und Abmessungen sind durch die in der Anlage aufgeführten Zeichnungen und Stückliste festgelegt.

Anforderungen an den Explosionsschutz:

Flammendurchschlagsicher bei Deflagrationen und stabilen Detonationen brennbarer Gase und Flüssigkeiten der Explosionsgruppe IIA mit einer Normspaltweite > 0,9 mm, im Zuge von Rohrleitungen ≤ DN 10, sowie einem maximalen Betriebsdruck von 1,1 bar absolut und bei max. Betriebstemperatur von 60 °C.

(16) Prüfbericht PTB Ex 99-40022 (bestehend aus 4 Seiten, 6 Zeichnungen und einer Stückliste)

Ergebnis: Das Baumuster entspricht den Bestimmungen der Richtlinie 94/9/EG für Schutzsysteme (Unterteilung IIA nach EN 50014). Die Sicherung erfüllt die Anforderungen an den Explosionsschutz wie unter Punkt (15) beschrieben.

(17) Besondere Bedingungen für den sicheren Gebrauch

Beim Einsatz der Detonationssicherung Typ „DS 20/MS" und „DS 20/VA" müssen folgende Bedingungen eingehalten, bzw. erfüllt werden:
- Der Innendurchmesser der Rohrleitung der ungeschützten Seite zwischen der möglichen Zündquelle und der Sicherung darf nicht größer als 10 mm sein.
- Die im Betrieb anfallenden brennbaren Gase bzw. Flüssigkeiten müssen der Explosionsgruppe IIA mit einer Normspaltweite > 0,9 mm angehören.
- Der max. Betriebsdruck darf nicht höher als 1,1 bar abs. sein.
- Die Betriebstemperatur darf 60 °C nicht überschreiten
- Der Umfang der durchgeführten Prüfungen erstreckt sich nicht auf stabilisiertes Brennen, dieser Sachverhalt ist bei der Verwendung zu berücksichtigen.

Physikalisch-Technische Bundesanstalt • Bundesallee 100 • D-38116 Braunschweig

Physikalisch-Technische Bundesanstalt

Braunschweig und Berlin

PTB

Anlage zur EG-Baumusterprüfbescheinigung PTB 99 ATEX 4022 X

(18) Grundlegende Sicherheits- und Gesundheitsanforderungen

Die grundlegenden Anforderungen der ATEX sind erfüllt.

Prüfungsunterlagen:

a) Baumuster der Detonationssicherung Typ „DS 20/MS"
b) Zeichnungen und Stückliste der Detonationssicherung

Zeichnungs-Nr.	Datum
50117	23.06.1996
50125	01.09.1998
50126	01.09.1998
50127	01.09.1998
50188	15.03.1999
2000-0001	30.09.1998
Stückliste	
Anlage 1 (DS 20)	21.01.1999

Zertifizierungsstelle Explosionsschutz
Im Auftrag

Braunschweig, 1999-04-06

Dr. H. Förster
Regierungsdirektor

5

LR Type Approval Certificate

This is to certify that the undernoted product(s) has/have been tested with satisfactory results in accordance with the relevant requirements of the LR Type Approval System.

This certificate is issued to:

PRODUCER	Auxitrol S.A.
PLACE OF PRODUCTION	5 Allée Charles Pathé F-18941 Bourges Cedex 9 France
DESCRIPTION	Automatic Gas Detection and Alarm System
TYPE	GASBAL Comprising: Analysing Unit Control Unit Gas Sampling Accessories
APPLICATION	Marine and offshore applications for ENV1, ENV2 environmental categories as defined in LR Test Specification No. 1: 1996.
ADDITIONAL TESTS	Safe Type Certification according to certificates listed on Design Appraisal Document No. 03/00049, with apparatus coding: Ex d 3n G5 EEx d IIC T4 EEx d IIC T5 EEx dia IIC T4 EEx ia IIC T4 EEx d IIC T6

Certificate No.	03/00049
Issue Date	2 May 2003
Expiry Date	1 May 2008
Sheet	1 of 2

A. Lough

A. Lough
Marine Support Group
London

Lloyd's Register of Shipping
71 Fenchurch Street, London EC3M 4BS

CORPORATE PAPER (01/2003)

"*This Certificate is not valid for equipment, the design, ratings or operating parameters of which have been varied from the specimen tested. The manufacturer should notify LR of any modification or changes to the equipment in order to obtain a valid certificate.*"

The Design Appraisal Document No. 03/00049 and its supplementary Type Approval Terms and Conditions form part of this Certificate.

Certificate No.	03/00049
Issue Date	2 May 2003
Expiry Date	1 May 2008
Sheet	2 of 2

A. Lough

A. Lough
Marine Support Group
London

Lloyd's Register of Shipping
71 Fenchurch Street, London EC3M 4BS

THIS DOCUMENT IS SUBJECT TO THE PROVISIONS ON THE REVERSE

Lloyd's Register of Shipping
71 Fenchurch Street, London, EC3M 4BS
Telephone 020 7709 9166 Fax 020 7488 4796

DESIGN APPRAISAL DOCUMENT

Date	Quote this reference on all future communications
02 May 2003	DAPP/MSG/ECE/TA/W00131087/AMA/O-33209

LLOYD'S REGISTER'S TYPE APPROVAL SYSTEM, 1996.

Issued to: AUXITROL AS
For: AUTOMATIC GAS DETECTION AND ALARM SYSTEM
Type: GASBAL
TYPE APPROVAL CERTIFICATE No. 03/00049

The undernoted documents have been reviewed for compliance with the requirements of Lloyd's Register's Type Approval System, 1996 and this Design Appraisal Document is a supplement to the Certificate.

APPROVAL DOCUMENTATION

Request form	Undated
Auxitrol fax	06.12.2002
LR Lyon memorandum	11.12.2002
Auxitrol letter	11.12.2002
Request for Services form	21.02.2003
Auxitrol fax	21.02.2003
LR Lyon fax	25.02.2003

Data Sheets / Technical Specifications

Functional Description Operating Characteristics ST 1080 Rev. 01	26.07.2001
Drawing M27848 Amd. A	14.09.2001
Drawing M24442 Rev. H	25.03.2002
Drawing M27816 Rev. A	03.07.2001
Drawing M28193 Rev. A	18.07.2001
Drawing M15134 Amd. A	18.07.2001
Drawing M27852 Rev. A	13.09.2001
Drawing M12276 Rev. A	12.07.2001
Drawing M29026 Rev. A	16.11.2001
Drawing M12247 Rev. A	14.01.1997
Drawing M12306 Rev. A	12.07.2001
Drawing M15005 Rev. B	16.11.2001
Drawing M29027 Rev. A	16.11.2001
Drawing M29028 Rev. A	16.11.2001
Drawing M27717 Rev. A	08.11.2002
Drawing M29066 Rev. A	20.11.2001
Drawing M29547 Rev. A	14.01.2002
Drawing M29548 Rev. A	21.01.2002
MT5094E Rev. 02 Technical Manual	undated
MI5094E Rev. 02 Installation Manual	undated
MM5094E Rev. 02 Maintenance Manual	undated
NT5092 Rev 00 Flamgard-4/20 Instruction Manual	undated
NT5095 Rev 00 TXgard-IS Instruction Manual	undated

Lloyd's Register of Shipping
71 Fenchurch Street, London, EC3M 4BS
Telephone 020 7709 9166 Fax 020 7488 4796

Document number	03/00049
Issue number	1

DESIGN APPRAISAL DOCUMENT

Date	Quote this reference on all future communications
02 May 2003	DAPP/MSG/ECE/TA/W00131087/AMA/O-33209

TEST REPORTS

PE 1012 Rev 01 Qualification Test Program	12.07.2001
RQ-01-60206/BLS/IM Edition 1 Climactical Test Report	29.08.2001
RQ-01-60206/1/FP/IM Edition 1 Mechanical Test Report	29.08.2001
RC-01-40179/FL/NG Edition 0 EMC Test Report	20.09.2001
PTB Certificate III B/E-30 173 U	04.02.1985
BASEEFA Certificate Ex 95C1028X	20.02.1995
BASEEFA Certificate BAS98ATEX2157X	06.10.1998
BASEEFA Certificate Ex 90C2357	08.10.1990
BASEEFA Certificate Ex 90C2356	08.10.1990
BASEEFA Certificate Ex 98D1103	07.05.1998
SCS Certificate Ex 97Y1052X	08.07.1997
Management System Certificate for ISO 9001 No. QUAL/1992/683D	04.10.2001
Laboratory Accreditation Certificate	undated

Supplementary Type Approval Terms and Conditions

LR Type Approval certifies that a representative sample of the product(s) referred to herein has/have been found to meet the applicable design criteria for the use specified herein. It does not mean or imply approval for any other use, nor approval of any product(s) designed or manufactured otherwise than in strict conformity with the said representative sample.

LR Type Approval is based on the understanding that the manufacturer's recommendations and instructions and any relevant requirements of the Rules of Lloyd's Register of Shipping are complied with.

This LR Type Approval does not eliminate the need for normal inspection and survey procedures required by the Rules and Regulations of Lloyd's Register of Shipping.

Lloyd's Register of Shipping reserves the right to cancel or withdraw this LR Type Approval Certificate in accordance with the LR Type Approval System Procedure.

A. Lough
AM Anderson

A. Lough
Electrical and Control Engineering
Tel. +44 (0)20 7423 1952
Fax. +44 (0)20 7423 2053

A. M. Anderson
Electrical and Control Engineering
Tel. +44 (0)20 7423 1948
Fax. +44 (0)20 7423 2053

Part 3
Fire and Gas Detection Alarm and Control Systems (Part 3)

Producer/Licence No.	Item Description	Technical Details	Category/Additional Tests	Remarks	Cert. No.
Auxitrol S.A., 5 Allée Charles Pathé, F-18941 Bourges Cedex 9, France.	Automatic Gas Detection and Alarm System Type : GASBAL Comprising: Analysing Unit Control Unit Gas Sampling Accessories	Power supply 115/230Vac	ENV1 ENV2 (1996)	Expires: 1 May 2008 Safe Type Certification according to the following certificates and apparatus coding: PTB III B/E-30 173 U Ex d 3n G5 BASEEFA Ex 95C1028X EEx d IIC T4 BASEEFA BAS98ATEX2157X EEx d IIC T5 BASEEFA Ex 90C2357 EEx dia IIC T4 BASEEFA Ex 90C2356 EEx ia IIC T4 BASEEFA Ex 98D1103 EEx d IIC T6 SCS Ex 97Y1052X EEx d IIC T6	03/00049

Date Issued: 2002-05-02 00:00:00
Certificate Number: 02-SN16894-X

ABS
TYPE APPROVAL PROGRAM

Certificate of Type Approval

This is to certify that Auxitrol, S.A.

has met the requirements of ABS Product Type Approval for

GASBAL Gas Detection System

Model Name(s): GASBAL

Presented to:

Auxitrol, S.A.
System & Sensors Division
5, allee Charles Pathe
Bourges Cedex 9
18941

Intended Service:
MArine and Offshore Application

Description:
The GASBAL system is based on suction process sampling towards one or more common detectors combined with analogue inputs from field gas detectors.

Ratings:
Power Supply: 115/230 VAC 50/60 Hz Control unit operating temperature: 0 to +70°C Enclosure Protection: IP 22 Suction channels number: Up to 48 Number of field detectors: 6 Output power supply for field detectors: 24VDC Control unit power consumption: 260 VA

Service Restrictions:
The GASBAL system control unit is for safe area installation only. Electrical equipment intended for installation in hazardous areas is to be of the certified safe type and is to be selected in accordance with 5-1-7/Table 1 based on the class of hazardous area at its location of installation.

Comments:
Approval Documentation EMC Test Report No. RC-01-40179/FL/NG Rec.0 Environmental Test Report No. RQ-01-60206/BLS/IM Rev.1 Mechanical Test Report No. RQ-01-60206/1/FP/IM Rev.1 Qualification Test

Program No. PE 1012 Rev.1

ABS Rules:
4-9-7/13.1, 13.3 & 5-1-7/17.1.4 of the Steel Vessel Rules, 2002

National Standards:

International Standards:

Government Authority:

Others:

Manager, ABS Programs

ABS has used due diligence in the preparation of this certificate and it represents the information on the product in the ABS Records as of the date and time the certificate was printed. Type Approval requires Drawing Assessment, Prototype Testing and assessment of the manufacturer's quality assurance and quality control arrangements. Limited circumstances may allow only Prototype Testing to satisfy Type Approval. The approvals of Drawings and Products remain valid as long as the ABS Rule, to which they were assessed, remains valid. ABS cautions manufacturers to review and maintain compliance with all other specifications to which the product may have been assessed. Further, unless it is specifically indicated in the description of the product; Type Approval does not necessarily waive witnessed inspection or survey procedures (where otherwise required) for products to be used in a vessel, MODU or facility intended to be ABS classed or that is presently in class with ABS. Questions regarding the validity of ABS Rules or the need for supplemental testing or inspection of such products should, in all cases, be addressed to ABS.

Printed on : 2002-06-17 10:01:18.985

MARINE DIVISION
17 Bis Place des Reflets - La Défense 2
92400 Courbevoie - France
Tel. 33 1 42 91 53 48
Fax 33 1 42 91 28 94

BUREAU VERITAS

Certificate N°:
11951/A1 BV
The attached Schedule forms part of the certificate

File Number : AP 3439
Product Code : 3791H

CERTIFICATE OF TYPE APPROVAL

This is to certify that the product identified below was found to be in compliance with the relevant hereunder stated Regulations & standards

FLAMMABLE GAS DETECTION AND ALARM SYSTEMS

Gas detection and alarm systems.

MANUFACTURED BY:

AUXITROL S.A.
Bourges - FRANCE

SPECIFIED REGULATIONS & STANDARDS :
BV Rules for Classification of Steel Ships, and MODU Code.

The Approval is valid until : 08/04/2007

BUREAU VERITAS NANTES

At Paris la Défense, on : 12/06/2002

J.BENOIT
For the Secretary

This Certificate remains valid until the date stated on, hereunder, unless cancelled or revoked, provided the conditions in the attached schedule are complied with and the equipement remains satisfactory into service.
This Certificate is not valid for equipment, the design or manufacture of which has been varied or modified from the specimen tested.
This Certificate is not valid without the stamp of the above mentioned by BUREAU VERITAS Inspection Center.
The manufacturer should notify BUREAU VERITAS of any modification or changes to the equipment in order to obtain a valid Certificate.
The latest published Regulations or Standards referred to, above, and the Marine Division General Conditions are applicable.

BUREAU
VERITAS
MARINE DIVISION

File Number : AP 3439

THE SCHEDULE OF APPROVAL

1. PRODUCT DESCRIPTION :

Automatic gas detection system **GASBAL** ™

Comprising:
- Gasbal Analysing Unit
- Gasbal Control Unit
- Regulating valve
- Shut-off valve
- Bulkhead penetrations
- Flame arrestor
- Non-return valve

The GASBAL ™ System is dedicated to the gas control in all tanks, void spaces or houses adjacent to cargo storage tanks and handling systems, in order to detect any gas concentration level over programmable limits, and to monitor visible and audible alarms consequently.

2. DESIGN DRAWINGS and/or SPECIFICATIONS

P/N M2442 H
P/N M 27816A, P/N 28193A
P/N M 15134A
P/N M 27852A
P/N M 12306A, P/N M290 27A
P/N M 15005 B, P/N 29028A
P/N M 12247A
P/N M 27717A, P/N M29066A.

3. TYPE TEST REPORTS / LABORATORY RECOGNITION STATUS

- Technical Specification	ST 1080, Rev. 1
- Technical Manual	MT 5094 E, Rev. 2
- Installation Manual	MI 5094 E, Rev. 2
- El. Bloc Diagram	Dated Dec. 01
- Maintenance Manual	MM - 5094 E, Rev. 1
- Various drawings filed in AP 3439.	

4. MATERIALS and/or COMPONENTS REQUIRED TO BE TYPE APPROVED

None.

5. OTHER MATERIALS and/or COMPONENTS

None.

6. APPLICATION / LIMITATION OF USE

6.1 - As per requirements of BV Rules.

6.2 - For each application, gas detectors are to be approved according to gas type and location.

6.3 - GASBAL ™ analysing and control unit should be located in a "non-hazardous area".

6.4 - For each application with equipment fitted in hazardous area, a specific study is to be carried out.

7. PRODUCTION SURVEY REQUIREMENTS

7.1 - Arrangements shall be made for a Society's Surveyor to carry out, on a periodic basis, visits of the Manufacturer's premises and product audits.

7.2 - Each equipment is to be supplied with manual(s) for installation, use and maintenance.

8. ON BOARD INSTALLATION & MAINTENANCE REQUIREMENTS

As per requirement of regulation stated on the front page of the certificate under the supervision of a Society's Surveyor.

9. MARKING FOR IDENTIFICATION

- Maker's name or trade mark.
- Serial number of the units.
- Equipment type or model identification under which it was type-tested.

10. OTHERS

10.1 -This approval is given with the understanding that the Society reserves the right to require check tests to be carried out on the gas detection system at any time, and that **AUXITROL S.A., Bourges - France,** will accept full responsibility for informing shipbuilders, shipowners or their sub-contractors of the proper methods of use and general maintenance of the units and the conditions of this approval.

10.2 - This certificate supersedes the Type Approval Certificate N° 11951/A0 BV issued on 08/04/2002 by the Society.

*** ***Last page (End of Document)*** ***

BUREAU VERITAS NANTES

le 13/06/02.

DET NORSKE VERITAS

TYPE APPROVAL CERTIFICATE

CERTIFICATE NO. A-8251
This Certificate consists of 3 pages

This is to certify that
Automatic Gas Detection System
with type designation(s)

GASBAL ™

Manufactured by
Auxitrol S.A.
BOURGES CEDEX 9, France

is found to comply with
Det Norske Veritas' Rules for Classification of Ships and Mobile Offshore Units

Application/limitations

Location classes:

Temperature	B
Humidity	B
Vibration	B
EMC	A
Enclosure	Required protection according to the Rules to be provided upon installation on board.

Place and date
Høvik, 2002-02-15
for DET NORSKE VERITAS AS

Knut-Helge Knutsen
Head of Section

DET NORSKE VERITAS 1864

Local Office
DNV Paris (Rueil Malmaison)

This Certificate is valid until
2004-06-30

Rolf O. Bjørvik
Surveyor

Notice: This Certificate is subject to terms and conditions overleaf. Any significant change in design or construction may render this Certificate invalid. The validity date relates to the Type Approval Certificate and not to the approval of equipment/systems installed.

Product description

The Type Approval covers the following products:

Products	Type Nos.
GASBAL™ Analysing Unit	P/N M24442F
GASBAL™ Control Unit	P/N M27816A, P/N 28193A
Regulating Valve	P/N M15134a
Shut-off Valve	P/N M27852A
Bulkhead Penetrations	P/N M12306A, P/N M29027A,
	P/N M15005B, P/N 29028A
Flame Arrestor	P/N M12247A
Non-return Valve	P/N M27717A, P/N M29066A

Software Version Nos.:	
Analysing Unit:	Soft 1016 Version V2
Control Unit:	Soft 1017 Version V2

Application/Limitation

System application to be in accordance with relevant parts of the Rules

For each installation, gas detectors are to be approved according to gas type and location.

Approval conditions

The Type Approval covers hardware and software listed under Product description.

The following documentation of the actual application is to be submitted for approval in each case:

Reference to this type approval certificate.
System Block diagram
Power Supply arrangement (may be part of the System block diagram)
Arrangement drawings showing location of suction points/detectors and central units, including external piping design data
Gas detectors Specifications
Instruction Manual covering each specific installation to be available on board.

As long as the units are covered by the Type Approval, no Product certificate will be required according to Pt.4 Ch.9 Sec.1 A 202.

Cert. No.: A-8251
File No.: 473.31

Clause for application software control.
All changes in software are to be recorded as long as the system is in use on board. Major changes in the software are to be approved before installed. Approval Test of Application Software may be required.

Type Approval documentation

Documentation	Documentation Reference Nos.
Technical Specification	ST 1080, Rev. 1
Technical Manual	MT5094E Rev. 00
Installation Manual	MI5094E Rev. 01
Mech. Drawings	M24442 Rev. F, M27816, Rev. A
	M28193 Rev. A, M15134 Rev. A2
	M12306 Rev A1, M29027 Rev. A
	M29028 Rev. A, M15005 Rev. B
	M12247 Rev. A, M27717 Rev. A
	M29066 Rev. A
El. Bloc Diagram	Dated December 2001
Certificate	PTB Nr. III B/E-30 173 U
Test Report EMITECH No.:	RQ-01-60206/BLS/IM, dtd. 2001-08-29
	RQ-01-60206/1/FP/IM, dtd. 2001-08-29
	RC-01-40179/FL/NG, dtd. 2001-09/20

Certificate retention survey

The scope of the retention/renewal survey is to verify that the conditions stipulated for the type are complied with, and that no alterations are made to the product design or choice of systems/components/materials.

The main elements of the survey are:

- Inspection of factory samples, selected at random from the production line (where practicable)
- Review of production and inspection routines, including test records from product sample tests and control routines.
- Ensuring that systems/components/materials used comply with type approved documents and/or referenced system/component/material specifications.
- Review of possible changes in design of systems/components/materials, software version and performance, and make sure that such changes do not affect the type approval given.
- Ensuring traceability between manufacturer's product type marking and the type approval certificate.
- Ensuring that type approved documentation is available.

Survey to be performed at renewal of this certificate.
END OF CERTIFICATE

Type Approval Certificate

Germanischer Lloyd

This is to certify that the undernoted product(s) has/have been tested in accordance with the relevant requirements of the GL Type Approval System.

Certificate No.	**42 545 - 02 HH**
Company	**Auxitrol S.A.** **5, Allee Charles-Pathe** **F-18941 Bourges Cedex 9**
Product Description	**GAS Detection and Alarm System** **for Ships and Mobile Offshore Units**
Type	**GASBAL™**
Environmental Category	**C, EMV2**
Technical Data / Range of Application	**GASBAL™ Analysing Unit P/N M24442H** **Power Supply : 115 / 230 V AC 50/60 Hz** **Power Consumption : 260 VA with 1 Control Unit; 270 VA with 3 Control Units** **GASBAL™ Control Unit P/N M27816A, P/N 28193A** **Power Supply : 24 V DC from Analysing Unit** **Regulating Valve P/N M15134A; Shut-off Valve P/N M27852A** **Bulkhead Penetrations P/N M12306A, P/N M29027A, P/N M15005B, P/N 29028A** **Flame Arrestor P/N M12247A** **Non-return Valve P/N M27717A, P/N M29066A** **Software Version** **Soft 1016 Version V2 Analysing Unit** **Soft 1017 Version V2 Control Unit**
Test Standard	**Regulations for the Performance of Type Tests Part 1, Edition 2001** **Regulations for the Use of Computers and Computer Systems**
Documents	**Test Reports : RQ-01-60206/BLS/IM dated 29-08-2001; RQ-01-60206/1/FP/IM dated 29-08-2001; RC-01-40179/FL/NG dated 20-09-2001;Software Questionnaire Requirement Class 3 dated 26-06-2002; Technical Manual MT5094 Rev. 02; Installation Manual MI5094E Rev. 02; Maintenance Manual MM5094E Rev. 02**
Remarks	**None**
Valid until	**2007-07-01**

Page **1** of **1**

Type Approval Symbol GL

File No. I.A.02

Hamburg, 2002-07-02

Germanischer Lloyd

i.V. J. Wittburg

i.A. A. Grün

This certificate is issued on the basis of "Regulations for the Performance of Type Tests, Part 0, Procedure".

[redacted]

TYPE APPROVAL CERTIFICATE
No. **MAC13002CS1**

This is to certify that the product below is found to be in compliance with the applicable requirement of the RINA type approval system.

Description	**Automatic gas detection system**
Type	**GASBAL TM**
Applicant	**AUXITROL S.A.** **5 allèe Charles Pathè, 18941 Bourges Cedex 9** **FRANCE**
Manufacturer	**AUXITROL S.A.**
Place of manufacture	**5 allèe Charles Pathè, 18941 Bourges Cedex 9** **FRANCE**
Reference standards	**Rules for the classification of ships.- Part C - Machinery, systems and fire protection. - Ch.3, Sect. 6, Table 1.**

Issued in **Genoa** on **June 17, 2002**. *This Certificate is valid until* **June 17, 2007**

RINA

This certificate consists of this page and 1 enclosure

RINA Società per azioni
Gruppo REGISTRO ITALIANO NAVALE
Via Corsica, 12 - 16128 Genova
Tel +39 010 53851
Fax +39 010 5351000

[redacted]

TYPE APPROVAL CERTIFICATE
No. MAC13002CS1
Enclosure - Page 1 of 1

GASBAL TM

System description:

GASBAL TM Gas detection system include the following products:

GASBAL TM Analysing Unit	P/N M24442F
GASBAL TM Control Unit	P/N M27816A, P/N 28193A
Regulating Valve	P/N M15134A
Shut-off Valve	P/N M27852A
Bulkhead penetrations	P/N M12306A, P/N M29027A, P/N M15005B, P/N 29028A
Flame Arrestor	P/N M12247A
Non - return valve	P/N M27717A, P/N M29066A

Software Versions:

Analysing unit	Soft 1016 Version V2
Control unit	Soft 1017 Version V2

Technical description:

Technical specification ST1080 Rev. 01
Technical manual - GASBAL TM System, doc. n. MT5094E Rev. 02
Installation manual - GASBAL TM System, doc. n. MI5094E Rev. 02
Maintenance manual - GASBAL TM System, doc. n. MM5094E Rev. 02
Drawing n. M24442 rev. H (2 pages)

Test Reports:

EMITECH EMC test: RC-01-40179/FL/NG
EMITECH Climatic test: RQ-01-60206/BLS/IM
EMITECH Mechanical test: RQ-01-60206/1/FP/IM

Limitations:

System application to be in accordance with relevant parts of the Rules
For each installation, gas detectors are to be approved according to gas type and location

Approval condition:

The type Approval covers hardware and software listed under product description

For each application, the following documentation is to be submitted for approval:
- Reference to this type approval certificate
- System block diagram
- Power Supply arrangement
- Arrangement drawings showing location of suction points/detectors and central units, including external piping design data
- Gas detectors specifications

Remarks:

This type approval certificate is also valid for Offshore applications.

Instruction Manual covering each specific installation to be available on board.
All changes in software are to be recorded as long as the system is in use on board.
Major changes in the software are to be approved before installed.
Approval test of Application Software may be required.

Genoa June 17, 2002

RINA Società per azioni
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